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Who Speaks Matters: Authority-Aware Multi-View Parliamentary RAG

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Abstract

Citizens seeking balanced information on political issues face a fundamental challenge: parliamentary proceedings are voluminous, fragmented, and difficult to navigate without bias. Consider a voter trying to understand where political parties stand on an upcoming referendum: they may find themselves overwhelmed by the volume of proceedings, uncertain which sources to trust, and unable to determine whether the information they encounter represents the full spectrum of political opinion. Large Language Models (LLMs) offer a way to obtain quick summaries of parliamentary debates, but risk favouring dominant actors, providing out-of-context quotes, and failing to represent all points of view.

This thesis addresses this challenge with a primary objective: to treat equally both **what is said** and **who is saying it**, allowing citizens to be presented with multiple viewpoints on parliamentary debate. This objective is reflected in the title, which combines a **Topic-Aware Speaker Authority Score**, that dynamically estimates speaker credibility given the topic, with a multi-view representation of political positions, built upon a Retrieval-Augmented Generation (RAG) system applied to Italian parliamentary transcripts.

The dissertation proposes a RAG system formed by four interconnected modules. First, a query-dependent authority scoring module estimates speaker credibility on the basis of six weighted components (profession, education, committee memberships, legislative acts, speech interventions, and institutional role) with a coalition-crossing invalidation mechanism. Second, a dual-channel retrieval module combines dense semantic search with knowledge graph traversal to identify speeches relevant to the query and representative of all ten parliamentary groups. Third, an offset-based citation module extracts citations deterministically from character-level spans in source transcripts, providing fully verifiable quotes. Fourth, a four-stage generation pipeline (Claim Analyst, Sectional Writer, Narrative Integrator, and Citation Surgeon) implements a citation-first principle to ensure semantic correspondence between the generated narrative and the quoted evidence.

The system is evaluated against Google NotebookLM on 15 policy topics through a two-level protocol combining automated metrics with a blind A/B evaluation by 6 domain experts. The system outperforms NotebookLM on parliamentary group coverage (97% vs. 95%) and on citation faithfulness (100% vs. 95%). Human evaluators rate both systems on a 1–5 Likert scale across 9 dimensions: evaluators rate the system higher on source and balance dimensions (Cohen's d up to 0.35), while overall satisfaction is essentially at parity (4.24 vs. 4.27 out of 5).

Contents

1 Introduction **10**

1.1 Motivation . . . 10
1.2 Research Questions . . . 12
1.3 Contributions . . . 13
1.4 Summary of Experimental Results . . . 15
1.5 Scope and Limitations . . . 17
1.6 Reproducibility and Public Access . . . 18
1.7 Thesis Outline . . . 18

2 Background **20**

2.1 Natural Language Processing and Information Retrieval . . . 20
2.1.1 Natural Language Processing . . . 20
2.1.2 Information Retrieval . . . 21
2.2 Retrieval-Augmented Generation (RAG) . . . 22
2.3 Query Expansion and Rewriting . . . 23
2.4 Dense Passage Retrieval . . . 24
2.5 Graph-Based Retrieval . . . 25
2.6 Hybrid Retrieval and Result Fusion . . . 25
2.7 Result Diversification and Fairness-Aware Ranking . . . 26
2.8 Dimensionality Reduction and Density Estimation . . . 27
2.8.1 Principal Component Analysis . . . 27
2.8.2 Kernel Density Estimation . . . 27
2.8.3 Contrastive Term Weighting . . . 28
2.9 Multi-Perspective Text Generation . . . 28
2.10 Citation Generation and Faithfulness . . . 29
2.10.1 Attribution in Language Models . . . 29
2.10.2 Hallucination in Language Models . . . 29

3 Related Work **31**

3.1 Parliamentary NLP: Corpora, Tasks, and Search Systems . . . 31

3.1.1 Corpora and Benchmarks . . . 31
3.1.2 Common Tasks . . . 32
3.1.3 Existing Search Systems . . . 32
3.2 Computational Political Text Analysis . . . 33
3.2.1 Ideological Scaling . . . 33
3.2.2 Political Bias in Language Models . . . 33
3.2.3 Multi-Perspective Generation in Political Contexts . . . 34
3.3 Authority and Expertise Modelling . . . 34
3.3.1 Link-Based Authority . . . 34
3.3.2 Expert Finding . . . 35
3.3.3 Source Reliability in RAG . . . 35
3.4 Domain-Specific RAG Systems . . . 35
3.4.1 Legal RAG . . . 35
3.4.2 Medical RAG . . . 36
3.4.3 Domain Adaptation of RAG . . . 36
3.5 Citation and Attribution in RAG . . . 36
3.6 Identified Gaps and Positioning . . . 37

4 System Architecture **39**
4.1 Design Principles and Requirements . . . 39
4.2 High-Level Overview . . . 40
4.2.1 Presentation Layer . . . 40
4.2.2 Application Layer . . . 44
4.2.3 Data Layer . . . 44
4.2.4 External Services . . . 45
4.3 Data Acquisition and Preprocessing . . . 45
4.3.1 Institutional Data Sources . . . 45
4.3.2 Text Preprocessing and Alignment Map . . . 45
4.3.3 Chunking Strategy . . . 46
4.4 Data Model and Knowledge Graph Schema . . . 46
4.4.1 Node Types . . . 47
4.4.2 Relationship Types . . . 49
4.4.3 Indexing Strategy . . . 50
4.5 Processing Pipeline . . . 50

5 System Components **53**
5.1 Pre-Retrieval Query Rewriting . . . 53
5.1.1 Problem Definition . . . 53

5.1.2 Proposed Approach . . . 53
5.2 Dual-Channel Retrieval . . . 54
5.2.1 Problem Definition . . . 54
5.2.2 Dense Channel: Vector Similarity Search . . . 55
5.2.3 Graph Channel: Structural Traversal . . . 55
5.2.4 Channel Merging and Ranking . . . 56
5.3 Topic-Dependent Authority Scoring . . . 57
5.3.1 Positioning and Design Rationale . . . 57
5.3.2 Component Definitions . . . 59
5.3.3 Temporal Decay . . . 61
5.3.4 Coalition Logic . . . 62
5.4 Ideological Compass . . . 62
5.4.1 Problem Definition . . . 62
5.4.2 Axis Discovery via Weighted PCA . . . 62
5.4.3 Projection and Filtering . . . 63
5.4.4 Group Clustering . . . 63
5.4.5 Evidence Binding . . . 63
5.4.6 Axis Labelling . . . 63
5.5 Four-Stage Generation Pipeline . . . 63
5.5.1 Problem Definition . . . 63
5.5.2 Stage 1: Claim Analyst . . . 64
5.5.3 Stage 2: Sectional Writer . . . 65
5.5.4 Stage 3: Narrative Integrator . . . 65
5.5.5 Stage 4: Citation Surgeon . . . 65
5.6 Citation Integrity System . . . 66
5.6.1 Citation Registry . . . 66
5.6.2 Citation-First Architecture . . . 67
5.6.3 Coherence Validation . . . 67
5.6.4 Semantic Deduplication . . . 67
5.7 A/B Evaluation Infrastructure . . . 68
5.7.1 Automated Metrics . . . 68
5.7.2 Human A/B Survey . . . 68

6 Experimental Evaluation 69
6.1 Research Questions . . . 69
6.2 Benchmark Construction . . . 70
6.2.1 Participatory Topic Collection . . . 71
6.2.2 Topic Selection and Justification . . . 71

6.3 Evaluation Metrics . . . 72
6.3.1 Level 1: Automated Metrics . . . 72
6.3.2 Level 2: Human Evaluation . . . 74
6.4 Baseline: Google NotebookLM . . . 74
6.5 Experimental Protocol . . . 75
6.5.1 Evaluator Population . . . 75
6.5.2 A/B Testing Procedure . . . 76
6.5.3 Statistical Analysis Plan . . . 76
6.5.4 Power Analysis . . . 77

7 Experimental Results **80**
7.1 Automated Metrics (Level 1) . . . 80
7.2 Human Evaluation Results (Level 2) . . . 82
7.2.1 Per-Dimension Ratings . . . 82
7.2.2 Preference Distributions . . . 83
7.2.3 Source Authority Assessment . . . 84
7.3 Statistical Tests . . . 86
7.4 Inter-Rater Agreement . . . 87
7.5 Qualitative Analysis . . . 88
7.5.1 Where ParliamentRAG Outperforms NotebookLM . . . 88
7.5.2 Where NotebookLM Outperforms ParliamentRAG . . . 89
7.5.3 Dimensions at Parity . . . 89
7.5.4 Per-Topic Consistency of Advantages . . . 89
7.5.5 Example: *Aiuti militari all'Ucraina* . . . 91
7.6 Synthesis per Research Question . . . 92
7.7 Limitations of the Experimental Evidence . . . 93

8 Conclusion **95**
8.1 Summary of Contributions . . . 95
8.2 Implications for Democratic Information Access . . . 96
8.3 Ethical Considerations . . . 98
8.4 Future Work . . . 99
8.4.1 Data-Driven Optimisation of Model Parameters . . . 100
8.4.2 Domain-Specific Embedding Models . . . 100
8.4.3 Integration of Heterogeneous Sources . . . 100
8.4.4 Temporal Extension: Beyond XIX Legislature . . . 100
8.4.5 Bicameral Extension: Integration of the Senate . . . 101
8.4.6 Cross-National Extension: European Parliamentary Families . . . 101

A Knowledge Graph Schema **115**
A.1 Node Types . . . 115
A.2 Relationship Types . . . 116
A.3 Indexes . . . 116

B Configuration Parameters **119**
B.1 Retrieval Parameters . . . 119
B.2 Authority Scoring Parameters . . . 120
B.3 Compass Parameters . . . 120
B.4 Generation Parameters . . . 121
B.5 Query Rewriting Parameters . . . 121
B.6 Coalition Definitions . . . 122

C Commissioni Topics Ontology **123**

D API Specification **124**
D.1 Chat and Query . . . 124
D.2 Evidence . . . 124
D.3 Search and Graph Exploration . . . 125
D.4 Authority Ranking . . . 125
D.5 Configuration . . . 125
D.6 History, Evaluation, and Baseline . . . 126

E System Prompt Templates **127**
E.1 Stage 1: Claim Analyst . . . 127
E.2 Stage 2: Sectional Writer . . . 127
E.3 Stage 3: Narrative Integrator . . . 128
E.4 Stage 4: Citation Surgeon . . . 128

F NotebookLM Baseline Prompt **129**

G Evaluation Dataset Schema **131**

H Example: Full Responses **132**
H.1 ParliamentRAG Response . . . 132
H.2 NotebookLM Response . . . 134
H.3 Key Differences . . . 136

List of Figures

1.1 ParliamentRAG interface overview 19

4.1 Three-layer system architecture of ParliamentRAG 41
4.2 Citation detail with verbatim excerpt 42
4.3 Ideological compass for justice reform 42
4.4 Expert card with authority score breakdown 43
4.5 A/B evaluation interface 43
4.6 Neo4j knowledge graph schema for parliamentary proceedings 48
4.7 End-to-end query processing pipeline 52

5.1 Dual-channel retrieval architecture with composite merging 54
5.2 Topic-dependent authority model architecture 57
5.3 Five-stage ideological compass pipeline 62
5.4 Four-stage generation pipeline 64

7.1 Automated metrics comparison 81
7.2 Authority utilisation per topic 82
7.3 Mean human ratings per dimension 84
7.4 Preference distribution per dimension 85
7.5 Radar chart of human evaluation dimensions 86

List of Tables

4.1 Service domains in the ParliamentRAG application layer. 44
4.2 Principal node types in the knowledge graph. 47
4.3 Principal relationship types in the knowledge graph. 49

5.1 Composite scoring components and default weights in the channel merger. 56
5.2 Authority score components and configured weights 60

6.1 From literature gaps to research questions, components, and evaluation . 70
6.2 List of 15 benchmark topics used in the evaluation 78
6.3 Nine dimensions of the human evaluation protocol 79
6.4 Architectural comparison: ParliamentRAG vs. NotebookLM 79
6.5 Profile of the six domain expert evaluators 79

7.1 Automated metrics: ParliamentRAG vs. NotebookLM. 80
7.2 Mean ratings (1–5) per dimension: ParliamentRAG vs. NotebookLM. . . 83
7.3 A/B preference counts per dimension (all evaluations). 83
7.4 Per-group authority preference votes 87
7.5 Statistical summary: Wilcoxon + Cohen's d per dimension. 87
7.6 Inter-rater agreement (Krippendorff's α, ordinal). 88
7.7 Per-topic rating differences 90
7.8 Speaker comparison on *Aiuti militari all'Ucraina* 91

A.1 Parliamentary actors nodes 115
A.2 Parliamentary proceedings nodes 116
A.3 Legislative acts nodes 116
A.4 Chat history and survey nodes 116
A.5 Relationship types in the knowledge graph with cardinalities. 117
A.6 Vector indexes. 117
A.7 Range indexes for property-based filtering. 118

B.1 Retrieval module parameters. 119
B.2 Authority scoring parameters. 120

B.3 Ideological compass parameters. 120
B.4 Generation pipeline parameters. 121
B.5 Query rewriting parameters. 121
B.6 Coalition definitions for the XIX Legislature 122

D.1 Chat and query endpoints. 124
D.2 Evidence endpoints. 124
D.3 Search and graph exploration endpoints. 125
D.4 Authority ranking endpoints. 125
D.5 Configuration endpoints. 125
D.6 History, evaluation, and baseline endpoints. 126

Chapter 1

Introduction

1.1 Motivation

Parliamentary debates constitute one of the richest records of democratic deliberation. Every floor speech, legislative act, and committee hearing represents an elected representative's position on a matter of public concern. For citizens, journalists, and researchers alike, access to this material is essential, yet the sheer volume of parliamentary proceedings has grown to a scale at which manual analysis is no longer feasible.

This thesis addresses the problem of enabling structured, multi-perspective access to parliamentary speech at scale, studying it in the context of the Italian parliamentary system. Italy is a parliamentary republic in which legislative power is vested in a bicameral Parliament: the Chamber of Deputies (*Camera dei Deputati*) and the Senate of the Republic (*Senato della Repubblica*). Following the 2020 constitutional reform, the Chamber comprises 400 deputies and the Senate 200 senators. Both chambers hold equivalent legislative powers.

The work focuses exclusively on data from the Italian Chamber of Deputies, for three reasons. First, its larger membership generates a substantially higher volume of speeches, providing a richer corpus. Second, political debate in the Chamber tends to be more vigorous and confrontational than in the Senate, making it a more demanding testbed for a system designed to balance competing political perspectives. Third, the Chamber's open data infrastructure (including `dati.camera.it`[1]) offers comprehensive and well-structured metadata accessible via SPARQL queries.

Four specific challenges arise when building an automated system for multi-view parliamentary topic summarisation.

Figure 1.1 illustrates the system in operation on a representative query.

[1] https://dati.camera.it/

Volume and fragmentation. As of February 2026, the XIX Legislature of the Italian Chamber has produced 608 plenary sessions containing 6,010 debates and 40,416 individual speeches, segmented into 151,073 retrieval chunks. A single policy issue (such as the separation of judicial careers) may be discussed across hundreds of speeches, spread over months, delivered by deputies from ten different parliamentary groups. No citizen can be expected to locate, read, and synthesise this material unaided.

Speaker heterogeneity. Not all speakers are equally authoritative on every topic. A deputy who sits on the Justice Committee, holds a law degree, and has co-signed multiple legislative acts on judicial reform carries a different weight on the separation of careers than one whose expertise lies in agricultural policy. Existing parliamentary search systems treat all speakers uniformly, ignoring the rich metadata that could inform a more nuanced assessment of topical authority.

Representational bias. Standard retrieval systems, including those based on dense passage retrieval [1], maximise relevance to the input query [2]. When applied to a parliamentary corpus, this returns the top-k most similar passages regardless of political provenance. Because the number of candidate chunks varies substantially across parliamentary groups (ranging from 1,295 interventions for Noi Moderati to 8,666 for Movimento 5 Stelle in the XIX Legislature), groups with fewer speeches have fewer opportunities to appear in the retrieved pool. Without an explicit balancing mechanism, the retriever's output reflects the distributional asymmetry of the corpus rather than the political landscape of the debate.

Citation integrity. Large language models (LLMs) are prone to hallucination: they may paraphrase quotes in ways that alter their meaning, attribute statements to the wrong speaker, or generate plausible-sounding claims with no basis in the source documents [3]. In political discourse, where precise attribution matters, such errors are potentially harmful.

Retrieval-Augmented Generation (RAG) [4] offers a principled framework for addressing volume and fragmentation: a retriever surfaces relevant passages and a generator synthesises them into a fluent response grounded in source material. In principle, a RAG system could retrieve speeches across the political spectrum and produce a balanced multi-view summary with citations. In practice, however, building a RAG system naïvely on parliamentary data inherits the representational bias, absence of speaker credibility signals, and citation integrity risks identified above. Dense retrieval models trained on general-purpose English corpora may not transfer well to Italian parliamentary proceedings. Language models tend toward moderate, consensus-seeking positions, thereby flatten-

ing genuine political disagreements [5]. The generation process can introduce further distortions: paraphrased quotes that drift from their source, misattributed citations, or hallucinated claims.

These considerations lead to the motivating question of this thesis:

> *How can a retrieval-augmented generation system over parliamentary records be designed to guarantee multi-perspective coverage across all parliamentary groups, verbatim citation verifiability, and topic-dependent modelling of speaker authority?*

1.2 Research Questions

The motivating question is decomposed into four research questions.

RQ1 Multi-View Retrieval. *How can a retrieval architecture ensure that the evidence it returns is representative of all parliamentary groups' positions, rather than biased toward the most vocal or most frequent speakers?*

Research on fairness in information retrieval has shown that standard ranking algorithms can systematically disadvantage certain groups [6]. In the parliamentary setting, the relevant notion of fairness is political: the retrieval output should provide balanced representation across all parliamentary groups, so that the generator can produce a genuinely multi-view synthesis.

RQ2 Speaker Authority. *How can a speaker's topical authority be modelled dynamically, depending on the subject of the user's query, so that the system surfaces the most authoritative voices for each specific topic?*

This question connects to two established research areas. First, *expert finding* [7] distinguishes between profile-based methods (constructing an expertise profile a priori) and document-based methods (inferring expertise at query time). Second, link-based ranking methods such as PageRank [8] model authority through structural relationships. However, PageRank computes a single, query-independent score, whereas parliamentary authority must be *query-dependent* (varying by topic) and *time-aware* (recent activity is more relevant). These requirements motivate the topic-dependent authority model developed in this thesis.

RQ3 Citation Integrity. *How can the system guarantee that every quotation in its output is a verbatim passage from an official parliamentary transcript, attributable to a specific speaker?*

Studies have documented that unfaithful citations are common in LLM-generated text and have proposed techniques for improving attribution [9, 10]. This thesis addresses the problem through a deterministic, offset-based mechanism that guarantees citation integrity by construction.

RQ4 Balanced Synthesis. *How can a language model generate a multi-view summary that fairly represents all parliamentary positions, including conflicting viewpoints, without flattening disagreements or omitting minority perspectives?*

Work on multi-perspective summarisation has developed methods for producing summaries that reflect multiple viewpoints [11]. The parliamentary setting introduces a stricter constraint: the *mandatory inclusion* of every parliamentary group's perspective, even when that group has produced fewer speeches. This requirement cannot be met through prompting alone; it demands architectural enforcement.

1.3 Contributions

This thesis makes four contributions, each corresponding to one research question. It is important to distinguish between the *modelling contribution* (the topic-dependent authority model, which constitutes the original scientific contribution) and the *engineering contributions*, which coordinate and adapt established techniques to the parliamentary domain.

C1 — Topic-Dependent Authority Scoring (RQ2). The primary and original scientific contribution is the formalisation of a query-dependent authority model:

$$\text{authority}(s, q) = \text{percentile}\left(\sum_{i=1}^{6} w_i \cdot c_i(s, q)\right)$$

where s is a speaker, q is the user's query, and the six component functions c_i capture profession, education, committee membership, time-decayed legislative acts, time-decayed speech frequency, and institutional role. The model incorporates exponential time decay with a 548-day half-life (Equation 5.9), a coalition-crossing invalidation mechanism that discards authority accumulated under a prior coalition affiliation, and percentile-based normalisation across all speakers evaluated for the current query. The authority score is computed *after* retrieval. This is a deliberate architectural choice motivated by three considerations. First, the broad retrieval pool (default $k = 200$ for the dense channel, augmented by graph-channel traversal) ensures that speakers who are topically active are already present in the evidence set before authority is ever evaluated, since parliamentary

authority on a subject presupposes having spoken about it. Second, an authority-first design (computing scores for all ∼387 deputies before retrieval and using them as a filter) would risk a feedback loop in which only established actors are retrieved, potentially excluding smaller parliamentary groups and violating the multi-view objective. Third, the six-component model requires per-query embedding comparisons for each evaluated speaker; in a live interactive system (Section 1.6), restricting this computation to the speakers in the retrieved pool bounds latency to acceptable levels. In the current pipeline, authority scores are used operationally downstream: to select the top-authority speaker per parliamentary group and, within the generation pipeline, to choose citing speakers and resolve semantic duplicates. The composite merging function (Equation 5.2) retains an authority term with weight 0.05 as an architectural placeholder for a future variant with pre-computed topic-cluster scores (see Section 5.2.4). Authority thus influences the final output as a ranking and selection signal without acting as a hard retrieval filter, preserving the multi-view objective. This model extends classical link-based authority [12] and expert finding [7] by introducing query dependence, temporal awareness, and coalition-aware logic. The combination of these three properties has no direct precedent in the literature. The formal specification is presented in Section 5.3.

C2 — Dual-Channel Retrieval with Diversity Constraints (RQ1). The retrieval module first applies an optional query rewriting step that uses a language model to expand short or ambiguous queries (e.g. acronyms like *SSN*, or proper names like *Bolkestein*) into richer keyword sequences [13]. The expanded query is then fed to both a *dense channel* (vector similarity search over speech chunk embeddings [1]) and a *graph channel* (traversal of the parliamentary knowledge graph from legislative acts to signatories to speeches). A five-component merging function (Equation 5.2) balances relevance, speaker diversity, party coverage, authority, and chunk salience. While the individual techniques are established, their coordination under coverage and diversity constraints adapted to the parliamentary domain constitutes an engineering contribution. The architecture is detailed in Section 5.2.

C3 — Offset-Based Citation Integrity (RQ3). Each speech is chunked during preprocessing and character offsets (`start_char_raw`, `end_char_raw`) are stored for every chunk relative to the original transcript. Citations are produced by $quote = speech.text[span_start : span_end]$, not by fuzzy string matching or LLM paraphrase [9]. A Citation Registry tracks every citation across all four generation stages (Section 5.6.1), a Coherence Validator checks semantic alignment between the integrated narrative and source sections (Section 5.6.3), and an Integrator Guard prevents citation loss during

narrative integration (Section 5.5.4). This approach eliminates citation hallucination by design; the verification procedure is described in Section 5.5.5.

C4 — Four-Stage Generation Pipeline with Citation-First Discipline (RQ4). The generation pipeline decomposes multi-view parliamentary topic summarisation into four stages: a Claim Analyst that decomposes the query into atomic claims and identifies evidence requirements per parliamentary group (Section E.1); a Sectional Writer that produces one section per group (up to ten), grounded in group-specific position briefs (Section 5.5.3); a Narrative Integrator that merges the sections into a coherent whole under coherence validation and citation guard constraints (Section 5.5.4); and a Citation Surgeon that deterministically replaces each placeholder with the verbatim quote extracted via character offsets (Section 5.5.5). The citation-first principle ensures structural alignment between the generated text and its supporting evidence. The full pipeline is presented in Section 5.5.

1.4 Summary of Experimental Results

ParliamentRAG is evaluated against Google NotebookLM, a commercially available RAG tool, on a benchmark of 15 policy topics spanning 8 thematic macro-areas. The evaluation combines automated metrics (Level 1) with a blind A/B human evaluation (Level 2) conducted by 6 domain experts (parliamentary journalists and parliamentary collaborators) who rated both systems on nine Likert-scale dimensions. The full methodology and results are discussed in Chapters 6 and 7; this section provides a concise overview by research question.

RQ1 Multi-View Retrieval. The dual-channel retrieval architecture achieves near-perfect citation-level coverage of all ten parliamentary groups ($GC = 0.97$ vs. 0.95 for NotebookLM). Human evaluators perceive ParliamentRAG as more balanced (Balance Perception: 4.66 vs. 4.48, Cohen's $d = 0.28$) and rate its coalition coverage more favourably (Source Coverage: 4.64 vs. 4.39, $d = 0.35$). Source Coverage shows the strongest preference asymmetry, with ParliamentRAG preferred at a ratio of 5.7:1.

RQ2 Speaker Authority. The core scientific contribution. The primary objective of the evaluation on this dimension is to assess whether the topic-dependent authority model produces speaker selections that domain experts recognise as more authoritative than those of a strong baseline without dedicated authority modelling. At the automated level, ParliamentRAG selects speakers with modestly higher mean authority scores (0.533

vs. 0.517). More importantly, blind human evaluation provides independent validation: evaluators consistently rate ParliamentRAG's cited speakers as more relevant (Source Relevance: 4.07 vs. 3.84, $d = 0.35$) and more institutionally authoritative (Source Authority: 4.21 vs. 4.00, $d = 0.28$). Per-group authority votes favour ParliamentRAG for 8 out of 10 parliamentary groups, ruling out the hypothesis that the advantage is concentrated on specific parties. These results are particularly meaningful given the strength of the baseline: NotebookLM is a commercial RAG product built on Google's Gemini model family and, in this evaluation, operated on manually curated high-recall source material (Section 6.4), providing it with near-optimal retrieval conditions. Outperforming such a system on speaker selection validates the authority model as a genuine contribution rather than a trivial improvement over a weak comparator.

RQ3 Citation Integrity. Citation Faithfulness achieves $CF = 1.00$ across all evaluated queries, compared to 0.95 for NotebookLM. This is an architectural invariant: the offset-based extraction mechanism makes citation hallucination impossible by construction. The 5% error rate observed in NotebookLM demonstrates that even a state-of-the-art commercial RAG system is not immune to citation misattribution when relying on generative paraphrasing rather than deterministic extraction.

RQ4 Balanced Synthesis. The four-stage generation pipeline translates the structural guarantees of the retrieval layer into responses that evaluators perceive as balanced and fair. ParliamentRAG achieves narrative completeness of 0.99 and receives favourable human ratings on Balance Perception ($\Delta = +0.18$) and Source Coverage ($\Delta = +0.25$). The trade-off is on Answer Quality ($\Delta = -0.25$) and Answer Clarity ($\Delta = -0.24$), reflecting an inherent tension between structural guarantees and generative fluency: NotebookLM, operating as a monolithic generation system with a powerful underlying language model, produces more fluid prose but lacks the architectural mechanisms that enforce balanced representation.

Overall, no dimension reaches statistical significance after Holm–Bonferroni correction for nine simultaneous tests ($N = 67$ paired observations), a consequence of the moderate sample size acknowledged in the a priori power analysis (Section 6.5.4). However, the effect sizes reveal a coherent pattern of *complementary strengths*: ParliamentRAG shows small positive effects on the dimensions targeted by its architectural innovations (source quality, political balance), while NotebookLM shows small positive effects on prose quality. The overall A/B preference is evenly split (31.3% ParliamentRAG, 26.9% NotebookLM, 41.8% equal; binomial test $p = 0.749$), confirming that neither system is globally preferred but that each excels on different facets. This parity in aggregate satisfaction should be

interpreted in light of two factors. First, the comparison is asymmetric: NotebookLM received manually curated high-recall source material and did not need to solve the retrieval problem, while ParliamentRAG operated on the full uncurated knowledge graph of 151,073 chunks (Section 6.4). Achieving equivalent overall satisfaction under these conditions, while simultaneously outperforming on source quality and balance, is a non-trivial result. Second, aggregate satisfaction compresses dimensions with opposing directions into a single score, masking the fact that the two systems serve different information needs. For a user seeking a fluent overview, NotebookLM's prose quality may suffice. For a parliamentary journalist who must verify speaker identity, check citation accuracy, and confirm that all political positions are represented, the structural guarantees of ParliamentRAG—which the evaluation shows are perceived as superior by domain experts—are not optional features but functional requirements that no amount of generative fluency can replace. The contribution of this thesis is not to produce a system that is universally preferred, but to demonstrate that authority-aware, structurally balanced parliamentary RAG is achievable, measurable, and valued by the professionals for whom it is designed.

1.5 Scope and Limitations

ParliamentRAG is designed and evaluated on data from the Italian Chamber of Deputies, specifically the XIX Legislature (2022–present). While the architectural principles are language- and institution-agnostic, adaptation to other parliaments would require new data ingestion pipelines, redesign of the knowledge graph schema (Section 4.4), redefinition of coalition logic (Section 5.3.4), and recalibration of the authority scorer's component weights (Section 5.3.2).

The evaluation focuses on multi-view parliamentary topic summarisation, that is, queries of the form "What is the position of the parliamentary groups on topic X?" The system is not intended to replace human judgment but to assist users in navigating a large documentary corpus.

The topic-dependent authority model relies on metadata available through parliamentary open data portals. Deputies for whom metadata is incomplete (i.e. those whose profession or education fields are not recorded in the Chamber's linked open data) may receive lower authority scores than warranted. This limitation can be mitigated as data quality improves.

The component weights of the authority model $(w_1, \ldots, w_6)$ and the merger coefficients are currently assigned heuristically, informed by domain knowledge and iterative experimentation. While these values have not been optimised against a formal objective, the A/B evaluation framework (Section 5.7) collects human preference signals that could

support data-driven weight optimisation in future work (Section 8.4).

Finally, ParliamentRAG depends on external LLM and embedding services (OpenAI GPT-4o and `text-embedding-3-small`), as described in Section 4.2.4. An outage of these services would prevent response generation, though retrieval and citation extraction, which operate on the local knowledge graph, would remain functional.

1.6 Reproducibility and Public Access

To support reproducibility and enable independent verification of the results reported in this thesis, the following resources are publicly available:

- **Source code**: https://github.com/Emeierkeio/thesis-ParliamentRAG.
- **Live system**: https://www.parliamentrag.it/.

1.7 Thesis Outline

The remainder of this dissertation is organised as follows.

Chapter 2 introduces the domain-agnostic techniques on which ParliamentRAG is built (RAG, dense and graph-based retrieval, result diversification, dimensionality reduction, multi-perspective generation, citation faithfulness). **Chapter 3** surveys parliamentary NLP, computational political text analysis, authority modelling, and domain-specific RAG in adjacent fields, identifying the gaps that justify the proposed contributions.

Chapter 4 presents the high-level architecture: the five design principles (Section 4.1), the three-layer structure (Section 4.2), the knowledge graph schema (Section 4.4), and the end-to-end processing pipeline (Section 4.5).

Chapter 5 details each module: pre-retrieval query rewriting (Section 5.1), the dual-channel retrieval engine with composite merging (Section 5.2), the topic-dependent authority model (Section 5.3), the ideological compass (Section 5.4), the four-stage generation pipeline (Section 5.5), the citation integrity system (Section 5.6), and the A/B evaluation infrastructure (Section 5.7).

Chapter 6 establishes the experimental framework: research questions, benchmark construction, metrics, baselines, and protocol. **Chapter 7** reports quantitative and qualitative findings, analyses the impact of the topic-dependent authority model, and discusses limitations. **Chapter 8** summarises the contributions and outlines directions for future research.

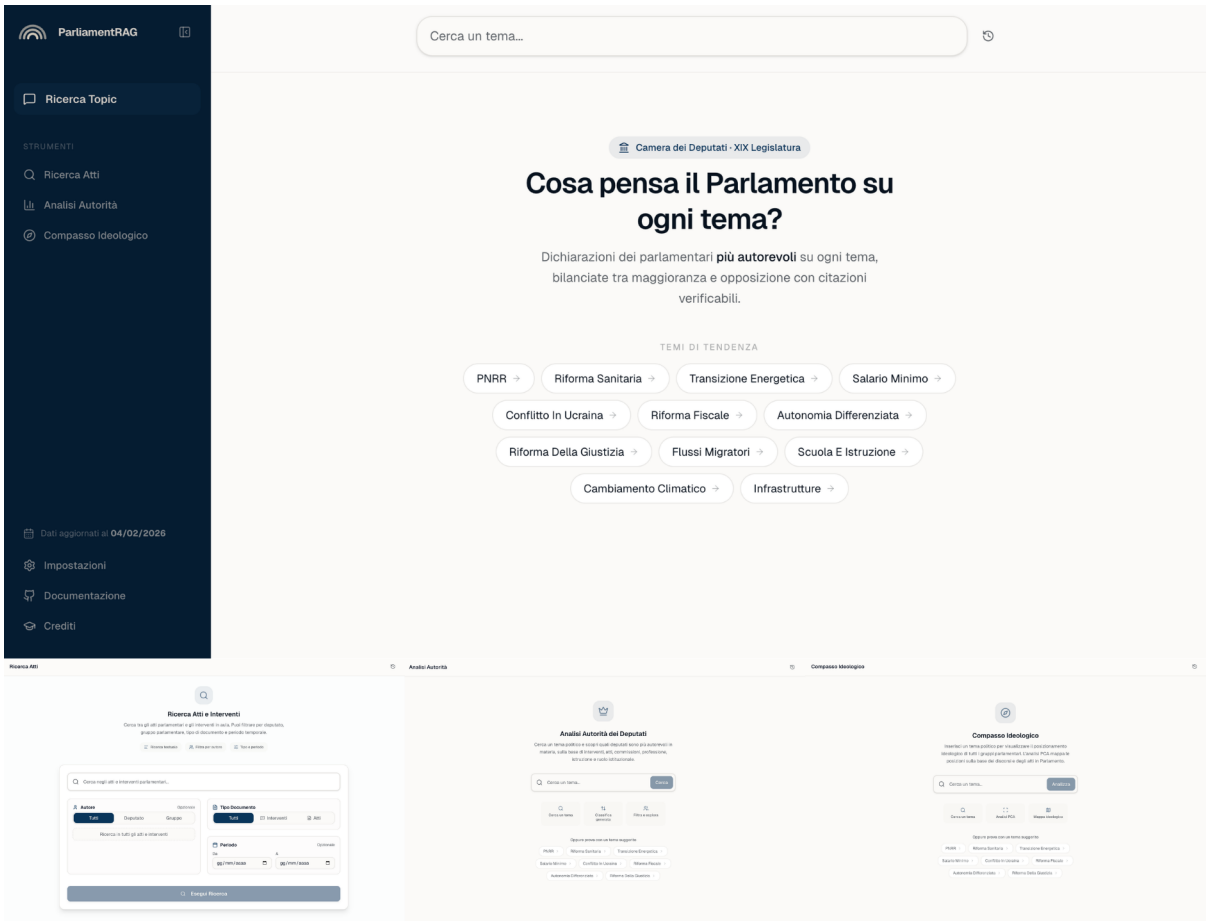

Figure 1.1: Overview of the ParliamentRAG interface. The main view (top) presents the landing page with a free-text search bar and twelve trending parliamentary topics. The sidebar provides access to three auxiliary tools, shown expanded at the bottom: *Ricerca Atti e Interventi* (left) allows filtered search over parliamentary acts and floor speeches by author, document type, and date range; *Analisi Autorità dei Deputati* (centre) identifies the most authoritative MPs on a given topic according to the topic-dependent authority model; and *Compasso Ideologico* (right) generates a PCA-based scatter plot of parliamentary group positions from speech embeddings. Data freshness is indicated in the sidebar footer (04/02/2026).

Chapter 2

Background

This chapter introduces the general-purpose techniques from Natural Language Processing (NLP) and Information Retrieval (IR) on which the system developed in this thesis is built. Each section presents a family of techniques in a domain-independent manner, explaining the core ideas, formalisms, and limitations that motivate the design choices made in later chapters.

Section 2.2 introduces Retrieval-Augmented Generation as a paradigm. Section 2.3 discusses query expansion and rewriting. Sections 2.4–2.6 detail the retrieval methods (dense, graph-based, and hybrid) that underpin the dual-channel architecture. Section 2.7 presents result diversification and fairness-aware ranking. Section 2.8 covers dimensionality reduction and density estimation. Section 2.9 discusses multi-perspective text generation. Section 2.10 addresses citation generation, attribution, and hallucination in language models.

2.1 Natural Language Processing and Information Retrieval

This section provides a brief historical overview of NLP and IR, tracing the conceptual shifts that made the current generation of retrieval-augmented systems possible.

2.1.1 Natural Language Processing

The computational study of natural language originated in the 1950s with rule-based approaches to machine translation. The Georgetown – IBM experiment of 1954 [14] demonstrated the first automatic translation of Russian sentences into English using hand-crafted grammar rules. For the following three decades, the field remained dominated by symbolic methods: formal grammars, parsers, and knowledge representations that achieved notable results on constrained tasks but proved brittle on unrestricted natural

language.

The 1990s brought a decisive shift toward statistical methods. Part-of-speech taggers based on Hidden Markov Models [15], statistical parsers trained on the Penn Treebank [16], and n-gram language models [17] demonstrated that data-driven approaches could match or surpass rule systems across a wide range of tasks, establishing the principle that linguistic structure can be learned from data rather than stipulated by experts.

A further advance occurred with distributed word representations. Bengio et al. [18] proposed learning continuous vector representations jointly with a neural language model, replacing discrete one-hot encodings. The practical breakthrough came with `Word2Vec` [19], which showed that lightweight neural architectures could learn high-quality embeddings from large corpora. Pennington et al. [20] complemented this with GloVe, combining global co-occurrence statistics with predictive training. These *static* embeddings became the standard input representation throughout the mid-2010s.

Static embeddings assign a single vector to each word type regardless of context. Peters et al. [21] addressed this with ELMo, generating token-level representations conditioned on the entire input via bidirectional LSTMs. Shortly thereafter, Vaswani et al. [22] introduced the Transformer architecture, replacing recurrence with multi-head self-attention.

The Transformer became the foundation for pre-trained language models. BERT [23] demonstrated that a Transformer encoder pre-trained on masked language modelling could be fine-tuned to achieve state-of-the-art results across NLP benchmarks with minimal task-specific architecture. The GPT family [24, 25] pursued a complementary strategy, training autoregressive decoders on increasingly large corpora and demonstrating that scale alone could unlock strong few-shot and zero-shot capabilities. The models used in this thesis (GPT-4o for generation and `text-embedding-3-small` for embedding) are direct descendants of this line of research.

2.1.2 Information Retrieval

Information Retrieval, defined as finding relevant material from a large collection in response to a user's information need [26], has moved from surface-level term matching toward semantic understanding.

Early IR systems represented documents and queries as vectors in a high-dimensional term space and computed relevance via cosine similarity. The SMART system [27] formalised this *vector space model*, and TF-IDF weighting provided a simple but effective way to distinguish informative terms from common ones.

Throughout the 1990s and 2000s, probabilistic models refined this framework. Robertson and Zaragoza [28] formalised BM25, a term-weighting function derived from the probabilistic relevance framework that remains widely deployed today. Like all lexical

methods, however, BM25 suffers from the *vocabulary mismatch problem*: a query and a relevant document expressing the same concept using different surface terms receive a low similarity score.

The convergence of NLP and IR became explicit with neural retrieval models. Huang et al. [29] proposed the Deep Structured Semantic Model (DSSM), projecting queries and documents into a shared semantic space. Karpukhin et al. [1] later introduced Dense Passage Retrieval (DPR), applying BERT-based dual encoders trained contrastively on query – passage pairs. DPR demonstrated that dense representations could substantially outperform BM25, establishing dense retrieval as the dominant paradigm. Subsequent developments, including late-interaction models such as ColBERT [30], learned sparse representations such as SPLADE [31], and retrieval-oriented embedding models such as the E5 family [32] and OpenAI's `text-embedding-3` family [33], have further blurred the boundary between the two fields.

This convergence culminated in Retrieval-Augmented Generation (RAG) [4], which couples a neural retriever with a neural generator, using retrieved passages as grounding context for the generated response. The following section introduces RAG in detail.

2.2 Retrieval-Augmented Generation (RAG)

Large language models possess broad linguistic competence but are limited by the knowledge encoded in their parameters at training time. They cannot access post-training information, may hallucinate plausible but unfounded claims, and provide no mechanism for source verification. Retrieval-Augmented Generation addresses these limitations by coupling a *retriever* with a *generator* [4].

Given a user query q, the retriever selects a set of k documents from a corpus $\mathcal{D}$:

$$D_q = \text{top-k}_{d \in \mathcal{D}} \; s(q, d) \tag{2.1}$$

where $s(q, d)$ is a relevance scoring function. The generator then produces a response r conditioned on both the query and the retrieved documents:

$$r = \text{LLM}(q, D_q) \tag{2.2}$$

The quality of the final response depends critically on the retriever: if the retrieved set D_q is incomplete, noisy, or biased, the generated response inherits those defects.

The canonical RAG architecture pairs a dense retriever with a sequence-to-sequence generator, originally BART [34], now typically a large language model. Izacard and Grave [35] introduced Fusion-in-Decoder (FiD), encoding each retrieved passage indepen-

dently before fusing representations in the decoder. More recent architectures introduce adaptive retrieval: Self-RAG [36] uses reflection tokens that allow the generator to assess passage relevance and decide when to invoke the retriever, while CRAG [37] evaluates retrieval quality and activates corrective strategies when initial retrieval is insufficient. Recent surveys provide comprehensive overviews of the evolving RAG landscape [38, 39].

The system developed in this thesis follows the retrieve-then-generate paradigm but departs from it in two respects. First, the retriever is a dual-channel architecture combining dense vector search with graph-based traversal (Section 5.2). Second, the generator is decomposed into a four-stage pipeline whose final stage is deterministic (Section 5.5).

2.3 Query Expansion and Rewriting

The effectiveness of a retrieval system depends not only on the quality of its index and scoring function but also on the quality of the query itself. Short or ambiguous queries often lack the specificity needed to surface relevant documents, particularly in specialised domains where acronyms, legislative proper names, and technical jargon are prevalent.

Classical approaches to query expansion fall into two families [40]. *Corpus-based* methods, such as pseudo-relevance feedback [41], expand the query using terms from an initial set of retrieved documents. *Knowledge-based* methods rely on external resources such as thesauri or ontologies. Both operate without supervision but are limited by the quality of the expansion source.

Recent work has explored large language models as query rewriters. Ma et al. [13] propose a framework in which an LLM rewrites the query to better align with documents in a RAG corpus. Query2Doc [42] generates a pseudo-document from the query and appends it to the original, providing the retriever with a richer semantic representation. These approaches require no task-specific training data and can leverage the broad linguistic knowledge of the model to resolve ambiguities and expand acronyms.

Formally, a query rewriting function ϕ transforms an input query q into an expanded query q':

$$q' = \phi(q) \tag{2.3}$$

such that the retriever applied to q' returns a more relevant document set than it would for q. A key design choice is whether the expansion is *generative* (free-form reformulation) or *controlled* (a short list of additional terms). Controlled expansion reduces the risk of hallucinated terms but may miss useful reformulations. The system developed in this thesis implements an optional, configurable LLM-based query rewriting step (Section 5.1).

2.4 Dense Passage Retrieval

Classical IR relies on sparse lexical representations such as TF-IDF and BM25 [28]. While efficient and interpretable, sparse methods struggle with semantic matching: a query about "fiscal policy" may fail to retrieve a document discussing "tax reform" if the two share no surface terms.

Dense Passage Retrieval (DPR) [1] addresses this by mapping queries and documents into a shared, low-dimensional embedding space using two encoder networks based on the Transformer architecture [22]:

$$\mathbf{q} = E_Q(q) \in \mathbb{R}^d, \qquad \mathbf{d} = E_D(d) \in \mathbb{R}^d \tag{2.4}$$

Relevance is computed as the inner product between the two vectors:

$$s(q, d) = \mathbf{q}^\top \mathbf{d} = E_Q(q)^\top E_D(d) \tag{2.5}$$

The encoders are trained contrastively on pairs of queries and relevant passages. At inference time, approximate nearest-neighbour algorithms such as HNSW [43] enable sub-linear retrieval over large corpora.

Subsequent work has refined dense retrieval through harder negative mining [44], knowledge distillation from cross-encoder rerankers [45], and late-interaction architectures like ColBERT [30]. More recently, embedding models trained specifically for retrieval, such as OpenAI's `text-embedding-3` family [33] and the open-source E5 family [32], have achieved state-of-the-art performance on the MTEB benchmark [46], providing strong multilingual capabilities. This thesis uses `text-embedding-3-small`, producing 1536-dimensional vectors, as the shared embedding model for all semantic operations (Section 4.2.4).

Despite these advances, dense retrieval has three limitations relevant to this thesis. First, performance degrades on out-of-domain data. Second, dense retrieval operates at the passage level and cannot exploit structured relationships between entities (e.g., committee membership or legislative act co-signatures). Third, dense methods provide no built-in mechanism for ensuring diversity or balance in the retrieved set. These limitations motivate the dual-channel architecture: the graph channel (Section 5.2.3) addresses the second, while the composite merging function (Section 5.2.4) addresses the third.

2.5 Graph-Based Retrieval

Knowledge graphs represent information as networks of entities (nodes) connected by typed relationships (edges), enabling retrieval strategies that complement textual similarity by exploiting structural relationships [47].

Within the RAG paradigm, three graph-based retrieval strategies have emerged. *Entity linking* identifies entities in the query and retrieves associated facts or documents [48]. *Subgraph retrieval* extracts a relevant neighbourhood around query entities and serialises it as additional context [49]. *Graph traversal* follows typed relationships to discover entities or documents through multi-hop paths.

GraphRAG [50], developed by Microsoft Research, constructs a hierarchical graph of document communities and answers queries by synthesising information at the community level, enabling answers to global questions that naive passage-level RAG cannot address.

Graph databases such as Neo4j[1] provide native support for graph-structured data. Recent versions (5.x and later) offer built-in vector indexes that enable hybrid queries combining graph traversal with vector similarity search in a single system. This capability is exploited in ParliamentRAG, where all data (text, embeddings, graph structure, and metadata) resides in a single Neo4j instance (Section 4.4).

2.6 Hybrid Retrieval and Result Fusion

Dense and graph-based methods have complementary strengths. Dense retrieval captures semantic similarity but ignores structural relationships; graph traversal exploits structure but may miss relevant documents not well-connected in the graph. Hybrid methods combine both paradigms.

One strategy is *retrieve-and-rerank*: a first-stage retriever generates a candidate set, and a second-stage reranker (typically a cross-encoder [51]) rescores candidates using richer features. An alternative is *multi-channel retrieval*: two or more retrievers operate in parallel, each tuned for a different dimension of relevance, and their outputs are merged. This approach has been explored for combining dense and sparse retrieval [52] as well as for fusing heterogeneous evidence sources. ParliamentRAG adopts the multi-channel strategy (Sections 5.2.2–5.2.3).

Reciprocal Rank Fusion (RRF) [53] is a widely used fusion method:

$$\mathrm{RRF}(d) = \sum_{r \in R} \frac{1}{k + \mathrm{rank}_r(d)} \tag{2.6}$$

[1] https://neo4j.com/

where R is the set of retrievers and k is a smoothing constant. RRF is simple, parameter-free (beyond k), and robust. However, neither RRF nor learned combination methods directly incorporate domain-specific signals such as political group coverage or speaker authority. The merging function adopted in this thesis (Section 5.2.4) extends multi-channel fusion with five weighted components (relevance, diversity, coverage, authority, and salience) that jointly optimise for semantic relevance and representational balance.

2.7 Result Diversification and Fairness-Aware Ranking

Standard retrieval optimises for relevance, which can lead to topically or perspectivally homogeneous result sets. In applications where multiple viewpoints must be represented, explicit diversification is necessary.

Maximal Marginal Relevance (MMR) [41] selects documents iteratively, balancing relevance against dissimilarity to already-selected documents:

$$\text{MMR} = \arg\max_{d_i \in \mathcal{D} \setminus S} \left[\lambda \cdot \text{sim}(d_i, q) - (1 - \lambda) \cdot \max_{d_j \in S} \text{sim}(d_i, d_j) \right] \tag{2.7}$$

where S is the set of already-selected documents and λ controls the relevance-diversity trade-off.

Beyond content diversity, *fairness* in ranking has received growing attention. Zehlike et al. [6] formalise the problem through FA*IR, ensuring that protected groups receive proportional representation in the top-k results. Fairness-aware ranking is relevant because standard authority-based mechanisms can amplify existing visibility disparities.

The diversification approach adopted in this thesis differs from both MMR and FA*IR. Rather than measuring inter-document dissimilarity or enforcing demographic proportionality, the system uses two dedicated components in the composite scoring function: a *diversity* component that penalises over-representation of individual speakers, and a *coverage* component that rewards evidence from underrepresented parliamentary groups (Section 5.2.4). This group-level coverage mechanism ensures that all ten parliamentary groups receive representation.

2.8 Dimensionality Reduction and Density Estimation

Several components of the system rely on techniques for reducing the dimensionality of embedding spaces and estimating probability densities in the reduced space. This section introduces Principal Component Analysis, Kernel Density Estimation, and contrastive term weighting, used in the ideological compass module (Section 5.4).

2.8.1 Principal Component Analysis

Principal Component Analysis (PCA) [54] is a linear dimensionality reduction technique that projects data onto the directions of maximal variance. Given a centred data matrix $X \in \mathbb{R}^{n \times d}$, PCA computes the eigendecomposition of the covariance matrix $C = \frac{1}{n} X^\top X$ and projects the data onto the top k eigenvectors:

$$Z = X\, V_k \in \mathbb{R}^{n \times k} \tag{2.8}$$

where $V_k \in \mathbb{R}^{d \times k}$ contains the k eigenvectors corresponding to the largest eigenvalues.

A *weighted* variant assigns different importances to observations. Given a diagonal weight matrix $W = \mathrm{diag}(w_1, \ldots, w_n)$ with $\sum_i w_i = 1$, the weighted covariance matrix becomes:

$$C_W = \sum_{i=1}^{n} w_i\, (\mathbf{x}_i - \bar{\mathbf{x}}_W)(\mathbf{x}_i - \bar{\mathbf{x}}_W)^\top \tag{2.9}$$

where $\bar{\mathbf{x}}_W = \sum_i w_i\, \mathbf{x}_i$ is the weighted mean. This variant is useful when some observations are more informative than others: for instance, when speech fragments with higher relevance scores should contribute more to the definition of semantic axes. The ideological compass module (Section 5.4.2) uses weighted PCA with a per-observation weight cap to prevent any single speaker from dominating the covariance structure.

2.8.2 Kernel Density Estimation

Kernel Density Estimation (KDE) [55] is a non-parametric method for estimating the probability density function of a random variable. Given n observations $x_1, \ldots, x_n$, the KDE estimate at a point x is:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right) \tag{2.10}$$

where $K(\cdot)$ is a kernel function (commonly Gaussian) and $h > 0$ is the bandwidth. KDE produces a continuous density estimate that can reveal multimodal structure, for example, distinct clusters of political positions along an ideological axis. In this thesis, KDE is applied to projected positions of a parliamentary group's speech fragments to identify the *modal* position (point of highest density), preferred over the mean for its robustness to outliers (Section 5.4.4).

2.8.3 Contrastive Term Weighting

TF-IDF (Term Frequency – Inverse Document Frequency) [56] assigns a weight to each term reflecting how characteristic it is of a document relative to a corpus:

$$\text{TF-IDF}(t, d, \mathcal{D}) = \text{tf}(t, d) \cdot \log \frac{|\mathcal{D}|}{|\{d' \in \mathcal{D} : t \in d'\}|} \tag{2.11}$$

When applied contrastively (comparing one group's vocabulary against all others), TF-IDF identifies terms most distinctive of a particular group. This contrastive usage, sometimes called class-based TF-IDF or c-TF-IDF [57], is used in the ideological compass to generate natural-language labels for each discovered axis endpoint (Section 5.4.6).

2.9 Multi-Perspective Text Generation

Multi-document summarisation aims to produce a concise synthesis of information distributed across multiple sources [58]. A key challenge is managing redundancy while ensuring comprehensiveness [59]. Standard summarisation, however, does not explicitly address *perspectival diversity*: a summary may be comprehensive yet systematically favour one viewpoint.

Multi-perspective summarisation extends the standard task by requiring that the output represent multiple viewpoints. In the news domain, conflicting accounts must be identified and contrasted [11]. Divergence metrics have been proposed to measure how closely a summary reflects the distribution of viewpoints in the source material [60].

Perspective representation becomes particularly challenging when viewpoints are unequally represented in the source corpus. Addressing this imbalance requires explicit constraints on the generation process or post-hoc rebalancing. The controlled generation literature [61] has explored conditioning outputs on desired attributes, but ensuring mandatory inclusion of specific perspectives remains an open challenge.

A related concern is the tendency of language models toward *consensus-seeking* behaviour. Santurkar et al. [5] show that LLMs exhibit systematic political biases

and tend to generate moderate positions that may not reflect the full range of opinions in a debate. This flattening effect is particularly problematic in parliamentary contexts, where explicit representation of disagreement is fundamental.

The generation pipeline developed in this thesis addresses these challenges through architectural decomposition. The four-stage pipeline (Section 5.5) enforces mandatory coverage of all ten parliamentary groups by generating one section per group in a dedicated stage, producing an explicit "no evidence" statement when a group has no relevant evidence rather than silently omitting it.

2.10 Citation Generation and Faithfulness

A critical requirement for RAG systems in high-stakes domains is that generated text be faithful to its sources and that claims be traceable to specific passages. This section introduces attribution, faithfulness, and hallucination as they apply to language model outputs.

2.10.1 Attribution in Language Models

Rashkin et al. [10] define attribution as the property that a generated statement is supported by an identifiable source. Their Attributable to Identified Sources (AIS) framework evaluates whether each sentence in a generated text can be attributed to a specific source passage, and has become a standard evaluation protocol for RAG systems.

Bohnet et al. [62] develop systems for attributed question answering in which every generated sentence is accompanied by a citation. Their work demonstrates that attribution quality improves with joint training of generator and attribution module, but also reveals a tension: strict attribution can constrain fluency and informativeness.

2.10.2 Hallucination in Language Models

Hallucination, the generation of fluent but unfounded text, is a well-documented problem [3]. In the RAG setting, it can manifest as fabricated citations, misattributed quotes, or unsupported claims.

Gao et al. [9] propose RARR (*Retrofit Attribution using Research and Revision*), a post-hoc approach using a second language model to verify and revise citations. While effective at reducing attribution errors, RARR cannot guarantee verbatim accuracy. Qian et al. [63] introduce VeriCite, a framework for rigorous citation verification in RAG systems.

An alternative to post-hoc verification is *citation integrity by construction*: designing the pipeline so that citations are deterministically extracted from source documents rather than generated by the language model. This eliminates citation hallucination by design but requires that source passages be pre-selected and that the generator write around pre-positioned citation anchors, a discipline referred to as *citation-first architecture* in this thesis (Section 5.6). The Citation Surgeon (Section 5.5.5), the deterministic final stage of the generation pipeline, implements this principle by extracting verbatim quotes via stored character-offset spans, without any fuzzy string matching or LLM involvement.

Chapter 3

Related Work

This chapter locates the present thesis within the existing research landscape. The exposition proceeds from the most domain-specific literature to the most general: Section 3.1 reviews parliamentary corpora, search systems, and NLP tasks; Section 3.2 discusses computational approaches to political text analysis; Section 3.3 surveys authority and expertise modelling; Section 3.4 examines domain-specific RAG systems in adjacent fields; Section 3.5 reviews citation and attribution systems; and Section 3.6 synthesises the identified gaps and positions the contributions of this thesis.

3.1 Parliamentary NLP: Corpora, Tasks, and Search Systems

Parliamentary speeches offer a unique NLP domain: speakers have known political affiliations, debates follow rigid topical structures, and positions are typically stated explicitly. This section surveys the data resources, analytical tasks, and existing search systems in this area.

3.1.1 Corpora and Benchmarks

ParlaMint [64] is a multilingual corpus of parliamentary speeches from 29 European countries, annotated in TEI-XML with speaker metadata, party affiliation, and temporal information. At over one billion words, it is the primary resource for cross-national comparative analysis. Europarl [65] contains speeches from the European Parliament in 21 languages, originally developed for machine translation but also used for cross-lingual political discourse analysis. Talk of Norway [66] covers all speeches in the Norwegian Storting with rich speaker metadata. IPSA [67] provides longitudinal coverage of the Italian Parliament spanning 175 years (1848–present), enabling diachronic investigations of Italian political discourse.

ParliaBench [68] is a recent benchmark designed for question-answering systems on parliamentary data, providing query-answer pairs with references to source documents.

For the Italian Chamber of Deputies, the official open data portal `dati.camera.it`[1] exposes structured linked open data on deputies, parliamentary groups, committees, legislative acts, and voting records via SPARQL queries. Stenographic transcripts are provided through the institutional website[2] in PDF and XML formats. ParliamentRAG uses `dati.camera.it` as its primary metadata source and the stenographic transcripts as its documentary corpus (Section 4.3.1).

3.1.2 Common Tasks

Several analytical tasks have been studied on parliamentary text. *Sentiment analysis* in parliamentary contexts differs from general-purpose sentiment analysis in that sentiment is directed toward a specific target (a bill, a government action, or a political opponent). Abercrombie and Batista-Navarro [69] study approval and disapproval patterns by classifying speech sentiment with respect to proposed legislation.

Stance detection identifies a speaker's position toward a given target and is closely related to the retrieval problem in this thesis: identifying speeches that express a clear position on a topic is a prerequisite for multi-view summarisation.

Topic modelling discovers the latent thematic structure of parliamentary debates. Standard LDA-based approaches [70] have been widely applied, and embedding-based models such as BERTopic [57] capture finer semantic distinctions.

Speaker profiling infers properties of speakers from their parliamentary activity. This task is conceptually related to the authority scorer of this thesis (Section 5.3), though existing work has focused on predicting political attributes (ideology, party loyalty) rather than estimating topical authority.

3.1.3 Existing Search Systems

Institutional portals (such as those of the Italian Chamber, the UK Parliament, and the European Parliament) provide keyword-based search over transcripts with filters for date, speaker, and debate topic. These systems return individual documents rather than synthesised multi-view summaries.

Academic prototypes have explored more sophisticated approaches. The Talk of Europe project linked European Parliament debates to biographical data and voting records. The LinkedPolitics project [71] created an RDF dataset of European Parliament debates for

[1] https://dati.camera.it/
[2] https://www.camera.it/leg19/207

SPARQL-based access. None of these systems offers the combination of features targeted by this thesis: authority-aware ranking, multi-view summarisation, and citation integrity within a RAG architecture.

3.2 Computational Political Text Analysis

3.2.1 Ideological Scaling

Estimating the ideological positions of political actors from textual data is a classical problem in computational political science. Wordfish [72] positions parties along a single dimension based on word frequencies. Wordshoal [73] extends this to individual speeches.

Embedding-based methods have demonstrated the ability to capture ideological dimensions beyond the traditional left-right axis. Rheault and Cochrane [74] show that word embeddings trained on parliamentary corpora can recover known ideological structures through PCA on party embeddings. Gentzkow, Shapiro, and Taddy [75] measure polarisation in congressional speech using high-dimensional text representations. These works provide the methodological foundation for the ideological compass module (Section 5.4), which applies weighted PCA to speech fragment embeddings to discover query-specific ideological axes.

The key difference between classical scaling methods and the approach adopted here is *query dependence*: classical methods compute a single, global positioning, whereas the ideological compass produces axes that vary with the topic. On a health-related query, the dominant axis might reflect a tension between public and private healthcare models; on an environmental query, it might capture a divide between ecological protection and industrial development.

3.2.2 Political Bias in Language Models

Language models trained on web-scale corpora exhibit systematic political biases. Santurkar et al. [5] find that most models lean toward moderate-progressive positions. Feng et al. [76] show that political bias varies across topics and is influenced by fine-tuning data and RLHF.

This bias has a direct implication for parliamentary RAG: when asked to synthesise multiple political perspectives, an LLM may flatten disagreements by gravitating toward centrist formulations. The four-stage generation pipeline (Section 5.5) mitigates this through structural constraints: each parliamentary group receives a dedicated section, written from evidence anchors selected by the authority-aware retriever rather than from the model's internal knowledge.

3.2.3 Multi-Perspective Generation in Political Contexts

Multi-perspective generation in the political domain requires ensuring the inclusion of specific, institutionally defined perspectives. Fabbri et al. [58] address multi-document summarisation of news articles, but their notion of perspective is emergent rather than predefined.

In the parliamentary setting, perspectives are defined by institutional structures: each parliamentary group holds a publicly stated position, and a balanced synthesis must include all of them. Li et al. [60] propose metrics for evaluating viewpoint representation but do not propose mechanisms for *enforcing* mandatory inclusion of predefined groups. The coverage component of the composite scoring function (Section 5.2.4) and the per-group Sectional Writer (Section 5.5.3) address this gap.

3.3 Authority and Expertise Modelling

The problem of determining which speakers are most authoritative on a given topic connects to established research traditions in both web search and enterprise information retrieval.

3.3.1 Link-Based Authority

Kleinberg's HITS algorithm [77] distinguishes between *hubs* and *authorities*, computing scores through iterative mutual reinforcement. PageRank [12] reduces this to a single authority score based on the stationary distribution of a random walk. Both share a foundational insight: authority is *relational*, not intrinsic, deriving from a node's position within a network rather than from its content alone. This insight transfers to the parliamentary domain: a deputy's authority on a given topic is a function of their position within the parliamentary knowledge graph (acts signed, committees served on, speeches delivered).

However, classical link-based authority has three limitations for parliamentary applications. First, PageRank computes a single, query-independent score, whereas parliamentary authority must vary by topic. Second, PageRank is static, whereas parliamentary authority evolves over time. Third, PageRank does not model political context: authority accumulated under one coalition alignment may not carry over when a deputy undergoes a *coalition crossing*, since political positioning and policy targets change fundamentally. These limitations motivate the query-dependent, time-aware, coalition-aware topic-dependent authority model developed in this thesis (Section 5.3).

3.3.2 Expert Finding

Expert finding has been extensively studied in enterprise and academic search [7]. Balog et al. formalise two approaches: *profile-based* methods construct an expertise profile a priori, while *document-based* methods first retrieve relevant documents and then identify associated candidates.

ParliamentRAG adopts a document-based approach: retrieval is performed first, and authority scores are computed only for speakers in the retrieved evidence set (Section 5.3.1). This avoids computing authority for all deputies on every query and ensures that deputies with high formal authority but no relevant speeches are not surfaced as experts.

Moreira et al. [78] extend expert finding to account for temporal dynamics. Their learning-to-rank approach informs a direction for future optimisation of the authority model's weights (Section 8.4).

3.3.3 Source Reliability in RAG

Hwang et al. [79] address source reliability estimation in RAG systems, learning reliability scores based on consistency with other retrieved documents. However, their approach assumes the existence of a factually correct reference answer. This assumption does not hold in parliamentary debate, where multiple legitimate viewpoints coexist. This distinction motivates a different conception of authority: one based on contextual expertise and institutional role rather than proximity to a reference answer.

3.4 Domain-Specific RAG Systems

This section reviews RAG systems developed for specialised domains adjacent to parliamentary analysis. Each illustrates design patterns that inform this thesis, though none addresses the specific combination of requirements that multi-view parliamentary topic summarisation demands.

3.4.1 Legal RAG

Legal RAG systems share structural similarities with parliamentary RAG: both operate on corpora with rich metadata, require precise citation, and serve domains where accuracy is paramount. Louis et al. [80] combine knowledge graphs encoding statutory relationships with dense retrieval over case law, demonstrating that hybrid retrieval outperforms pure dense retrieval on complex legal queries. Their work validates the multi-channel strategy adopted in this thesis (Section 5.2), though legal applications do not require representational balance across parties.

3.4.2 Medical RAG

Medical RAG systems such as those surveyed by Yasunaga et al. [49] integrate biomedical knowledge graphs to improve retrieval precision. The requirement for source traceability parallels the citation integrity requirement in parliamentary RAG (Section 5.6). However, medical applications assume the existence of a single correct answer, whereas parliamentary RAG must synthesise multiple, potentially conflicting perspectives.

3.4.3 Domain Adaptation of RAG

Siriwardhana et al. [81] demonstrate that domain-adaptive pre-training of the retriever produces significant improvements in biomedical and legal domains, suggesting that fine-tuning embedding models on parliamentary corpora could improve retrieval quality (Section 8.4).

The common pattern across these systems is that specialised domains impose requirements *beyond* relevance: authority of sources, recency, representational balance, and citation verifiability. Standard RAG architectures must be augmented with domain-specific components to meet these requirements.

3.5 Citation and Attribution in RAG

Rashkin et al. [10] propose the AIS (Attributable to Identified Sources) framework for evaluating whether generated statements can be traced to specific source passages. Liu et al. [82] introduce methods for verifying correspondence between citations and claims. Gao et al. [9] develop RARR, a post-hoc system using a second language model to verify and revise citations. Qian et al. [63] propose VeriCite for rigorous citation verification.

These approaches share a common architecture: the generator produces text with citations, and a separate verification module checks accuracy. This design improves citation quality but cannot *guarantee* verbatim accuracy, because the generator remains free to paraphrase or fabricate source material. The citation integrity system in ParliamentRAG (Section 5.6) takes a fundamentally different approach: citations are extracted deterministically from source transcripts using character offsets stored at preprocessing time. The language model never generates quotation text; it only positions placeholders that the Citation Surgeon (Section 5.5.5) resolves via $speech.text[span_start : span_end]$. This by-construction guarantee eliminates the class of errors that post-hoc systems aim to detect.

Studies of commercial AI search systems have documented significant attribution failures. Maynez et al. [83] find that abstractive summarisation models frequently produce

unfaithful outputs. These findings underscore the importance of citation integrity by construction for politically sensitive domains, where a single misattributed quote can have disproportionate consequences.

3.6 Identified Gaps and Positioning

The survey conducted in this chapter reveals four gaps that define the contribution space of this thesis.

Gap 1: Multi-view retrieval for parliamentary applications. Existing RAG architectures optimise for relevance, not representational balance. Fairness-aware ranking algorithms [6] have not been integrated into RAG pipelines for political applications. GraphRAG [50] exploits document structure but does not enforce coverage of predefined groups. No existing retrieval system guarantees that all parliamentary groups are represented. The dual-channel retrieval architecture with coverage-weighted composite merging addresses this gap (Section 5.2).

Gap 2: Query-dependent, temporally aware authority scoring. Expert finding formalises the problem of identifying authoritative individuals [7], but existing approaches are either static (PageRank) or learned end-to-end from downstream performance (source reliability estimation). No system computes interpretable, query-dependent authority scores from observable parliamentary features while accounting for temporal decay and coalition-crossing invalidation. The topic-dependent authority model addresses this gap (Section 5.3).

Gap 3: Guaranteed citation integrity in RAG. Attribution frameworks [10] and post-hoc verification systems [9] improve citation quality but cannot guarantee verbatim accuracy. No existing RAG system provides a by-construction guarantee that every quotation is a verbatim passage from an identified source document. The offset-based citation integrity system addresses this gap (Section 5.6).

Gap 4: Mandatory perspective inclusion in generation. Multi-perspective summarisation [11, 60] treats perspectives as latent or optional. In parliamentary debate, perspectives are institutionally defined (one per parliamentary group), and their inclusion is mandatory. No existing generation pipeline enforces per-group section generation with explicit coverage constraints. The four-stage generation pipeline addresses this gap (Section 5.5).

This thesis addresses these four gaps in an integrated manner. The topic-dependent authority model (Gap 2) constitutes the primary scientific contribution; the dual-channel retrieval with coverage constraints (Gap 1), the offset-based citation system (Gap 3), and the four-stage generation pipeline with per-group sections (Gap 4) constitute engineering contributions that coordinate established techniques under a principled authority-aware framework. The following chapters present the system architecture (Chapter 4) and the algorithmic detail of each component (Chapter 5).

Chapter 4

System Architecture

This chapter describes the architecture of ParliamentRAG. It states the design principles (4.1), presents the three-layer system decomposition (4.2), describes data acquisition and preprocessing (4.3), formalises the knowledge graph schema (4.4), and details the end-to-end processing pipeline (4.5).

4.1 Design Principles and Requirements

The architecture is shaped by three design principles derived from the research objectives stated in Chapter 1.

Principle 1: Citation integrity by construction. Every quotation attributed to a parliamentary speaker must be verbatim and traceable to a specific character-offset span in the original speech transcript. The raw speech text must be preserved without modification, and citation extraction must be deterministic, without reliance on fuzzy string matching.

Principle 2: Mandatory multi-group coverage. All ten parliamentary groups in the XIX Italian legislature must receive representation in every system response. This requirement must be embedded in the retrieval ranking mechanism, ensuring that groups with lower speech volume are not systematically excluded from the evidence pool.

Principle 3: Query-dependent authority. Speaker authority is not a static biographical attribute but a function of the current query topic. The architecture must support per-query authority computation over the retrieved speaker pool, with access to biographical embeddings, legislative activity histories, and committee assignments stored in the knowledge graph. This principle is the architectural manifestation of the topic-dependent authority model formalised in Section 5.3.

These principles collectively impose constraints on storage format (graph database with stored embeddings and character offsets), retrieval design (dual-channel with coverage weighting), and generation architecture (multi-stage with deterministic final stage).

4.2 High-Level Overview

The system follows a three-layer architecture: a presentation layer (frontend), an application layer (backend), and a data layer (knowledge graph), with external services providing embedding computation and text generation. Figure 4.1 provides an overview.

4.2.1 Presentation Layer

The frontend is a Next.js application (version 16.1.4) that communicates with the backend exclusively via HTTP and SSE and performs no domain logic. It delivers content through five principal view components.

Chat interface. Renders the conversational interaction, displaying generated multi-perspective responses with inline citation markers. Each marker is interactive and links to the corresponding evidence item. Figure 1.1 shows the landing page.

Citation display. Presents structured citation objects including the verbatim quote, speaker metadata (name, parliamentary group, coalition), speech date, similarity score, and authority score. As shown in Figure 4.2, clicking a citation marker opens a detail panel where the verbatim excerpt is highlighted within the full parliamentary speech.

Compass visualisation. Renders the ideological compass output as a two-dimensional scatter plot with group position markers and TF-IDF-generated axis labels (Section 5.4). Figure 4.3 illustrates the compass for justice reform.

Expert cards. Displays the expert panel (one top-authority speaker per parliamentary group) with biographical information and the authority score component breakdown. Figure 4.4 shows an example card.

Evaluation panel. Implements the A/B evaluation survey interface, presenting two responses (system and baseline) in blind, randomised order and collecting ratings across nine dimensions. Figure 4.5 shows the interface.

The frontend proxies all API calls through Next.js server-side routes, manages SSE streaming state, and handles progressive rendering.

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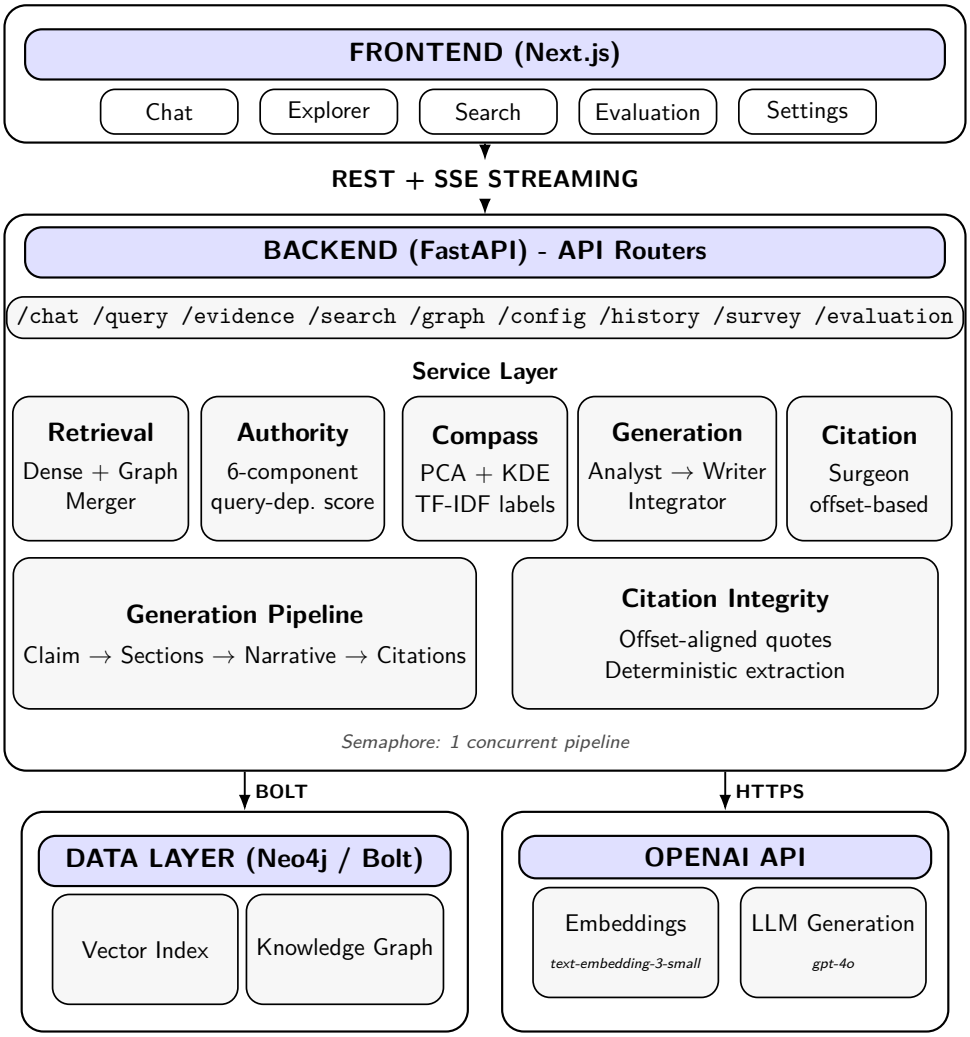

Figure 4.1: Three-layer architecture of ParliamentRAG, a Retrieval-Augmented Generation system over Italian parliamentary proceedings (XIX Legislature). The **presentation layer** is a Next.js frontend delivering incremental pipeline results via REST and Server-Sent Events. The **application layer** is a FastAPI backend with nine API endpoints and five modular service domains: dual-channel retrieval with composite merging, topic-dependent authority scoring, ideological compass computation, four-stage response generation, and deterministic citation extraction via character-offset spans. A semaphore enforces single-pipeline concurrency. The **data layer** combines a Neo4j knowledge graph (232,755 nodes, 488,487 relationships) with a native vector index over 151,073 chunk embeddings. Embedding computation (`text-embedding-3-small`) and text generation (`gpt-4o`) are delegated to the OpenAI API.

Figure 4.2: Citation detail panel. The panel shows the speaker's identity, coalition role, parliamentary group, and speech date. A contextual header identifies the legislative proceeding (here, the constitutional bill on judicial career separation, A.C. 1917). The verbatim excerpt is highlighted in yellow within the full speech text, enabling the user to verify the quoted passage in its original context.

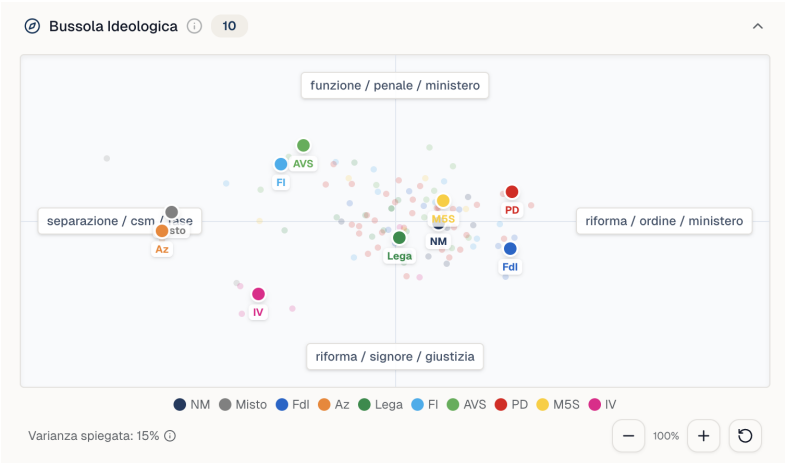

Figure 4.3: Ideological compass for justice reform. Each dot represents an MP's position derived via PCA from parliamentary speech embeddings; larger labeled dots indicate party centroids. Axis labels are generated from the most discriminative terms along each principal component (e.g. *funzione / penale / ministero* vs. *riforma / signore / giustizia*). Explained variance: 15%. Colour encodes parliamentary group. The visualisation reveals, for instance, that FdI and PD occupy opposite positions along the horizontal axis, while IV is isolated in the lower-left quadrant.

Figure 4.4: Expert card for a selected parliamentary source. The card displays the speaker's identity, parliamentary group, and coalition role (here, opposition), followed by the composite topic-specific authority score (71/100). Below, the six contributing criteria are shown: floor speeches (80%), legislative acts (80%), committee membership (58%), professional background (71%), education (76%), and institutional role (50%). One such card is generated for each top-authority speaker per parliamentary group.

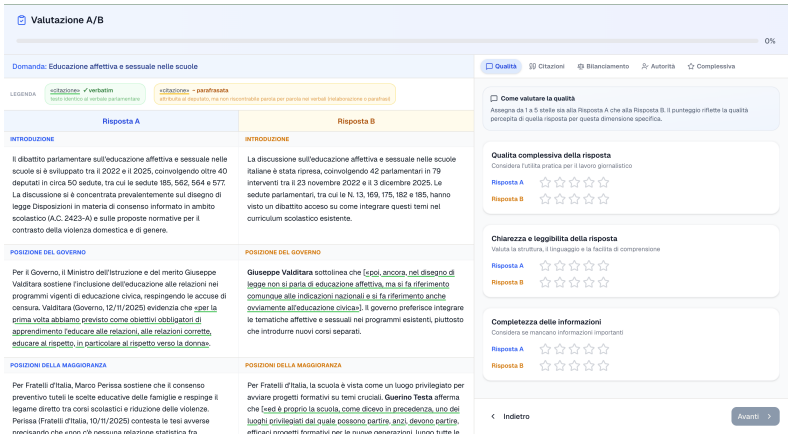

Figure 4.5: A/B evaluation interface for human assessment. Two anonymised responses (Risposta A and Risposta B) are displayed side by side in randomised order. A legend distinguishes verbatim citations (green) from paraphrased ones (orange). The right panel collects 1–5 star ratings across multiple dimensions organised under five tabs. Assignment order is counterbalanced across sessions.

4.2.2 Application Layer

The backend is a FastAPI application structured around routers (HTTP interface) and services (business logic). The services are organised into five independently modular domains, summarised in Table 4.1.

Table 4.1: Service domains in the ParliamentRAG application layer.

Domain	Location	Responsibility
Retrieval	`services/retrieval/`	Dual-channel evidence retrieval, merging, reranking
Authority	`services/authority/`	Topic-dependent authority scoring
Compass	`services/compass/`	Ideological axis discovery and group positioning
Generation	`services/generation/`	Four-stage response generation with citation management
Citation	`services/citation/`	Salience computation and verbatim quote extraction

Each domain exposes a clean interface to the router layer and does not directly depend on other domains. This modularity enables swapping the embedding model or replacing the generation model without modifying downstream consumers.

The query router (`routers/query.py`) is the primary orchestration point. It accepts a `QueryRequest` (containing the query string, optional date filters, and a configurable `top_k`), enforces a semaphore-based concurrency limit (one active pipeline), and coordinates the full processing pipeline described in Section 4.5. Results are delivered incrementally via SSE with distinct event types for progress updates, expert panel data, citation objects, compass output, and the final completion signal.

A secondary router (`routers/evaluation.py`) computes automated evaluation metrics, and a survey router (`routers/survey.py`) handles submission and aggregation of human A/B evaluation responses.

4.2.3 Data Layer

All persistent data (text, embeddings, graph structure, and metadata) is stored in a single Neo4j 5.15+ instance accessed via the Bolt protocol. Parliamentary data is inherently relational: speeches belong to phases, phases to debates, debates to sessions; speakers sign acts, sit on committees, and belong to parliamentary groups with temporal validity. A graph database enables retrieval queries to combine vector similarity (via Neo4j's native vector index) with graph traversal in a single query execution context.

The formal specification of the graph schema is presented in Section 4.4.

4.2.4 External Services

The system integrates two external services, both provided by OpenAI.

The embedding service, `text-embedding-3-small` (1536-dimensional), is the shared representation backbone. Every semantic similarity computation (retrieval scoring, authority component evaluation, ideological projection) operates on embeddings from this model, ensuring geometric coherence across all comparisons.

Large language models are invoked exclusively within the generation module and the optional query rewriting stage. The generation pipeline uses GPT-4o across all stages with a low temperature setting to favour factual fidelity. The query rewriter uses GPT-4o-mini for lightweight expansion (Section 5.1).

4.3 Data Acquisition and Preprocessing

4.3.1 Institutional Data Sources

The primary data source is the open data infrastructure of the Italian Parliament (`dati.camera.it`), which publishes full transcripts of plenary and committee debates, structured information on legislative acts and their signatories, deputy biographical data, and committee membership records with temporal validity intervals. These records are provided under an open licence and updated incrementally.

The transcripts constitute the evidential core of the system. Each transcript is associated with a session date, a debate identifier, and a phase marker. Deputy profiles contain professional and educational history in free-text form, which must be embedded for use in authority computations. Legislative acts carry thematic descriptors that are similarly embedded for semantic matching.

A secondary thematic resource is a curated ontology of committee topics (`commissioni_topics.yaml`), mapping each parliamentary committee to domain-specific keywords. This mapping supplements structural committee membership data with semantic content, enabling the authority scorer to assess the topical relevance of committee roles with greater precision.

4.3.2 Text Preprocessing and Alignment Map

Raw parliamentary transcripts contain a mixture of substantive speech content and procedural markers (presiding officers' announcements, voting declarations, quorum calls, formal greetings) that carry no informational value for topic summarisation. These are identified and removed through structural heuristics and textual pattern matching. Text

normalisation resolves encoding inconsistencies, Unicode variation forms, and whitespace irregularities.

A critical design decision concerns the relationship between the cleaned representation used for indexing and the original raw text used for citation extraction. Rather than discarding the raw text, the system maintains an *alignment map* for each speech that records, for every position in the cleaned text, the corresponding byte offset in the original source. This bidirectional mapping allows retrieval over processed text while extracting citations from the verbatim original. The alignment map is the architectural mechanism that makes Principle 1 a structural guarantee: no post-hoc string-matching is necessary, because quoted strings are retrieved deterministically from stored offset pairs.

4.3.3 Chunking Strategy

Each speech is segmented into fixed-size chunks, stored as `Chunk` nodes in Neo4j. Each chunk records:

`text`: The chunk's text content (may reflect preprocessing).

`start_char_raw`, `end_char_raw`: Character offsets into the original raw speech text.

`index`: Sequential chunk index within the speech.

`embedding`: A 1536-dimensional dense vector produced by `text-embedding-3-small` at ingestion time.

`char_count`: The chunk's character count.

The distinction between `text` (potentially preprocessed) and the raw span `speech.text[start_char_raw:end_char_raw]` is critical. Retrieval and display operations use `text`; citation verification uses exclusively the raw span. This dual representation ensures that preprocessing decisions do not compromise citation integrity.

The chunk length involves a trade-off between retrieval precision and context preservation. The adopted parameters represent an empirically motivated balance, validated through retrieval quality experiments (Chapter 6).

4.4 Data Model and Knowledge Graph Schema

The Neo4j graph schema organises parliamentary data hierarchically, following the institutional structure. Figure 4.6 illustrates the principal node types and relationships. The vertical hierarchy Session → Debate → Phase → Speech → Chunk follows the structure

of parliamentary proceedings. Lateral relationships connect speakers to parliamentary groups (with temporal validity), parliamentary acts (with signatory role), and committees.

4.4.1 Node Types

Table 4.2 summarises the principal node types and their key properties. The complete property list is provided in Appendix A.

Table 4.2: Principal node types in the knowledge graph.

Node Type	**Key Properties**
Session	`id`, `date`, `number`, `complete_date`, `chamber`, `day`, `month`, `year`, `legislature`
Debate	`id`, `title`, `order`
Phase	`id`, `order`, `title`
Speech	`id`, `text`, `text_embedding`, `surname_name`, `preprocessed_text`, `char_count`
Chunk	`id`, `text`, `start_char_raw`, `char_count`, `end_char_raw`, `embedding`, `index`
Deputy	`id`, `deputy_card`, `gender`, `photo`, `term_of_office`, `term_of_office_start`, `first_name`, `last_name`, `profession`, `education`, `profession_embedding`, `education_embedding`
GovernmentMember	`id`, `first_name`, `last_name`, `institutional_role`, `is_government`, `committee_role`, `role_type`
ParliamentaryGroup	`name`, `acronym`
ParliamentaryAct	`id`, `title`, `description`, `description_embedding`, `eurovoc`, `number`, `presentation_date`, `recipient`, `uri`, `type`, `title_embedding`, `eurovoc_embedding`
Committee	`id`, `name`, `embedding`

Deputy nodes carry biographical attributes alongside precomputed embeddings of free-text professional and educational fields, which are the primary mechanism for integrating expertise signals into the topic-dependent authority model (Section 5.3).

GovernmentMember nodes represent members of the executive branch (ministers, vice-ministers, undersecretaries) who intervene in parliamentary proceedings in an institutional capacity.

ParliamentaryGroup nodes encode the political formations to which deputies belong. Group membership is temporal: the relationship carries explicit start and end date attributes, enabling affiliation resolution at any historical point.

Speech and *Chunk* nodes form the evidential core. Each speech carries the full raw text; each chunk stores its content, dense vector embedding, and byte-level offsets mapping back

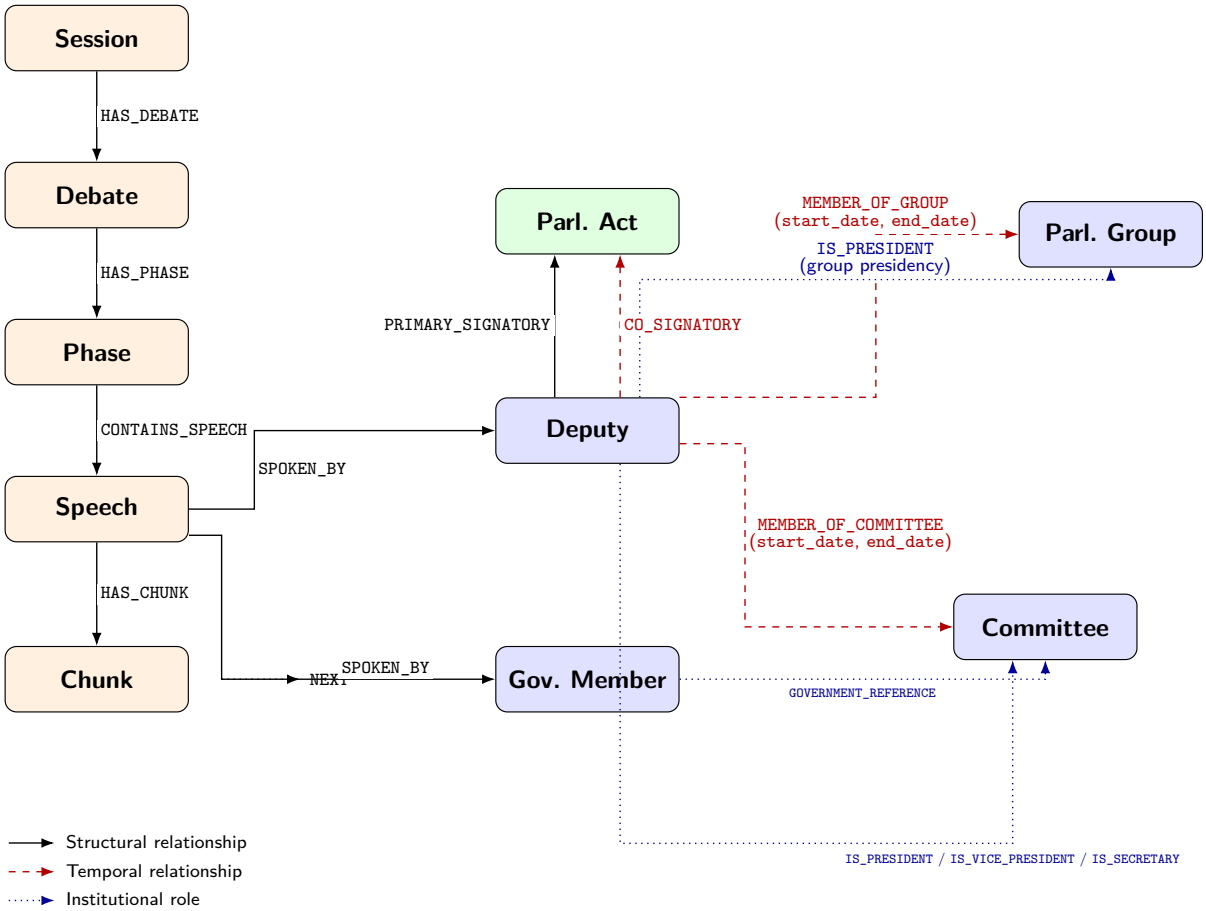

Figure 4.6: Knowledge graph schema of ParliamentRAG, modelling the proceedings of the Italian Chamber of Deputies (XIX Legislature). The vertical hierarchy (orange nodes) reflects the institutional structure: each `Session` contains `Debates` organised into `Phases`; each phase includes `Speeches` segmented into vector-indexed `Chunks` linked by a sequential `NEXT` chain. Speeches are attributed to `Deputy` or `Gov. Member` nodes via `SPOKEN_BY`. Deputies connect to `Parl. Act` nodes through `PRIMARY_SIGNATORY` and `CO_SIGNATORY` edges. Dashed red edges represent time-bounded memberships (`MEMBER_OF_GROUP`, `MEMBER_OF_COMMITTEE`), where `start_date` and `end_date` enable resolution of party affiliation at any point and detection of coalition crossings. Dotted blue edges denote committee role assignments (`IS_PRESIDENT`, `IS_VICE_PRESIDENT`, `IS_SECRETARY`), parliamentary group presidency (`IS_PRESIDENT` from Deputy to Parl. Group), and government-to-committee references. This schema supports dual-channel retrieval, topic-dependent authority scoring, and temporal validation of political affiliations.

to the original speech. The chunk is simultaneously an index unit, a preview representation, and a citation anchor.

ParliamentaryAct nodes carry thematic descriptors and precomputed embeddings for semantic matching during authority scoring and graph-channel retrieval.

Committee nodes carry precomputed name embeddings used in authority scoring to measure the topical relevance of a deputy's committee assignment. *Session*, *Debate*, and *Phase* nodes form the hierarchical schema of parliamentary proceedings.

4.4.2 Relationship Types

Table 4.3 summarises the principal relationship types. The full list with cardinalities is provided in Appendix A.2.

Table 4.3: Principal relationship types in the knowledge graph.

Relationship	**Between**	**Properties**
`HAS_DEBATE`	Session → Debate	
`HAS_PHASE`	Debate → Phase	
`CONTAINS_SPEECH`	Phase → Speech	
`HAS_CHUNK`	Speech → Chunk	
`NEXT`	Chunk → Chunk	(sequential)
`SPOKEN_BY`	Speech → Deputy / GovMember	
`MEMBER_OF_GROUP`	Deputy / GovMember → Parl. Group	`start_date`, `end_date`
`MEMBER_OF_COMMITTEE`	Deputy / GovMember → Committee	`start_date`, `end_date`
`PRIMARY_SIGNATORY`	Deputy → ParliamentaryAct	
`CO_SIGNATORY`	Deputy → ParliamentaryAct	
`IS_PRESIDENT`	Deputy → Committee	
`IS_PRESIDENT`	Deputy → Parl. Group	
`IS_VICE_PRESIDENT`	Deputy → Committee	
`IS_SECRETARY`	Deputy → Committee	
`GOVERNMENT_REFERENCE`	GovMember → Committee	

The `MEMBER_OF_GROUP` and `MEMBER_OF_COMMITTEE` relationships carry temporal validity properties (`start_date`, `end_date`). This temporal modelling supports three capabilities: identifying the group a speaker belonged to at the time of a specific speech; detecting coalition crossings (movement between *majority* and *opposition*), which triggers authority invalidation (Section 5.3.4); and flagging evidence items where the speaker's current group differs from their group at speech time.

The role relationships (`IS_PRESIDENT`, `IS_VICE_PRESIDENT`, `IS_SECRETARY`) record institutional positions within committees and are used by the role component of the

topic-dependent authority model to assign differentiated weights based on leadership. The `IS_PRESIDENT` relationship from Deputy to ParliamentaryGroup records parliamentary group presidency, a distinct institutional role that confers high political visibility and is weighted accordingly in the authority scorer. The `GOVERNMENT_REFERENCE` relationship links government members to their reference committees.

4.4.3 Indexing Strategy

Vector index (`chunk_embedding_index`). A Neo4j vector index on the `embedding` property of `Chunk` nodes, using cosine similarity, supporting the dense retrieval channel (Section 5.2.2).

Uniqueness constraints. Node-level uniqueness constraints on the `id` property of all principal node types, ensuring referential integrity.

Full-text index. An optional full-text index on `ParliamentaryAct.title` supporting keyword matching in the graph channel (Section 5.2.3).

No secondary vector indexes are maintained for act or speaker embeddings; these are accessed via property lookups during graph traversal.

4.5 Processing Pipeline

The end-to-end processing pipeline transforms a user query into a multi-perspective response with verified citations. It consists of eight stages, depicted in Figure 4.7.

0. **Query rewriting (optional).** Short or ambiguous queries (up to five words by default) are expanded by a lightweight LLM call (GPT-4o-mini) that adds parliamentary terminology. Longer queries bypass this stage. Details in Section 5.1.

1. **Query embedding.** The (possibly rewritten) query is encoded into $E(q) \in \mathbb{R}^{1536}$ using `text-embedding-3-small`. This embedding is computed once and shared by all downstream stages.

2. **Dual-channel retrieval.** The dense channel performs vector similarity search over the chunk index; the graph channel traverses parliamentary acts to reach signatories' speech chunks. Both use $E(q)$ as input. Details in Section 5.2.

3. **Channel merging and reranking.** Results are deduplicated and reranked using a composite scoring function with five weighted components: relevance, diversity, coverage, authority, and salience. The authority component ($w_a = 0.05$) is structurally present in the scoring function; however, at merge time the topic-dependent authority scorer has not yet executed, because it requires the identifiers of the speakers

present in the evidence pool—which are known only after retrieval. Consequently, all chunks receive a uniform authority value (0.5), making the authority term a constant bias that does not alter the relative ranking. The weight is retained in the merger configuration as an architectural placeholder: should pre-computed or periodically updated authority scores become available (e.g., static scores indexed alongside chunk embeddings), the merger could incorporate them directly without waiting for post-retrieval computation, realising the feedback loop illustrated by the dashed arrow in Figure 4.7. Details in Section 5.2.4.

4. **Authority scoring.** For each speaker in the merged evidence pool, a topic-dependent authority score $authority(s, q) \in [0, 1]$ is computed as a weighted sum of six components. In the current pipeline, these query-specific scores are *not* fed back into the merger (see Stage 3); instead, they are used downstream in two ways: (i) to select the top-authority speaker per parliamentary group (Stage 5), and (ii) within the generation pipeline, to choose the citing speaker for each evidence group and to resolve semantic duplicates. Details in Section 5.3.

5. **Expert computation.** For each parliamentary group $g \in G$, the speaker with the highest authority score in the evidence pool is selected as the group's expert representative.

6. **Concurrent execution.**

   (a) The compass pipeline discovers ideological axes and computes group positions (Section 5.4).

   (b) The generation pipeline produces the multi-perspective response with verified citations (Section 5.5).

   Both pipelines receive the same evidence set and execute concurrently.

7. **Expert–citation reconciliation.** The expert panel is updated so that each group's expert is the speaker actually cited in the final response, ensuring consistency between the expert display and cited content.

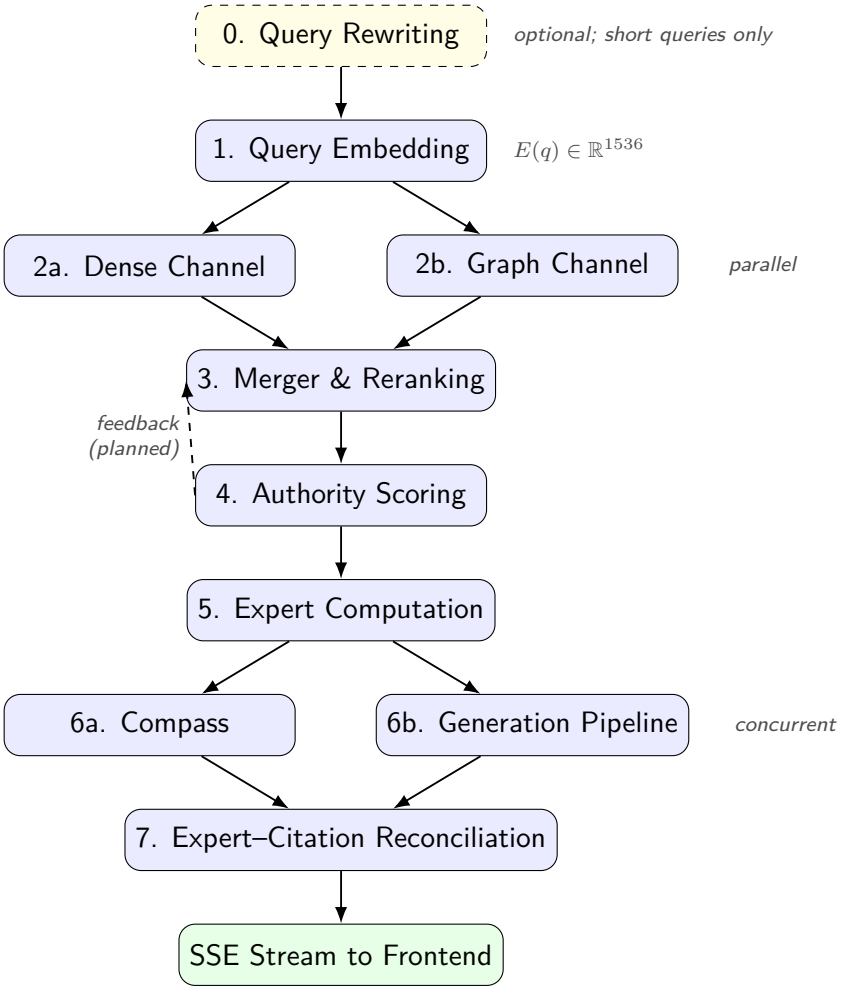

Figure 4.7: End-to-end processing pipeline of ParliamentRAG. Stage 0 optionally rewrites short queries via a lightweight LLM call. Stage 1 encodes the query into a 1536-dimensional embedding. Stages 2a–2b run in parallel: dense channel (vector search) and graph channel (legislative act traversal). Stage 3 deduplicates and reranks using a five-component composite score; the authority component is present in the scoring function but receives a uniform constant value (0.5) at this stage, since query-specific authority scores are not yet available (see text). Stage 4 computes topic-dependent authority scores for each speaker in the evidence pool. The dashed arrow from Stage 4 to Stage 3 represents an *architectural provision* for a future feedback loop: if pre-computed or cached authority scores became available, the merger could incorporate them directly; in the current implementation this path is not active. Stage 5 selects one top-authority speaker per group. Stages 6a–6b execute concurrently: compass (PCA-based ideological axes) and generation pipeline (four stages: claim analysis, per-group writing, narrative integration, deterministic citation extraction). Stage 7 reconciles the expert panel with actually cited speakers. Results are delivered via SSE.

Chapter 5

System Components

This chapter presents the algorithmic content of each module in the ParliamentRAG processing pipeline. Whereas Chapter 4 described the high-level architecture (design principles, layer decomposition, knowledge graph schema, and pipeline flow), this chapter formalises the internal logic of each component: its problem definition, mathematical specification, and design rationale.

The chapter follows pipeline execution order: pre-retrieval query rewriting (5.1), dual-channel retrieval (5.2), topic-dependent authority scoring (5.3), ideological compass (5.4), four-stage generation pipeline (5.5), citation integrity system (5.6), and the A/B evaluation infrastructure (5.7).

5.1 Pre-Retrieval Query Rewriting

5.1.1 Problem Definition

User queries about parliamentary topics are often formulated in everyday language that differs from the register of parliamentary speech. A query such as *"immigration"* may fail to retrieve chunks containing institutional terms like *permesso di soggiorno*, *protezione internazionale*, or *decreto flussi*. The pre-retrieval stage addresses this vocabulary mismatch.

5.1.2 Proposed Approach

The `query_rewriter.py` module implements optional query expansion using GPT-4o-mini. When activated, it prompts the model to: (1) expand acronyms and proper names into their full institutional forms (e.g., *"Bolkestein"* → *"direttiva Bolkestein concessioni balneari stabilimenti balneari liberalizzazione servizi"*); (2) add domain-specific parliamentary terminology in Italian; and (3) produce a semantically enriched version (capped at 15 words) preserving the original intent.

The rewriter activates only for short queries (at most five words by default), on the assumption that longer queries are already sufficiently descriptive. GPT-4o-mini is used rather than GPT-4o because query rewriting requires only lexical expansion, not complex reasoning, and the lighter model provides adequate quality at lower cost and latency.

The expanded query may produce a richer embedding $E(q')$ or generate additional keyword variants for the graph channel. The original query q is preserved alongside q' for audit purposes. Query rewriting is configurable via `default.yaml`, enabling ablation experiments that isolate its contribution to retrieval quality.

5.2 Dual-Channel Retrieval

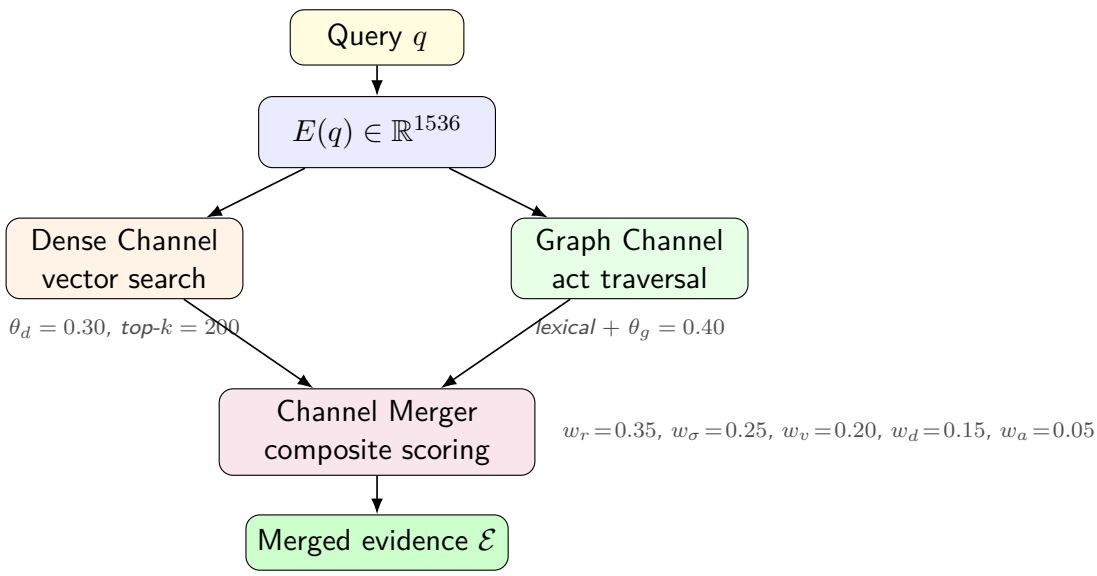

Figure 5.1: Dual-channel retrieval architecture. The query is encoded into $E(q) \in \mathbb{R}^{1536}$, shared by two parallel channels. The *dense channel* performs approximate nearest-neighbour search over a Neo4j vector index (cosine similarity threshold $\theta_d = 0.30$, top-k = 200). The *graph channel* matches parliamentary acts by combining lexical keyword overlap with semantic similarity ($\theta_g = 0.40$), then traverses from matched acts to signatories' speech chunks. The merger deduplicates and reranks using a five-component composite score: relevance ($w_r = 0.35$), salience ($w_\sigma = 0.25$), group coverage ($w_v = 0.20$), speaker diversity ($w_d = 0.15$), and authority ($w_a = 0.05$). In the current pipeline the authority component receives a uniform constant value at merge time, as query-specific authority scores are computed post-retrieval by design (see Sections 5.3.1 and 5.2.4).

5.2.1 Problem Definition

Given a query q about a policy topic, the retrieval stage must identify a set of speech chunks $\mathcal{E} \subseteq \mathcal{C}$ that are (a) semantically relevant to q, (b) representative of all ten parliamentary

groups, and (c) inclusive of speakers whose expertise is evidenced by legislative activity rather than solely by speech content. No single retrieval channel satisfies all three requirements; a dual-channel architecture is therefore employed.

5.2.2 Dense Channel: Vector Similarity Search

The dense channel performs approximate nearest-neighbour search over the Neo4j native vector index (`chunk_embedding_index`). Given $E(q) \in \mathbb{R}^{1536}$, it retrieves the top-$k$ chunks (default $k = 200$) with cosine similarity exceeding a threshold θ_d:

$$\mathcal{C}_{\text{dense}} = \left\{ c \in \mathcal{C} \;\middle|\; \text{sim}\Big(E(q),\, E(c)\Big) \geq \theta_d \right\}, \tag{5.1}$$

where $\theta_d = 0.30$ by default. For each retrieved chunk, the channel resolves the associated speech, speaker, parliamentary group, and session metadata via graph traversal. The channel also detects group changes: if the speaker's current group differs from their group at speech time, the `party_changed` flag is set.

The dense channel captures thematically relevant contributions not associated with formal legislative activity (opposition critiques, committee testimonies, speeches addressing legislation sponsored by other parties). Its limitation is that it cannot distinguish a tangential mention from substantive legislative engagement.

5.2.3 Graph Channel: Structural Traversal

The graph channel retrieves evidence through parliamentary acts in three steps:

1. *Keyword extraction.* Italian stopwords (including articulated prepositions) are removed; unigrams and bigrams are extracted.

2. *Act matching.* Acts are matched via: (a) lexical overlap between extracted keywords and act titles (minimum one match), and (b) semantic similarity between $E(q)$ and the act's precomputed embeddings (threshold $\theta_g = 0.40$). A post-traversal chunk-level filter discards chunks with embedding similarity below 0.30.

3. *Graph traversal.* Matched acts are traversed to their primary and co-signatories (`PRIMARY_SIGNATORY`, `CO_SIGNATORY`), and from there to speakers' speech chunks.

When the query maps to a parliamentary commission via the curated ontology (`commissioni_topics.yaml`), chunks from commission members are additionally privileged. The graph channel explicitly captures expert-authored legislative activity as a retrieval signal, complementing the dense channel's content-based approach.

5.2.4 Channel Merging and Ranking

The merger (`merger.py`) combines outputs from both channels. Items appearing in both are deduplicated by chunk identifier; the higher score is retained. The composite scoring function assigns each evidence item e a score:

$$\text{score}(e) = w_r \cdot r(e) + w_d \cdot d(e) + w_v \cdot v(e) + w_a \cdot \mathit{authority}(s_e, q) + w_\sigma \cdot \sigma(e), \tag{5.2}$$

where s_e denotes the speaker of evidence e. Table 5.1 summarises the components.

Table 5.1: Composite scoring components and default weights in the channel merger.

Symbol	Component	Weight	Definition
$r(e)$	Relevance	0.35	Normalised cosine similarity from embedding search
$d(e)$	Diversity	0.15	1 − speaker_dominance; penalises over-represented speakers
$v(e)$	Coverage	0.20	Rewards evidence from underrepresented parliamentary groups
$\mathit{authority}(s_e, q)$	Authority	0.05	Topic-dependent authority score (Section 5.3)†
$\sigma(e)$	Salience	0.25	Political substance score; procedural language receives lower salience

Weight rationale. Relevance (0.35) is the primary retrieval objective. Salience (0.25) prioritises substantive political content over procedural utterances. Coverage (0.20) ensures multi-group representation (Principle 2). Diversity (0.15) prevents single-speaker dominance. Authority (0.05) is structurally present in the scoring function but is not operationally active in the current pipeline.[1]

[1]The authority component is set to $w_a = 0.05$ in the merger configuration. However, the topic-dependent authority scorer (Section 5.3) requires as input the identifiers of the speakers present in the evidence pool, which are known only after retrieval completes. Computing authority scores before retrieval would require scoring the entire parliamentary corpus for every query—computationally impractical with the current architecture. Consequently, all chunks receive a uniform authority value (0.5) during merging, making the authority term a constant additive bias ($0.05 \times 0.5 = 0.025$) that does not alter the relative ranking. The weight is retained as a configuration placeholder for a possible future development: if pre-computed or periodically updated authority scores were available (e.g., static scores indexed alongside chunk embeddings), the merger could incorporate them directly without awaiting post-retrieval computation, realising the feedback loop illustrated in Figure 4.7. In the current pipeline, the effective authority-informed ranking occurs downstream: query-specific authority scores are used in the generation pipeline to select the citing speaker for each evidence group and to resolve semantic duplicates (Section 5.6.4).

5.3 Topic-Dependent Authority Scoring

This section presents the topic-dependent authority model, the primary and original modelling contribution of this thesis. The model computes a speaker's domain credibility as a function of the current query topic. Figure 5.2 provides an overview.

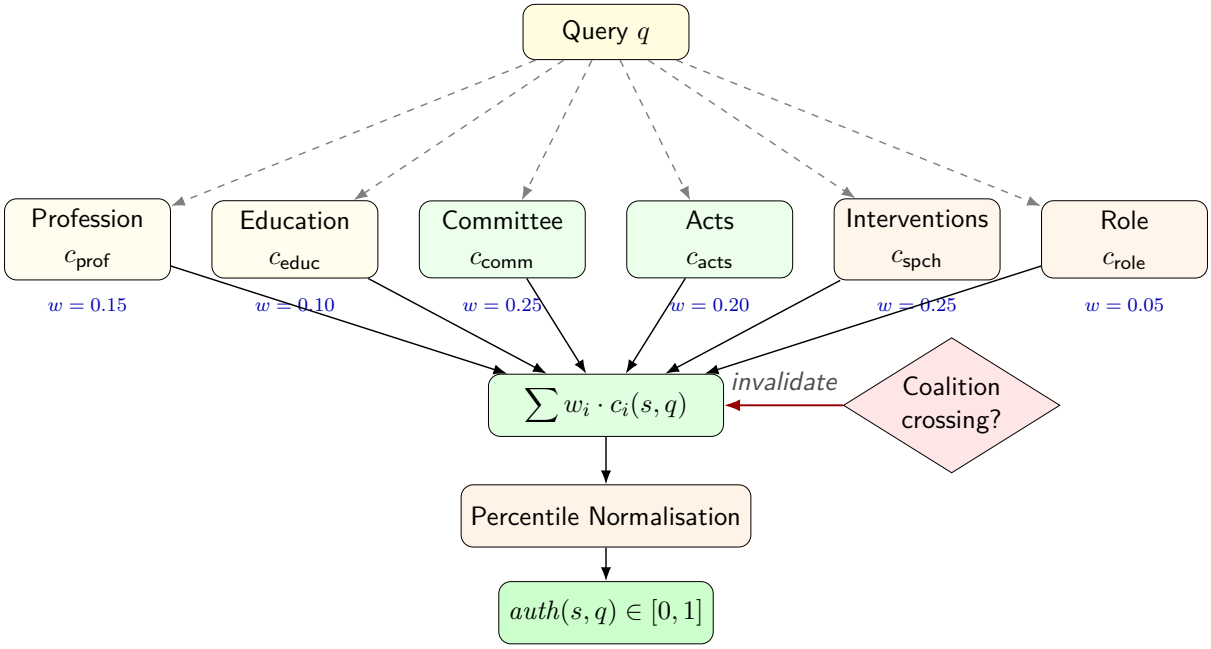

Figure 5.2: Topic-dependent authority model architecture. Given a query q, six component scores are computed for each speaker: profession relevance ($w = 0.15$), education relevance ($w = 0.10$), committee membership relevance ($w = 0.25$), time-decayed relevant parliamentary acts ($w = 0.20$), time-decayed relevant speech interventions ($w = 0.25$), and institutional role ($w = 0.05$). Component colours reflect signal type: yellow for biographical, green for legislative activity, orange for parliamentary activity. The weighted sum is subject to coalition-crossing invalidation (diamond gate). The resulting raw score is percentile-normalised over all speakers evaluated for the current query, yielding a final authority score $auth(s, q) \in [0, 1]$.

5.3.1 Positioning and Design Rationale

Problem definition. Given a query q and a set of speakers $\mathcal{S} = \{s_1, \ldots, s_n\}$ in the retrieved evidence pool, assign each speaker an authority score $authority(s, q) \in [0, 1]$ reflecting their domain credibility on the topic of q.

Rejected alternatives. A static authority approach (e.g., total speeches or acts) fails to capture domain-specificity: a prolific speaker on economic policy is not necessarily authoritative on healthcare. An authority-first retrieval design (computing scores before retrieval and filtering) was considered and rejected for three reasons.

1. *Feedback-loop risk.* Pre-filtering by authority would create a self-reinforcing cycle in which only established actors are retrieved, potentially violating Principle 2 if high-authority speakers are concentrated in a subset of groups. Smaller parliamentary groups whose deputies have lower aggregate visibility—but may hold genuine domain expertise on specific topics—would be systematically excluded before their contributions could be evaluated.

2. *Empirical redundancy given broad retrieval.* The dense channel retrieves up to $k = 200$ chunks (Equation 5.1), and the graph channel adds further chunks via legislative-act traversal (Section 5.2.3). Parliamentary authority on a subject presupposes having spoken about it; consequently, speakers with high topical authority are retrieved by semantic similarity before authority is ever computed. The experimental results confirm this: in all evaluated queries, the top-authority speaker per group is drawn from the retrieval pool (Section 7.6), indicating that the broad retrieval strategy provides sufficient recall for post-retrieval authority scoring to operate effectively.

3. *Computational feasibility.* The six-component model requires per-query embedding comparisons (profession, education, committee, acts, speeches) for each evaluated speaker. In a live interactive system (Section 1.6), computing this for all ∼387 deputies at every query would add unacceptable latency; restricting computation to the speakers present in the retrieved pool bounds the cost proportionally to retrieval breadth.

Proposed approach. Authority is computed after retrieval, over the speakers present in the merged evidence pool. The merger's composite scoring function (Equation 5.2) includes an authority term with weight $w_a = 0.05$; however, since the authority scorer requires the evidence pool as input, query-specific scores are not yet available at merge time. In the current implementation all chunks therefore receive a uniform authority value during merging, and the authority weight acts as a constant bias that does not affect the relative ranking. The effective use of authority scores occurs downstream in two ways: (i) expert selection, where the top-authority speaker per group is chosen as the group's representative (Stage 5 of the pipeline); and (ii) the generation pipeline, where authority scores guide the selection of the citing speaker for each evidence group and the resolution of semantic duplicates. This design ensures authority informs the final output as a ranking and selection signal without acting as a hard retrieval filter, preserving political balance.

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Novelty. The model departs from existing expert-finding approaches (Section 3.3) in three respects: (i) authority is recomputed per query based on semantic relevance of biographical attributes; (ii) temporal decay on behavioural components; and (iii) coalition-crossing invalidation, a domain-specific mechanism without precedent in the literature.

5.3.2 Component Definitions

The authority score is defined as a weighted sum of six components:

$$\mathit{authority}(s, q) = \text{percentile}\left(\sum_{i=1}^{6} w_i \cdot c_i(s, q)\right), \tag{5.3}$$

where $c_i(s, q)$ denotes the i-th component score, w_i is its weight, and percentile$(\cdot)$ denotes rank-based percentile normalisation over all speakers evaluated for the current query.[2] Table 5.2 summarises the components.

Profession and education components. Each speaker s is associated with precomputed embeddings $\mathbf{p}_s$ (profession) and $\mathbf{u}_s$ (education), stored in Neo4j. Each component computes cosine similarity with ReLU clipping:

$$c_{\text{prof}}(s, q) = \max\left(0,\ \frac{\mathbf{p}_s \cdot E(q)}{\|\mathbf{p}_s\| \cdot \|E(q)\|}\right), \qquad c_{\text{educ}}(s, q) = \max\left(0,\ \frac{\mathbf{u}_s \cdot E(q)}{\|\mathbf{u}_s\| \cdot \|E(q)\|}\right). \tag{5.4}$$

If either embedding is absent, the component returns zero. The two components are kept separate because educational formation reflects foundational disciplinary competence, whereas professional background reflects applied expertise.

[2]Weights are loaded from `default.yaml`. The code fallback defaults in `scorer.py` differ (profession = 0.10, committee = 0.20, acts = 0.25, interventions = 0.30) and represent legacy values never active during normal operation.

Table 5.2: Authority score components and configured weights (loaded from `default.yaml`).

Component	Symbol	Weight	Signal
Profession	c_{prof}	0.15	$\text{sim}\big(E(q),\, \mathbf{p}_s\big)$
Education	c_{educ}	0.10	$\text{sim}\big(E(q),\, \mathbf{u}_s\big)$
Committee	c_{comm}	0.25	$\text{sim}\big(E(q),\, \mathbf{k}_m\big)$, max over active memberships
Acts	c_{acts}	0.20	Time-decayed count of relevant parliamentary acts
Interventions	c_{spch}	0.25	Time-decayed count of relevant speeches
Role	c_{role}	0.05	Institutional role weight

Committee component. Committee relevance is computed in two stages. The primary mechanism uses a curated ontology (`commissioni_topics.yaml`) mapping each committee to domain-specific keywords, embedded at runtime using `text-embedding-3-small` to produce $\mathbf{k}_m$. For committees not in the ontology, a fallback uses the committee name embedding from Neo4j. The component selects the maximally relevant active membership:

$$c_{\text{comm}}(s, q) = \max_{m \in \mathcal{M}_s(t)} \max\left(0,\ \frac{\mathbf{k}_m \cdot E(q)}{\|\mathbf{k}_m\| \cdot \|E(q)\|}\right), \tag{5.5}$$

where $\mathcal{M}_s(t)$ is the set of committees to which s holds a valid membership at time t. The max-over-memberships aggregation reflects that authority is determined by the most relevant assignment, not the average.

Acts component. The acts component aggregates legislative activity with role weighting, topical relevance, and temporal decay, compressed via a logarithmic function:

$$c_{\text{acts}}(s, q) = \min\left(1,\ \frac{\ln\Big(1 + \sum_{a \in \mathcal{A}_s} \rho_a \cdot \delta_a \cdot e^{-\lambda_{\text{acts}}(t - t_a)} \cdot \max\Big(0,\ \frac{\mathbf{a} \cdot E(q)}{\|\mathbf{a}\| \cdot \|E(q)\|}\Big)\Big)}{\ln(1 + \tau_{\max})}\right), \tag{5.6}$$

where $\mathcal{A}_s$ is the set of acts co-signed by s, ρ_a distinguishes primary sponsorship ($\rho = 1.0$) from co-signatorship ($\rho = 0.3$), δ_a zeroes contributions below a relevance threshold ($\theta_{\min} = 0.25$), λ_{acts} is the decay rate, and $\tau_{\max} = 20$ is a calibration ceiling.[3] The threshold gate prevents prolific legislators from accumulating high scores through off-topic activity. Acts without a precomputed embedding receive a neutral relevance of 0.5.

[3] $\tau_{\max} = 20$ was calibrated on the observation that approximately 30 relevant acts with mean decay ≈ 0.7 and mean relevance ≈ 0.4 represent a high-activity speaker.

Interventions component. The same aggregation logic applies to the speaker's speech history:

$$c_{\text{spch}}(s,q) = \min\left(1,\ \frac{\ln\Bigl(1 + \sum_{v\in\mathcal{V}_s} \delta_v \cdot e^{-\lambda_{\text{spch}}(t-t_v)} \cdot \max\Bigl(0,\ \frac{\mathbf{v}\cdot E(q)}{\|\mathbf{v}\|\cdot\|E(q)\|}\Bigr)\Bigr)}{\ln(1+\tau_{\max})}\right), \tag{5.7}$$

where $\mathcal{V}_s$ is the set of speeches by s, $\mathbf{v}$ is the embedding of each speech (first 2000 characters), and $\lambda_{\text{spch}} = \lambda_{\text{acts}}$ (548-day half-life). Speeches without embeddings receive neutral relevance of 0.5; those below $\theta_{\min} = 0.25$ are excluded.

Role component. The role component assigns a baseline authority weight to speakers in formal institutional positions, with an optional semantic bonus when the associated committee or group is topically relevant. Two categories are distinguished.

Group or committee roles receive base weights α_r of 0.7 (president), 0.5 (vice-president), and 0.4 (secretary). These values are sourced from the role relationships (`IS_PRESIDENT`, `IS_VICE_PRESIDENT`, `IS_SECRETARY`) in the knowledge graph. The `IS_PRESIDENT` relationship applies both to committee presidencies and to parliamentary group presidencies; the latter confers high political visibility and is weighted identically ($\alpha_r = 0.7$).

Government roles receive base weights of 0.9 (minister), 0.8 (vice-minister), and 0.7 (undersecretary).

Deputies without any institutional role receive a base score of 0.3. The role score is:

$$c_{\text{role}}(s,q) = \alpha_r \cdot \left(1 + \beta_r \cdot \max\left(0,\ \frac{\mathbf{r}_s \cdot E(q)}{\|\mathbf{r}_s\| \cdot \|E(q)\|}\right)\right), \tag{5.8}$$

where α_r is the base weight, $\beta_r = 0.3$ is the relevance bonus coefficient, and $\mathbf{r}_s$ is the embedding of the associated committee or group. The bonus applies only when cosine similarity exceeds 0.5; below this, $c_{\text{role}}(s,q) = \alpha_r$.

Individual component scores are capped at a configurable maximum (default: 0.8) to prevent any single component from dominating the composite authority score.

5.3.3 Temporal Decay

The exponential decay in the acts and interventions components follows:

$$\delta(t,t_0) = e^{-\lambda(t-t_0)}, \tag{5.9}$$

with half-life $h = \ln 2/\lambda$. The configured half-life is 548 days (approximately 18 months), yielding $\lambda \approx 0.00126$ day^{-1}. A common half-life for acts and interventions is an empirically

motivated simplification. The decay prevents authority scores from being dominated by deputies whose prominence peaked in earlier legislative cycles.

5.3.4 Coalition Logic

Group affiliation is resolved at query time via the temporal validity attributes of `MEMBER_OF_GROUP` (Section 4.4.2). The active group $g_s(t)$ is the unique group g such that $t_{\text{start}}(s, g) \leq t \leq t_{\text{end}}(s, g)$.

Coalition-crossing invalidation. If a speaker has moved between the governing coalition (*maggioranza*) and the opposition, authority contributions from the pre-crossing period are invalidated. Only activities under the current coalition affiliation are counted. This prevents a deputy who was in the governing coalition during a previous legislature from inheriting that institutional advantage after moving to the opposition, or vice versa.

5.4 Ideological Compass

5.4.1 Problem Definition

Given the merged evidence set $\mathcal{E}$, the ideological compass computes a two-dimensional visualisation of political positioning on the query topic. No a priori left–right spectrum is imposed; axes emerge from dimensionality reduction (Section 2.8.1) and are interpreted via contrastive term weighting (Section 2.8.3). The pipeline consists of five stages, shown in Figure 5.3.

Figure 5.3: Five-stage ideological compass pipeline. Weighted PCA discovers up to two axes of maximal semantic variation (inverse group-frequency weighting, cap 0.05 per fragment). Projection maps fragments onto axes with SCR filtering (< 0.15) and z-score normalisation (soft-clipped at $\pm 4\sigma$). KDE Clustering estimates group positions (Scott's bandwidth, mode-based, mean fallback for < 3 fragments). Evidence Binding selects 3–5 extreme fragments per pole. TF-IDF Labelling generates axis endpoint labels via contrastive TF-IDF (purity ≥ 0.30).

5.4.2 Axis Discovery via Weighted PCA

The evidence fragment embeddings $\{E(c_i)\}_{c_i \in \mathcal{E}}$ undergo weighted PCA with a per-fragment weight cap of 0.05 and inverse group-frequency weighting, ensuring smaller groups con-

tribute proportionally to axis discovery.

The first two principal components capture the dominant axes of semantic variation. If the ratio of the second eigenvalue to the first falls below 0.40, only a single axis is reported. The analysis is refused if total explained variance falls below 5%. Axis orientation is determined by skewness: negative skewness triggers a flip for deterministic orientation.

5.4.3 Projection and Filtering

Each fragment is projected onto the principal components. A Squared Cosine Residual (SCR) filter excludes fragments with alignment below 0.15. Coordinates are z-score normalised and soft-clipped at $\pm 4\sigma$. An optional soft anchoring mechanism (currently disabled) allows axis orientation guidance without enforcing a priori ideological positions.

5.4.4 Group Clustering

Group positions are estimated via per-group KDE (Section 2.8.2) with Scott's bandwidth. The mode serves as each group's representative position, providing robustness to outliers. For groups with fewer than three fragments, a mean fallback is used.

5.4.5 Evidence Binding

At each axis pole, the three to five most extreme fragments are selected as *pole evidence*, providing interpretable support for the axis's semantic content.

5.4.6 Axis Labelling

Contrastive TF-IDF (Section 2.8.3) is applied to pole fragments, comparing vocabulary at each pole against the opposite. The top terms (default: 3 per pole) generate natural-language axis labels. A minimum pole purity threshold of 0.30 ensures labels are generated only for semantically coherent poles.

5.5 Four-Stage Generation Pipeline

5.5.1 Problem Definition

Given a query q and evidence set $\mathcal{E}$, the generation stage must produce a response satisfying: (a) *party coverage*, with a dedicated section per represented group; (b) *citation grounding*, with every claim supported by a verbatim quotation; and (c) *coherence* across the integrated narrative.

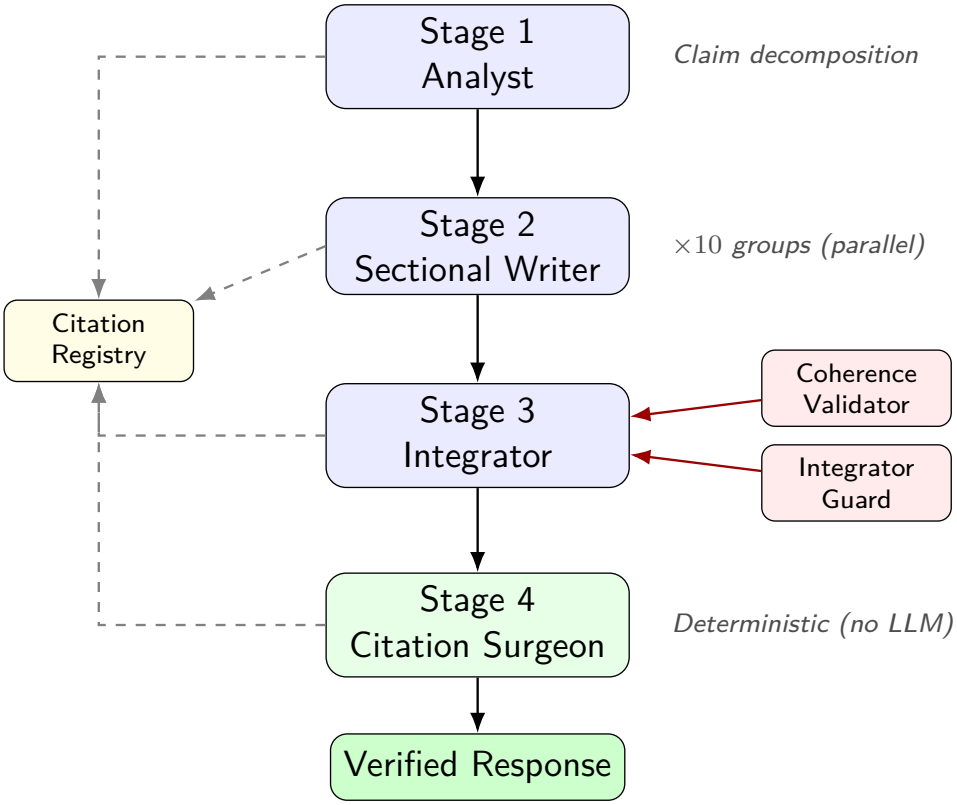

Figure 5.4: Four-stage generation pipeline. Stage 1 (Analyst) decomposes the query into atomic claims with per-group evidence requirements. Stage 2 (Sectional Writer) generates one section per group from position briefs (top-5 chunks, 400 chars each, 500 chars context); an explicit no-evidence statement is produced for groups without relevant material. Stage 3 (Integrator) merges sections into a coherent narrative, constrained by a Coherence Validator ($\tau_{\text{coh}} = 0.30$) and an Integrator Guard that prevents citation loss. Stage 4 (Citation Surgeon) is deterministic: it resolves placeholders to verbatim quotes via stored character offsets. The Citation Registry tracks citations across all stages.

Monolithic generation cannot reliably enforce all three constraints. The pipeline decomposes the problem into four stages. Stages 1–3 invoke GPT-4o (temperature 0.3, top-p 1.0, max tokens 4000); Stage 4 is deterministic. Figure 5.4 illustrates the structure.

5.5.2 Stage 1: Claim Analyst

The Analyst (`analyst.py`) receives query q and evidence set $\mathcal{E}$. It produces a structured decomposition of q into *atomic claims* (discrete factual or positional assertions) with a mapping from claims to per-group evidence requirements. This constrains all downstream stages to address the same set of issues, preventing the Sectional Writer from producing generic party descriptions unrelated to the query.

5.5.3 Stage 2: Sectional Writer

The Sectional Writer (`sectional.py`) generates one section per parliamentary group. Each invocation receives the atomic claims and a *position brief* summarising the top-5 evidence chunks (up to 400 characters each, with 500 characters of surrounding context).

All ten groups are processed (`generation.require_all_parties: true`). If a group has no relevant evidence, the writer produces an explicit statement rather than omitting the group or fabricating content. Position briefs use a consistent template for all groups, ensuring symmetric treatment.

Alternative: Evidence-First Writer. An alternative implementation (`evidence_first_writer.py`) inverts the direction: it begins with the strongest evidence items and generates narrative around them, prioritising citation density over fluency. Both variants are implemented; the default is the standard Sectional Writer for readability.

5.5.4 Stage 3: Narrative Integrator

The Integrator (`integrator.py`) merges per-group sections into a coherent narrative. Two validation mechanisms constrain this stage:

Coherence Validator. After integration, embedding similarity between the integrated narrative and each source section is measured. If alignment falls below $\tau_{\text{coh}} = 0.30$ for any section, re-processing is triggered.

Integrator Guard. A placeholder-counting mechanism verifies that citation count in the output matches the total across all sections. If citations are dropped, the guard reinjects affected content as fallback paragraphs.

5.5.5 Stage 4: Citation Surgeon

The Citation Surgeon (`surgeon.py`) is deterministic and invokes no LLM. It resolves each citation reference into a verified verbatim quotation from the parliamentary record. Three citation formats are processed in priority order: (1) *citation-integrated* format `«inline verbatim» [CIT:id]`, where the LLM writes quoted text between guillemets; (2) bare placeholders `[CIT:id]`; and (3) a legacy markdown format `["text"](id)`.

Primary path: inline verbatim verification. The Surgeon verifies inline text by checking whether it appears as a literal substring of the stored `quote_text`. If successful, two contextual adjustments are applied: (1) *sentence-boundary expansion*, extending

leftward to the nearest sentence boundary (up to 160 characters) to prevent citations from beginning mid-sentence; and (2) *rhetorical question expansion*, appending the immediately following answer sentence (up to 120 characters) when the quote ends with a question mark, preventing semantic inversion.

Nested quote detection. Before accepting an inline quote, the Surgeon checks for third-party quotations embedded within the speaker's discourse (e.g., a deputy quoting a minister). Nested quotes are flagged to preserve correct attribution.

Fallback path. When inline text is not a literal substring (typically due to LLM paraphrasing), the Surgeon falls back to semantic sentence extraction, selecting the most query-relevant sentence from `quote_text`. For bare placeholders, `quote_text` is retrieved directly; if unavailable, stored character offsets (`span_start`, `span_end`) are used on the full speech text; a final fallback uses `chunk_text` (truncated to 300 characters).

Salience gate. Citations with salience ≤ 0.2 (purely procedural language) trigger re-extraction to find a more substantive sentence. The degradation hierarchy is: (1) verified inline substring; (2) sentence-boundary-expanded quote; (3) semantically extracted sentence; (4) truncated `chunk_text`; (5) *"Citazione non disponibile"*. Only level 5 constitutes a true failure.

Post-processing. Resolved citations undergo normalisation: removal of parenthetical interruptions (*applausi*, *interruzioni*), stripping of trailing punctuation, closure of unmatched parentheses, and trimming of dangling function words.

5.6 Citation Integrity System

5.6.1 Citation Registry

The Citation Registry (`citation_registry.py`) tracks citation presence across all four generation stages. At each stage boundary it records which citations were identified (Stage 1), incorporated (Stage 2), survived integration (Stage 3), and successfully extracted (Stage 4). At completion, a check compares Stage 1 citations with those in the final output; unresolved citations are logged for diagnostics.

5.6.2 Citation-First Architecture

The citation integrity system implements a *citation-first* principle: the citation's verifiable text is the immutable anchor around which the narrative is constructed. This manifests in three design decisions:

1. Character-offset spans are stored at ingestion time and never modified.

2. The Citation Surgeon operates on raw speech text, not preprocessed versions.

3. The language model never generates quotation text; it only positions placeholders that the Surgeon resolves via `speech.text[span_start : span_end]`.

This inversion narrows the generation task to *guided selection and arrangement* rather than open-ended synthesis, reducing the surface area for hallucination and grounding the output in the parliamentary record.

5.6.3 Coherence Validation

For each citation, cosine similarity between the surrounding claim context and the cited chunk is computed:

$$\mathrm{coh}(x, c_i) = \frac{E(x) \cdot E(c_i)}{\|E(x)\| \cdot \|E(c_i)\|}, \tag{5.10}$$

where $E(x)$ is the claim context embedding and $E(c_i)$ is the chunk embedding. Scores below $\tau_{\mathrm{coh}} = 0.30$ indicate semantic mismatch (e.g., from Integrator reorganisation). Mismatched citations are flagged in metadata rather than silently corrected, preserving output transparency.

5.6.4 Semantic Deduplication

Citations with pairwise cosine similarity exceeding 0.85 are considered duplicates; only the highest-scoring is retained. This reduces visual clutter without sacrificing evidential breadth.

The combination of citation registry, citation-first architecture, coherence validation, and semantic deduplication constitutes a multi-layered defence of citation faithfulness that is structurally stronger than any post-hoc verification approach.

5.7 A/B Evaluation Infrastructure

The system includes an integrated evaluation infrastructure supporting both automated metrics and human A/B surveys. The evaluation protocol and results are presented in Chapters 6 and 7.

5.7.1 Automated Metrics

The evaluation router (`evaluation.py`) computes metrics by comparing stored chat histories against a ground-truth dataset (`evaluation_set.json`) containing predefined topics, baseline responses, and baseline authority scores. Metrics include party coverage, citation relevance, coalition balance, and authority distribution statistics.

5.7.2 Human A/B Survey

The survey router (`survey.py`) implements a blind comparative protocol. For each query, system and baseline responses are presented in randomised order. Evaluators rate each pair across seven dimensions: answer quality, clarity, completeness, citation relevance, citation accuracy, balance perception, and balance fairness. The survey does not disclose which response is system-generated.

Chapter 6

Experimental Evaluation

This chapter presents the experimental methodology. Section 6.1 formalises the research questions. Section 6.2 describes the benchmark construction process. Section 6.3 defines the evaluation metrics at two levels: automated (deterministic) and human (perceptual). Section 6.4 specifies the baseline system. Section 6.5 details the experimental protocol.

6.1 Research Questions

The evaluation is structured around four research questions, each derived from a gap in the literature (Section 3.6).

RQ1 Multi-View Retrieval. *Does the dual-channel retrieval architecture with coverage-weighted composite merging ensure that evidence is representative of all parliamentary groups?*
This question addresses Gap 1: no existing RAG system structurally guarantees balanced political representation in retrieval results.

RQ2 Speaker Authority. *Does the topic-dependent authority model surface more authoritative speakers than those selected by a general-purpose RAG system without domain-specific authority modelling?*
This is the central research question, corresponding to the core scientific contribution. It tests whether the six-component model (Section 5.3) produces meaningfully differentiated authority scores and whether human experts validate the speaker selections.

RQ3 Citation Integrity. *Does the offset-based citation extraction guarantee verbatim correspondence with the original parliamentary transcript?*
This addresses the citation hallucination problem [10, 83], verifying that every citation is an exact substring of the source speech.

RQ4 Balanced Synthesis. *Does the four-stage generation pipeline produce responses that represent all parliamentary groups and that are perceived as balanced by human evaluators?*
This tests whether the architectural guarantees (per-group sections, Integrator Guard, coverage-weighted merging) translate into responses perceived as fair and complete.

Table 6.1 summarises the mapping between literature gaps, research questions, system components, and primary evaluation metrics.

Table 6.1: Mapping from literature gaps to research questions, system components, and primary evaluation metrics.

Gap	RQ	Component	Primary Metrics
G1: No RAG balances across political groups	RQ1	Dual-channel retrieval, composite merging	Groups with Citation, Completeness
G2: Authority is static and query-independent	RQ2	Authority scorer (6 components)	Mean Authority, Authority Std. Dev.
G3: No verbatim citation guarantee	RQ3	Citation Surgeon (offset-based)	Citation Faithfulness
G4: Multi-perspective treated as optional	RQ4	Four-stage generation pipeline	Completeness, human balance ratings

6.2 Benchmark Construction

No existing benchmark is adequate for evaluating ParliamentRAG. Standard RAG benchmarks (Natural Questions [84], HotpotQA [85], MS MARCO [86]) measure factoid retrieval accuracy, not multi-perspective balance, speaker authority, or citation integrity across politically defined groups. Parliamentary NLP corpora such as ParlaMint [64] provide annotated speech data but no evaluation queries or gold-standard summaries. A dedicated evaluation set was therefore constructed.

6.2.1 Participatory Topic Collection

The benchmark was constructed through a participatory process to avoid introducing the system designer's biases. A structured questionnaire[1] was distributed to potential evaluators and politically informed individuals, asking each respondent to propose three parliamentary topics. For each topic, the respondent provided a thematic category (free-text), a debate intensity rating (Likert 1–5), a polarisation rating (Likert 1–5), a prior knowledge self-assessment, and an optional description of expected party positioning.

Respondent population. The questionnaire received $N = 17$ responses, yielding 51 topic proposals. Profiles: 8 professionals in parliamentary journalism, law, policy analysis, or public administration (47.1%); 6 politically engaged citizens (35.3%); 2 university students (11.8%); 1 accountant (5.9%). Self-reported familiarity with parliamentary activity had a median of 4 and mean of 3.82 (SD = 1.07) on a Likert scale 1–5.

6.2.2 Topic Selection and Justification

From the 51 proposals, 15 topics were selected. This size balances evaluation breadth against evaluator burden (4–5 minutes per topic, 75 minutes total). Selection was guided by four criteria:

1. **Frequency of proposal.** Topics proposed independently by multiple respondents were prioritised (e.g., immigration 4×, AI policy 6×, justice reform 4×).
2. **Thematic coverage.** The final set spans 8 distinct macro-areas.
3. **Variance in debate properties.** Topics with varying debate intensity and polarisation ensure testing under both consensual and adversarial conditions.
4. **Stress-testing.** Certain topics exercise specific system features: coalition crossings (authority invalidation logic), clearly competent committees (committee-based authority), low debate volume (scarce evidence).

Table 6.2 presents the 15 selected topics.

Query formulation. Each topic is converted into a standardised query: *"Qual è la posizione dei gruppi parlamentari su [topic]?"*. This formulation forces the system to identify relevant claims across all ten groups, eliminates query phrasing as a confounding variable, and matches the core use case of multi-view parliamentary topic summarisation.

[1] Administered via Google Forms at https://forms.gle/D66JKZWd91c9UfLh7, open from 20 to 23 February 2026.

6.3 Evaluation Metrics

The evaluation employs two complementary levels: *automated metrics* (Level 1), which are deterministic and fully reproducible, and *human ratings* (Level 2), which capture perceptual quality through a structured A/B protocol. Level 1 verifies structural properties (citation integrity, group coverage); Level 2 measures whether responses are perceived as informative, balanced, and trustworthy by domain experts.

6.3.1 Level 1: Automated Metrics

Automated metrics are computed programmatically from ParliamentRAG's output by the evaluation module (`evaluation.py`), requiring no human judgement. They are deterministic: given the same stored chat history and evaluation dataset, they produce identical values. For each benchmark query q, five metrics are computed.

Groups with Citation

The number of parliamentary groups with at least one citation in the response:

$$GC(q) = |\{g \in \mathcal{G} : \exists\, c \in \mathcal{C}(q),\ \text{group}(c) = g\}| \tag{6.1}$$

where $\mathcal{G}$ is the set of ten parliamentary groups and $\mathcal{C}(q)$ is the citation set. A group is counted only if represented by a *verified citation*, not merely a textual reference. This adapts the concept of *aspect coverage* [87, 88] to the parliamentary domain. For ParliamentRAG, $GC = 10$ is an architectural *design target* pursued through coverage-weighted merging and mandatory per-group sections, but it is not an invariant: a group may receive a dedicated narrative section yet lack a verified citation if the Citation Surgeon cannot extract a verbatim quote from the available evidence. The metric becomes informative when comparing against NotebookLM.

Completeness

The fraction of groups with a dedicated section:

$$Compl(q) = \frac{|\{g \in \mathcal{G} : \text{section}(g) \in \text{response}(q)\}|}{|\mathcal{G}|} \tag{6.2}$$

This generation-side counterpart of GC adapts *aspect recall* [87]. The `generation.require_all_parties:true` flag ensures that the Sectional Writer (Stage 2) produces a dedicated section for every group. However, the downstream Narrative Integrator (Stage 3) may occasionally merge or condense sections during narrative integration, so $Compl = 1.0$

is a design target rather than an architectural invariant. In contrast, Citation Faithfulness (Section 6.3.1) is a true invariant, guaranteed by deterministic offset-based extraction.

Citation Faithfulness

The fraction of citations verified as exact substrings of the source speech:

$$CF(q) = \frac{|\{c \in \mathcal{C}(q) : \text{speech}_c[\text{span_start}_c : \text{span_end}_c] = \text{quote_text}(c)\}|}{|\mathcal{C}(q)|} \tag{6.3}$$

This strict character-level condition is analogous to span extraction in extractive QA [89] and substantially stricter than ROUGE-L [90]. It relates to the *faithfulness* dimension in RAGAS [91] and the *attributability* criterion of Rashkin et al. [10]. For ParliamentRAG, $CF = 1.0$ is guaranteed by the offset-based Citation Surgeon (Section 5.5.5).

Mean Authority

The mean authority score of cited speakers:

$$MA(q) = \frac{1}{|\mathcal{S}(q)|} \sum_{s \in \mathcal{S}(q)} \text{authority}(s, q) \tag{6.4}$$

where $\mathcal{S}(q)$ is the set of distinct cited speakers and $\text{authority}(s, q)$ is the topic-dependent authority score (Section 5.3). This metric has no direct precedent in RAG evaluation; the closest constructs are *source credibility* [92] and *topical authority* [93]. Comparison with NotebookLM is enabled by the baseline expert extraction mechanism (Section 6.4): speakers cited in NotebookLM responses are identified in Neo4j and scored using the same topic-dependent authority model.

Authority Standard Deviation

$$ASD(q) = \text{std}\Big(\{\text{authority}(s, q) : s \in \mathcal{S}(q)\}\Big) \tag{6.5}$$

This tests whether the authority scorer produces differentiated scores. A ranker that does not discriminate is uninformative regardless of its mean output [94].

Interpretation. These metrics occupy two distinct categories. $CF = 1.0$ is the only true architectural invariant: offset-based extraction is deterministic and cannot produce a non-verbatim citation. $GC = 10$ and $Compl = 1.0$ are architectural *design targets* enforced by coverage-weighted merging and mandatory per-group generation, but downstream stages (citation extraction for GC, narrative integration for $Compl$) may introduce marginal

shortfalls. The experimental value of all three lies in the Δ against NotebookLM and, for *GC* and *Compl*, in verifying that the design targets are met at near-ceiling levels.

6.3.2 Level 2: Human Evaluation

Each evaluator rates both ParliamentRAG and NotebookLM on nine dimensions (Likert 1–5). The dimensions provide at least one perceptual measure per research question while keeping evaluation burden manageable.

In addition, evaluators provide an overall preference (A, B, or equal), an overall satisfaction score (1–5), and a would-recommend binary indicator.

The multi-dimensional Likert protocol follows the PARADISE framework [95] and SummEval [96]. The A/B preference follows Chatbot Arena [97] and WMT Direct Assessment [98].

6.4 Baseline: Google NotebookLM

ParliamentRAG is evaluated against Google NotebookLM[2], a commercially available RAG tool, as the sole external baseline. The choice of a single strong baseline was deliberate: NotebookLM is widely regarded as one of the strongest commercially available systems for document-grounded RAG, making it a demanding comparison target. An external baseline also answers the practically relevant question: *does ParliamentRAG produce better results than a tool a user could readily access?* Internal ablations were deferred to future work (Section 7.7).

Source material and retrieval asymmetry. A key methodological difference between the two systems concerns source selection. ParliamentRAG operates on the full knowledge graph (151,073 chunks from 40,416 speeches) and performs automatic dual-channel retrieval. NotebookLM cannot operate on a corpus of this scale; instead, for each benchmark topic, a curated set of approximately 300 speech chunks was uploaded: 150 topically relevant chunks (selected to maximise recall across all ten parliamentary groups) and 150 distractor chunks from adjacent but off-topic debates. This *high-recall curated retrieval* setup provides NotebookLM with a favourable retrieval environment (all relevant perspectives are present in its source material), but at reduced precision compared to a perfectly curated set (due to the distractors). The asymmetry means that NotebookLM does not need to solve the retrieval problem, only the generation problem, while ParliamentRAG must solve

[2]NotebookLM (https://notebooklm.google.com/) is a generative AI tool by Google that allows users to upload documents and query them conversationally.

both. Any advantage of ParliamentRAG on the retrieval-dependent dimensions (coverage, authority) is therefore conservative.

Baseline prompt design. NotebookLM receives a structured prompt specifying the same output requirements as ParliamentRAG: an introduction with concrete data, per-group sections covering all ten parliamentary groups, direct citations with speaker attribution, balanced length between coalition and opposition, and explicit no-evidence statements when applicable. The full prompt is reproduced in Appendix F. This ensures that observed differences reflect architectural capabilities rather than output specification.

Fixed baseline. NotebookLM responses were pre-generated at a fixed point in time and stored in `evaluation_set.json`, so all evaluators judge the same text.

Table 6.4 summarises the key architectural differences between ParliamentRAG and NotebookLM.

Baseline expert extraction. Speakers cited in NotebookLM responses do not have pre-computed authority scores. A dedicated endpoint extracts speaker names from the baseline text, identifies them in Neo4j, and computes their authority scores using the same topic-dependent authority model (Section 5.3). Results are stored in `evaluation_set.json`, ensuring methodological comparability.

Limitations. NotebookLM is proprietary and non-reproducible: its internal mechanisms are opaque and may change over time. Pre-computing responses at a fixed point mitigates this but does not eliminate the limitation.

6.5 Experimental Protocol

6.5.1 Evaluator Population

The evaluation is conducted by 6 domain experts: parliamentary journalists and parliamentary collaborators (*collaboratori parlamentari*), all with direct professional experience with Italian parliamentary proceedings. Domain experts were chosen over crowd-workers because dimensions such as Citations Relevance and Balance Fairness require familiarity with Italian parliamentary dynamics.

6.5.2 A/B Testing Procedure

A blind A/B comparison protocol is used. The paired design controls for evaluator-level bias and enables within-evaluator comparisons. Randomised label assignment prevents order effects. For each benchmark query:

1. **Response generation.** ParliamentRAG processes the query at runtime. The pre-computed NotebookLM response is retrieved from `evaluation_set.json`.
2. **Randomised assignment.** The two responses are randomly labelled "Response A" and "Response B", stored on the `ChatHistory` node. The evaluator does not know which is which.
3. **Side-by-side presentation.** Both responses are shown simultaneously in rendered form, including citations and speaker attributions.
4. **Rating.** The evaluator rates each response on nine dimensions (Table 6.3), expresses an overall preference, and provides a satisfaction score.
5. **Submission.** The evaluation is submitted via `POST /api/surveys` and stored as a `SurveyEvaluation` node linked to the `ChatHistory` node.

Each evaluator assesses all 15 queries (estimated 60–75 minutes). NotebookLM responses are pre-generated, eliminating waiting time.

6.5.3 Statistical Analysis Plan

The analysis plan is specified *a priori* to avoid post-hoc test selection [99].

Level 1 metrics. Mean and standard deviation across 15 queries, with 95% confidence intervals (CIs) via bootstrap resampling (10,000 iterations). Bootstrap is used because normality cannot be reliably verified with 15 queries. By-design metrics deviating from expected values are flagged as software defects. For Mean Authority and Authority Standard Deviation, $\Delta = \bar{m}_{\text{ParliamentRAG}} - \bar{m}_{\text{NotebookLM}}$ is reported.

Level 2 metrics. Each evaluator provides paired ratings $(r_{\text{ParliamentRAG}}, r_{\text{NotebookLM}})$ per dimension. The **Wilcoxon signed-rank test** is the primary significance test, following Graham et al. [100, 101]. With up to $N = 90$ paired observations per dimension (6 evaluators $\times$ 15 queries), the test has adequate power for medium-to-large effects.

Multiple comparisons. Nine simultaneous tests require **Holm–Bonferroni correction** [99] on p-values.

Effect size. **Cohen's** d is reported per dimension. An effect of $d \geq 0.5$ is medium, $d \geq 0.8$ is large [102].

Overall preference. The preference distribution is tested against $H_0 : p_{\text{ParliamentRAG}} = p_{\text{NotebookLM}} = 0.5$ via a **binomial test** (excluding "equal" responses).

Inter-rater agreement. **Krippendorff's** α [103] is computed per dimension, preferred over Cohen's κ for handling more than two raters and ordinal scales.

Variance decomposition. Intra-topic variance (evaluator disagreement on the same material) is separated from inter-topic variance (inherent topic difficulty), as they have different implications for interpreting results.

6.5.4 Power Analysis

With 6 evaluators and 15 queries, the maximum is $N = 90$ paired observations per dimension. The Wilcoxon signed-rank test achieves power ≥ 0.80 for detecting a medium effect ($d \approx 0.5$) at $\alpha = 0.05$ with $N \geq 35$. After Holm–Bonferroni correction for 7 tests, the effective threshold for the last-ranked test is $\alpha/7 \approx 0.007$, requiring larger effects or N closer to the upper bound. Small effects may not reach significance; the evaluation should be interpreted primarily in terms of effect sizes.

Table 6.2: The 15 benchmark topics, selected from 51 proposals ($N = 17$ respondents). Topics span 8 thematic macro-areas. **Deb.** = median debate intensity (Likert 1–5); **Pol.** = median polarisation (Likert 1–5). Each topic is queried as *"Qual è la posizione dei gruppi parlamentari su [topic]?"*.

#	Topic	Thematic Area	Deb.	Pol.	Selection Rationale
1	Riforma della giustizia	Justice & Civil Rights	5	4	High frequency (4×); tests committee authority (II Commissione)
2	Riforma del premierato	Institutional Reforms	3	4	High frequency (3×); constitutional reform, sharp coalition divide
3	Gestione dei flussi migratori	Immigration	5	5	Highest frequency (4×); maximum polarisation
4	Autonomia differenziata	Institutional Reforms	4	5	Intra-coalition tensions (Lega vs. rest of majority)
5	Salario minimo	Labour & Welfare	5	5	Clear coalition–opposition divide; high debate volume
6	Ritorno al nucleare	Environment & Energy	3	4	Cross-cutting: internal splits within PD
7	Politiche sull'intelligenza artificiale	Technology & Digital	2	3	Highest frequency (6×); emerging topic, sparse evidence
8	Aiuti militari all'Ucraina	Defence & Foreign Policy	5	5	Cross-cutting cleavages within both coalitions
9	Legge di Bilancio	Economy & Finance	4	4	Annual macro-topic; broad committee coverage
10	Golden Power	Economy & Finance	2	2	Low debate intensity; tests scarce evidence
11	Riarmo europeo	Defence & Foreign Policy	4	4	Tests temporal recency of authority scoring
12	Bolkestein (riforma concessioni)	Economy & Finance	3	3	Tests query rewriting; evolving party positions
13	Educazione affettiva e sessuale	Education	3	5	High polarisation; non-economic cleavage axis
14	Sgravi sul costo del lavoro	Economy & Finance	2	3	Low prior knowledge; technical topic
15	Bioetica	Health / Civil Rights	2	4	No respondent knew party positions; cross-cutting moral issues

Table 6.3: Nine dimensions of the human evaluation protocol, rated on a Likert scale 1–5.

Dimension	**Description**	**RQ**
Answer Quality	Overall quality of the response	RQ1–4
Answer Clarity	Clarity and readability of the exposition	RQ4
Answer Completeness	Completeness of informational coverage	RQ1, RQ4
Citations Relevance	Pertinence of the cited parliamentary passages	RQ2, RQ3
Balance Perception	Perceived balance across political groups	RQ1, RQ4
Balance Fairness	Perceived fairness in the representation of positions	RQ1, RQ4
Source Relevance	Pertinence of cited speakers to the query topic	RQ2
Source Authority	Institutional authority of cited speakers	RQ2
Source Coverage	Coverage of parliamentary groups among cited sources	RQ1, RQ2

Table 6.4: Architectural comparison between ParliamentRAG and the NotebookLM baseline.

Feature	**ParliamentRAG**	**NotebookLM**
Source selection	Graph-augmented hybrid retrieval over full knowledge graph (151K chunks)	High-recall curated chunks (150 relevant + 150 distractors per topic)
LLM	GPT-4o (OpenAI)	Gemini Pro (Google)
Authority model	Topic-dependent, 6-component scorer	None (implicit semantic selection)
Coverage enforcement	Architectural (per-group sections, coverage-weighted merging)	Prompt-based
Citation integrity	By construction (offset-based extraction)	Post-hoc (LLM-generated)

Table 6.5: Profile of the six domain expert evaluators. n_{topics}: topics assessed out of 15. Overall completion: 67 out of 90 possible evaluations (74.4%).

ID	**Role**	**Exp.**	n_{topics}	**Completion**
E1	Policy analyst	6 years	11	73.3%
E2	Parliamentary journalist	12 years	15	100%
E3	Parliamentary collaborator	8 years	10	66.7%
E4	Parliamentary journalist	3 years	9	60.0%
E5	Parliamentary collaborator	10 years	15	100%
E6	Policy analyst	4 years	7	46.7%

Chapter 7

Experimental Results

This chapter reports the results of the experimental evaluation described in Chapter 6. Section 7.1 presents the automated metrics (Level 1). Section 7.2 presents the human evaluation (Level 2). Section 7.3 reports the statistical tests. Section 7.4 analyses inter-rater agreement. Section 7.5 provides qualitative analysis. Section 7.6 synthesises findings per research question. Section 7.7 discusses limitations.

7.1 Automated Metrics (Level 1)

Table 7.1 reports the automated metrics computed over the 15 benchmark topics. Figure 7.1 provides a visual comparison of ParliamentRAG against NotebookLM.

Table 7.1: Automated metrics: ParliamentRAG vs. NotebookLM.

Metric	**ParliamentRAG** ($\mu \pm \sigma$)	[95% CI]	**NotebookLM** (μ)	[95% CI]	Δ
Groups with Citation (GC)	0.97 ± 0.07	[0.93, 1.00]	0.95	[0.91, 0.98]	+0.02
Completeness	0.99 ± 0.04	[0.97, 1.00]	1.00	[1.00, 1.00]	−0.01
Citation Faithfulness (CF)	1.00 ± 0.00	[1.00, 1.00]	0.95	[0.91, 0.99]	+0.05
Mean Authority (MA)	0.53 ± 0.16	[0.45, 0.62]	0.52	[0.49, 0.55]	+0.02
Authority Std Dev (ASD)	0.05 ± 0.02	[0.04, 0.06]		—	—

Groups with Citation ($\mu = 0.97$ vs. 0.95; $\Delta = +0.02$). ParliamentRAG achieves near-perfect citation-level coverage of all ten parliamentary groups (CI $= [0.94, 1.00]$). NotebookLM reaches 0.95, occasionally omitting one or more groups. The difference is architecturally meaningful: ParliamentRAG pursues full group coverage as an architectural design target through coverage-weighted merging and mandatory per-group sections (Section 5.2.4), achieving near-ceiling results. The marginal shortfall (0.97) arises when a group receives a narrative section but the Citation Surgeon cannot extract a verified verbatim quote from the available evidence. NotebookLM, by contrast, achieves coverage as an emergent property of its prompt, which may fail for groups with weaker signal.

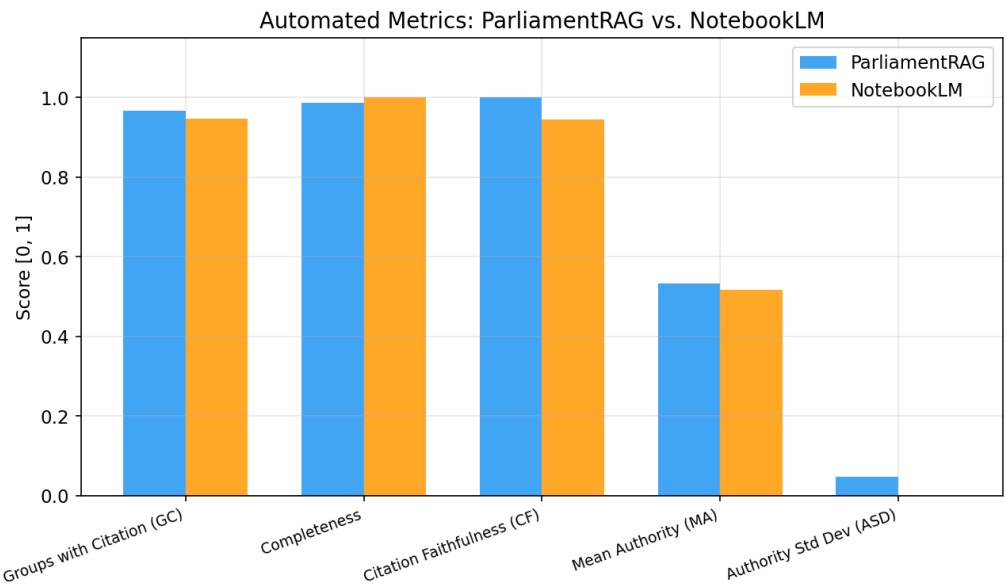

Figure 7.1: Automated metrics: ParliamentRAG (blue) vs. NotebookLM (orange). Error bars indicate ±1 standard deviation. The three leftmost metrics (GC, Completeness, CF) are near-ceiling for both systems; the difference in CF ($\Delta = +0.05$) reflects ParliamentRAG's architectural guarantee. Mean Authority shows a modest advantage ($\Delta = +0.016$).

Completeness ($\mu = 0.99$ vs. 1.00; $\Delta = -0.01$). Both systems achieve near-perfect narrative completeness. ParliamentRAG's marginal shortfall (0.99) reflects a single case where a group was mentioned without a full section. NotebookLM achieves 1.00 because its structured prompt requires per-group sections and the model complies reliably.

Citation Faithfulness ($\mu = 1.00$ vs. 0.95; $\Delta = +0.05$). The most architecturally significant difference. ParliamentRAG achieves $CF = 1.00$ by design: every citation is extracted via character offsets (Section 5.5.5). NotebookLM achieves 0.95, meaning approximately 5% of its citations do not correspond verbatim to the source text. In a journalistic context, even a small rate of misattributed citations can undermine credibility.

Mean Authority ($\mu = 0.533$ vs. 0.517; $\Delta = +0.016$). ParliamentRAG selects speakers with modestly higher mean authority scores ($\text{CI}_{\text{ParliamentRAG}} = [0.47, 0.62]$; $\text{CI}_{\text{NotebookLM}} = [0.49, 0.55]$). As noted in Section 6.3.1, this metric is partially circular. The independent human validation comes from Source Authority (Section 7.2.3).

Authority Standard Deviation ($\mu = 0.05$; CI $= [0.04, 0.06]$). The low but non-zero standard deviation confirms that the authority scorer produces differentiated scores across topics. Figure 7.2 shows the variation: topics with a clearly competent committee (e.g., *Riforma della giustizia*) produce higher authority scores, while emerging topics (e.g.,

Golden Power) yield lower scores for both systems.

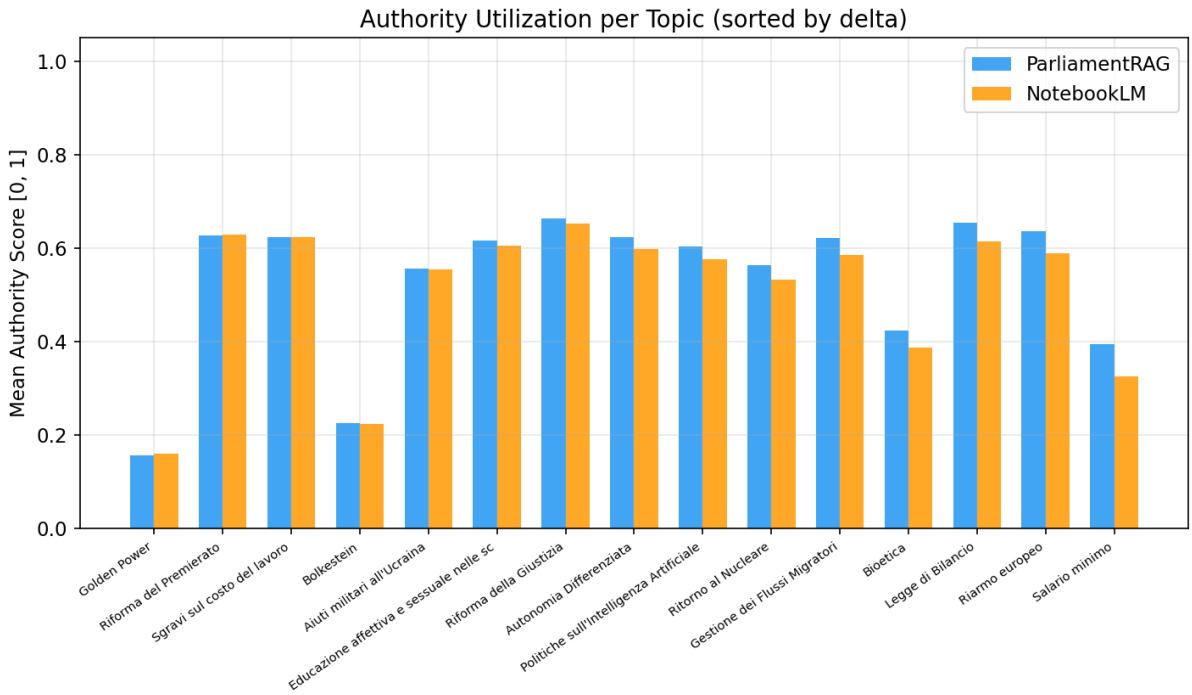

Figure 7.2: Mean authority score per topic: ParliamentRAG (blue) vs. NotebookLM (orange), sorted by Δ. The variation confirms the query-dependence of the topic-dependent authority model.

7.2 Human Evaluation Results (Level 2)

A total of $N = 67$ paired evaluations were collected from 6 domain experts across 15 topics.

7.2.1 Per-Dimension Ratings

Table 7.2 reports the mean Likert ratings (1–5). Figure 7.3 visualises the comparison.

The data reveal *complementary strengths*. ParliamentRAG outperforms NotebookLM on dimensions related to its architectural innovations: Balance Perception ($\Delta = +0.18$), Source Relevance ($\Delta = +0.24$), Source Authority ($\Delta = +0.21$), and Source Coverage ($\Delta = +0.25$). NotebookLM outperforms on prose quality: Answer Quality ($\Delta = -0.26$) and Answer Clarity ($\Delta = -0.24$). The remaining dimensions show negligible differences ($|\Delta| \leq 0.03$).

This pattern reflects the design trade-offs: ParliamentRAG's four-stage pipeline prioritises structural guarantees (per-group coverage, citation integrity, authority-weighted selection) at the cost of fluency that a monolithic generation approach achieves.

Table 7.2: Mean ratings (1–5) per dimension: ParliamentRAG vs. NotebookLM.

Dimension	ParliamentRAG ($\mu \pm \sigma$)	NotebookLM ($\mu \pm \sigma$)	n
Answer Quality	4.04 ± 0.86	4.30 ± 0.78	67
Answer Clarity	4.27 ± 0.93	4.51 ± 0.68	67
Answer Completeness	4.12 ± 0.86	4.09 ± 0.85	67
Citations Relevance	4.37 ± 0.76	4.36 ± 0.71	67
Balance Perception	4.66 ± 0.54	4.48 ± 0.73	67
Balance Fairness	4.67 ± 0.59	4.66 ± 0.54	67
Source Relevance	4.07 ± 0.72	3.84 ± 0.64	67
Source Authority	4.21 ± 0.75	4.00 ± 0.72	67
Source Coverage	4.64 ± 0.64	4.39 ± 0.82	67
Overall Satisfaction	4.24 ± 0.80	4.27 ± 0.73	67

7.2.2 Preference Distributions

Table 7.3 and Figure 7.4 report A/B preference counts.

Table 7.3: A/B preference counts per dimension (all evaluations).

Dimension	ParliamentRAG preferred	Equal	NotebookLM preferred
Answer Quality	20 (30%)	17 (25%)	30 (45%)
Answer Clarity	15 (22%)	32 (48%)	20 (30%)
Answer Completeness	16 (24%)	37 (55%)	14 (21%)
Citations Relevance	16 (24%)	36 (54%)	15 (22%)
Balance Perception	14 (21%)	48 (72%)	5 (7%)
Balance Fairness	8 (12%)	52 (78%)	7 (10%)
Source Relevance	22 (33%)	32 (48%)	13 (19%)
Source Authority	20 (30%)	38 (57%)	9 (13%)
Source Coverage	17 (25%)	47 (70%)	3 (4%)
Overall Satisfaction	21 (31%)	28 (42%)	18 (27%)

Overall preference. ParliamentRAG preferred in 21 evaluations (31.3%), NotebookLM in 18 (26.9%), tied in 28 (41.8%). Binomial test on non-tied comparisons: $p = 0.749$, no evidence of overall preference. Overall satisfaction: $\mu_{\text{ParliamentRAG}} = 4.24$, $\mu_{\text{NotebookLM}} = 4.27$ ($\Delta = -0.03$).

This parity is noteworthy given the asymmetry of the comparison. NotebookLM received high-recall curated source material (Section 6.4; Table 6.4), did not need to solve the retrieval problem, and is powered by Google's Gemini model family—a state-of-the-art commercial LLM. ParliamentRAG achieves equivalent overall satisfaction despite operating on the full uncurated knowledge graph (151K chunks) with automatic retrieval. The high tie rate (41.8%) indicates comparable overall quality, with differentiation at the dimension level: the two systems serve complementary information needs. Aggregate satisfaction, as a

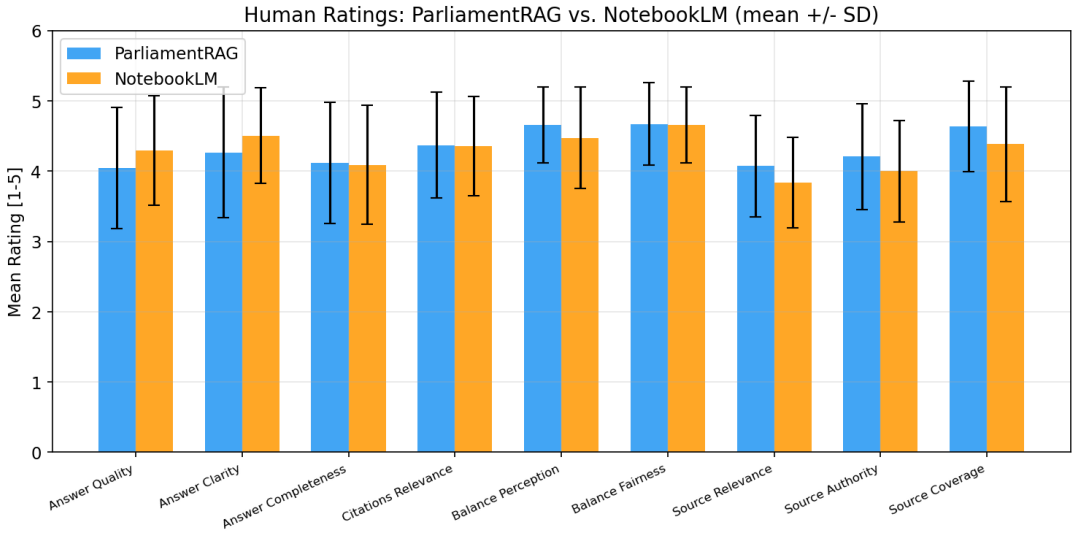

Figure 7.3: Mean human ratings (1–5 Likert): ParliamentRAG (blue) vs. NotebookLM (orange). Error bars: ±1 SD. ParliamentRAG leads on Balance Perception, Source Relevance, Source Authority, and Source Coverage. NotebookLM leads on Answer Quality and Answer Clarity.

single scalar, compresses these complementary strengths into an uninformative mean. The dimension-level analysis (Sections 7.2.1–7.2.3) reveals that the value of ParliamentRAG's architectural complexity is concentrated precisely where it was designed to be: source authority, source coverage, and political balance.

Dimension-level preferences. On Source Coverage, ParliamentRAG is preferred 25% vs. 4% (ratio 6:1 among non-tied). On Balance Perception, 21% vs. 7% (3:1). On Source Authority, 30% vs. 13% (2.3:1). Conversely, on Answer Quality, NotebookLM is preferred 45% vs. 30%.

7.2.3 Source Authority Assessment

The three source dimensions were introduced to operationalise RQ2 at the perceptual level.

ParliamentRAG consistently outperforms NotebookLM: Source Relevance (4.07 vs. 3.84, $\Delta = +0.24$), Source Authority (4.21 vs. 4.00, $\Delta = +0.21$), Source Coverage (4.64 vs. 4.39, $\Delta = +0.25$). Preference counts reinforce this: Source Authority 20 vs. 9 (2.2:1); Source Relevance 22 vs. 13 (1.7:1); Source Coverage 17 vs. 3 (5.7:1).

These results constitute *independent human validation* of the topic-dependent authority model, which is the core scientific contribution of this thesis. The evaluators, blind to which system used authority-aware ranking, consistently perceive ParliamentRAG's cited

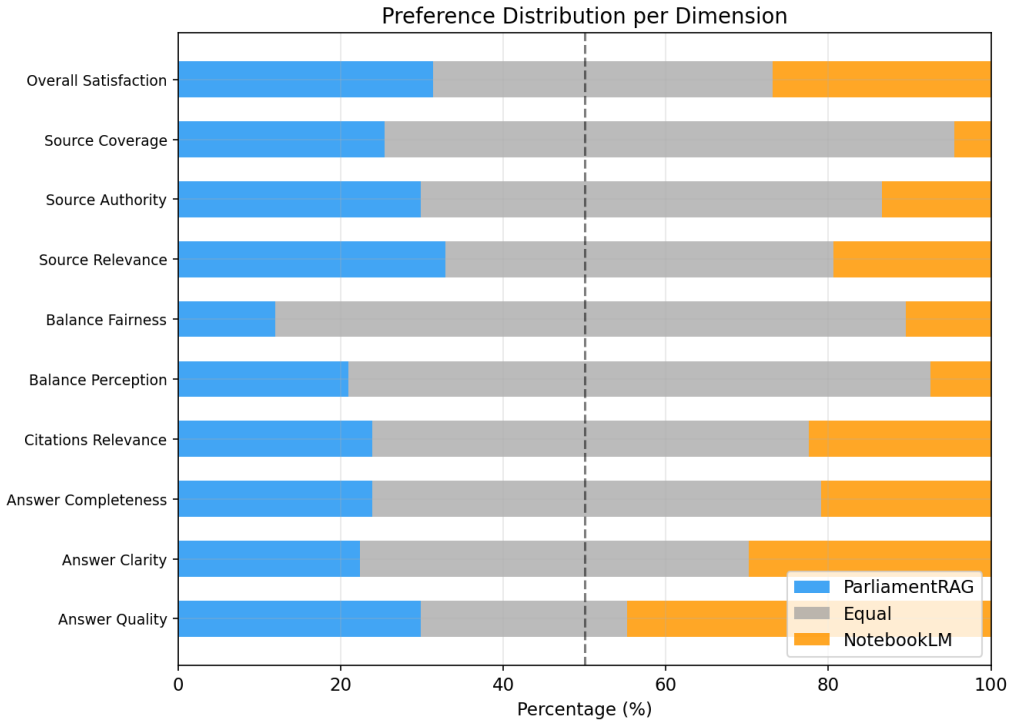

Figure 7.4: Preference distribution per dimension. Blue = ParliamentRAG preferred, grey = equal, orange = NotebookLM preferred. ParliamentRAG's strongest advantages: Source Coverage (25% vs. 4%), Source Authority (30% vs. 13%), Balance Perception (21% vs. 7%). NotebookLM's advantage: Answer Quality (30% vs. 45%).

speakers as more relevant, more authoritative, and better distributed across coalitions. It is worth emphasising the strength of the baseline against which this validation is obtained: NotebookLM is a production-grade RAG system developed by Google, powered by the Gemini model family, and represents a state-of-the-art commercial offering in the RAG space. In this evaluation it further benefited from manually curated source material with high recall (Section 6.4; Table 6.4). The consistent advantage of ParliamentRAG on all three source dimensions, under these conditions, provides strong evidence that dedicated authority modelling adds measurable value beyond what general-purpose semantic retrieval can achieve.

Per-group authority votes. Table 7.4 summarises per-group votes.

The strongest advantages appear for Fratelli d'Italia (3.4:1), Forza Italia (3.0:1), and Misto (7.0:1). The only group where NotebookLM is slightly preferred is Azione – Renew Europe (0.8:1). For Movimento 5 Stelle, the ratio is 1:1. The strong Misto performance likely reflects ParliamentRAG's ability to identify the most active individual voices within a structurally diverse group through fine-grained, per-speaker authority scoring.

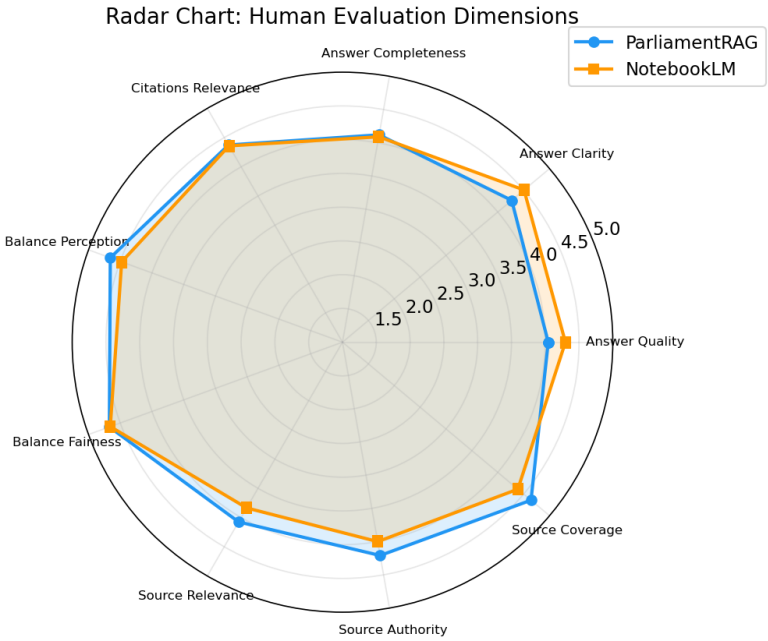

Figure 7.5: Radar chart: ParliamentRAG (blue) vs. NotebookLM (orange). ParliamentRAG extends further on Balance Perception, Source Coverage, Source Authority, Source Relevance. NotebookLM extends further on Answer Clarity and Answer Quality.

7.3 Statistical Tests

Table 7.5 reports the full statistical analysis: Wilcoxon signed-rank test, raw and Holm–Bonferroni-corrected p-values, and Cohen's d.

No dimension reaches significance after correction. The closest are Source Coverage ($p_{\text{raw}} = 0.016$, $p_{\text{Holm}} = 0.141$), Source Authority ($p_{\text{raw}} = 0.023$, $p_{\text{Holm}} = 0.181$), and Source Relevance ($p_{\text{raw}} = 0.027$, $p_{\text{Holm}} = 0.188$). Before correction, three dimensions would have reached significance at $\alpha = 0.05$. The correction for nine simultaneous tests absorbs these effects, as anticipated in the power analysis (Section 6.5.4).

Effect sizes reveal a coherent pattern. ParliamentRAG shows small positive effects on Source Relevance ($d = 0.35$), Source Coverage ($d = 0.35$), Balance Perception ($d = 0.28$), and Source Authority ($d = 0.28$). NotebookLM shows small positive effects on Answer Quality ($d = -0.31$) and Answer Clarity ($d = -0.29$). The remaining dimensions have negligible effect sizes ($|d| \leq 0.03$). A user prioritising source quality and political balance would tend to prefer ParliamentRAG; one prioritising prose fluency would tend to prefer NotebookLM.

Table 7.4: Per-group authority preference votes. S = ParliamentRAG preferred, B = NotebookLM preferred, Eq. = tied. ParliamentRAG achieves a favourable ratio for 8 out of 10 groups.

Parliamentary Group	**S**	**Eq.**	**B**	**Ratio**
Partito Democratico	**30**	25	11	2.7:1
Alleanza Verdi e Sinistra	**26**	29	11	2.4:1
Fratelli d'Italia	**24**	36	7	3.4:1
Noi Moderati	**19**	36	12	1.6:1
Lega – Salvini Premier	**17**	43	6	2.8:1
Movimento 5 Stelle	17	32	17	1.0:1
Forza Italia	**15**	46	5	3.0:1
Misto	**14**	50	2	7.0:1
Italia Viva	**13**	46	7	1.9:1
Azione – Renew Europe	9	47	**11**	0.8:1

Table 7.5: Statistical summary: Wilcoxon + Cohen's d per dimension.

Dimension	μ_{sys}	μ_{base}	Δ	W	p (raw)	Cohen's d	Effect
Answer Quality	4.04	4.30	-0.25	455.0	0.0629	-0.31	small
Answer Clarity	4.27	4.51	-0.24	214.0	0.0895	-0.29	small
Answer Completeness	4.12	4.09	+0.03	228.0	0.9242	0.03	negligible
Citations Relevance	4.37	4.36	+0.01	241.5	0.8916	0.02	negligible
Balance Perception	4.66	4.48	+0.18	49.0	0.0492	0.28	small
Balance Fairness	4.67	4.66	+0.01	56.5	0.8296	0.03	negligible
Source Relevance	4.07	3.84	+0.24	188.5	0.0268	0.35	small
Source Authority	4.21	4.00	+0.21	121.5	0.0226	0.28	small
Source Coverage	4.64	4.39	+0.25	42.5	0.0157	0.35	small
Overall Satisfaction	4.24	4.27	-0.03	—	—	—	n/a

Overall preference is not significant. Binomial test: 21 vs. 18 (excluding 28 ties), $p = 0.749$.

7.4 Inter-Rater Agreement

Table 7.6 reports Krippendorff's α (ordinal) per dimension.

All dimensions show low agreement. Values range from -0.203 (Answer Completeness) to 0.083 (Source Relevance), well below the conventional threshold of $\alpha \geq 0.667$ [103].

Three factors explain the low agreement. First, the evaluation requires domain-specific subjective judgement: what constitutes an "authoritative" speaker is inherently a matter of professional interpretation. Amidei et al. [104] show that inter-rater agreement in NLG evaluation is systematically lower for subjective dimensions. Second, the incomplete design

Table 7.6: Inter-rater agreement (Krippendorff's α, ordinal).

Dimension	Krippendorff α	Interpretation
Answer Quality	-0.003	tentative
Answer Clarity	-0.050	tentative
Answer Completeness	-0.203	tentative
Citations Relevance	0.021	tentative
Balance Perception	0.002	tentative
Balance Fairness	0.026	tentative
Source Relevance	0.083	tentative
Source Authority	0.052	tentative
Source Coverage	-0.040	tentative

matrix (only two evaluators completed all 15 topics) produces unstable, downward-biased α estimates [103]. Third, a ceiling effect compresses the rating distribution: most ratings fall between 4 and 5.

The low agreement does not invalidate the evaluation but requires cautious interpretation. Effect sizes (Section 7.3) provide a more robust basis than p-values, as they are less sensitive to individual rating distributions [105].

7.5 Qualitative Analysis

7.5.1 Where ParliamentRAG Outperforms NotebookLM

Source Authority and Source Relevance. The advantage ($d = 0.28$–0.35) is attributable to the topic-dependent authority model (Section 5.3), which incorporates committee membership, legislative acts, and prior speeches as signals of topic-specific expertise. NotebookLM selects speakers based on semantic similarity alone, which may surface eloquent but non-expert speakers. The per-group authority votes (Table 7.4) confirm the advantage is broad (8 out of 10 groups), ruling out query-specific artefacts.

Balance Perception and Source Coverage. The advantage reflects the multi-view retrieval architecture (Section 5.2) and mandatory per-group sectional writing (Section 5.5.3). The composite merging function includes a coverage component (weight 0.20) that penalises underrepresentation. NotebookLM achieves balance as an emergent prompt property, which is fragile when source material is unevenly distributed across groups.

7.5.2 Where NotebookLM Outperforms ParliamentRAG

Answer Quality and Answer Clarity. The advantage ($d = -0.29$ to -0.31) reflects the inherent trade-off in ParliamentRAG's generation architecture. The four-stage pipeline and citation-first discipline constrain the narrative flow. NotebookLM's monolithic generation benefits from the full generative capacity of its underlying model. This trade-off is by design: ParliamentRAG prioritises verifiability and structural guarantees over generative elegance.

The trade-off in context. The prose quality gap does not imply that ParliamentRAG's architectural complexity is without practical value. The two systems target different points in the design space: NotebookLM optimises for a single objective (fluent, grounded response), while ParliamentRAG simultaneously enforces multiple constraints (multi-group coverage, authority-aware speaker selection, verbatim citation integrity) that are functional requirements in the parliamentary domain. Domain experts confirm this: on the three source dimensions and on Balance Perception—the dimensions most directly relevant to the professional use case—ParliamentRAG is consistently preferred (Section 7.2.3). The prose quality disadvantage is a cost of these guarantees, not evidence of their irrelevance. Future work on generation quality (e.g., replacing the four-stage pipeline with a single constrained-decoding pass, or upgrading the underlying LLM) could narrow this gap without sacrificing structural properties.

7.5.3 Dimensions at Parity

Answer Completeness ($d = 0.03$), Citations Relevance ($d = 0.02$), and Balance Fairness ($d = 0.03$) show negligible differences. Both systems cover all groups in their responses. The authority-weighted selection and semantic-similarity-based selection produce equally pertinent excerpts. Both systems are perceived as fair, despite different mechanisms.

7.5.4 Per-Topic Consistency of Advantages

Table 7.7 reports per-topic mean rating differences on six key dimensions. The abbreviations correspond to: AQ = Answer Quality, AC = Answer Clarity, SA = Source Authority, SC = Source Coverage, SR = Source Relevance, BP = Balance Perception.

The source advantage is broad. ParliamentRAG achieves higher Source Coverage on 9/15 topics (1 for NotebookLM), Source Relevance on 8/15 (4 for NotebookLM), and Source Authority on 7/15 (3 for NotebookLM). These topics span the full range of thematic areas and debate intensities, ruling out benchmark artefacts. The only topic

Table 7.7: Per-topic mean rating difference ($\Delta = \mu_{\text{ParliamentRAG}} - \mu_{\text{NotebookLM}}$) on six key dimensions: AQ (Answer Quality), AC (Answer Clarity), SA (Source Authority), SC (Source Coverage), SR (Source Relevance), BP (Balance Perception). The three dimensions with negligible aggregate effects ($|d| \leq 0.03$) are omitted. Positive values (**bold**) indicate ParliamentRAG advantage. In the summary rows, PRAG = ParliamentRAG, NLM = NotebookLM.

Topic	n	**AQ**	**AC**	**SA**	**SC**	**SR**	**BP**
Aiuti militari all'Ucraina	6	**+0.50**	−0.33	**+0.83**	**+0.50**	**+1.00**	**+0.33**
Autonomia Differenziata	5	**+0.20**	**+0.80**	**+0.80**	**+0.40**	**+0.80**	±0.00
Bioetica	5	**+0.40**	±0.00	**+0.60**	**+0.80**	**+0.40**	**+0.40**
Bolkestein	6	−0.50	−0.83	**+0.17**	**+0.50**	**+0.33**	−0.17
Educazione affettiva	6	−0.83	−1.00	−0.17	−0.33	−0.17	±0.00
Gestione dei Flussi Migratori	4	**+0.25**	**+0.25**	**+0.50**	**+1.50**	**+1.00**	**+0.75**
Golden Power	2	−0.50	−0.50	±0.00	±0.00	±0.00	±0.00
Intelligenza Artificiale	6	−0.67	−0.67	±0.00	**+0.33**	±0.00	**+0.17**
Legge di Bilancio	5	−0.40	−0.20	**+0.40**	±0.00	**+0.40**	**+0.40**
Riarmo europeo	6	**+0.33**	±0.00	**+0.33**	**+0.17**	**+0.67**	±0.00
Riforma della Giustizia	3	−1.00	−0.33	±0.00	±0.00	−0.33	±0.00
Riforma del Premierato	4	−0.75	−0.50	±0.00	**+0.75**	−0.50	**+0.75**
Ritorno al Nucleare	4	−0.25	**+0.50**	−0.75	±0.00	−0.50	**+0.25**
Salario minimo	3	**+0.33**	**+0.33**	±0.00	±0.00	**+0.33**	**+0.33**
Sgravi sul costo del lavoro	2	−0.50	−0.50	−0.50	**+0.50**	±0.00	**+0.50**
Topics: PRAG > NLM		6	4	7	9	8	9
Topics at parity		0	2	5	5	3	5
Topics: NLM > PRAG		9	9	3	1	4	1

where all three source dimensions favour NotebookLM is *Educazione affettiva*, where party lines are less defined and institutional signals less discriminating.

The prose quality disadvantage is equally broad. NotebookLM achieves higher Answer Quality on 9/15 topics and Answer Clarity on 9/15 topics, confirming this as a structural consequence of the four-stage pipeline, not a query-specific artefact.

Balance Perception is consistently non-negative. ParliamentRAG achieves higher or equal Balance Perception on 14/15 topics, the strongest per-topic result. This directly supports RQ1.

Robustness check. Removing the two topics with largest Δ on Source Relevance (*Flussi Migratori*, *Aiuti militari*) still yields positive means on Source Relevance (+0.20), Source Authority (+0.17), and Source Coverage (+0.26).

7.5.5 Example: *Aiuti militari all'Ucraina*

This section examines the speaker selections for *Aiuti militari all'Ucraina*, where ParliamentRAG outperformed NotebookLM on five of six key dimensions. Table 7.8 compares the cited speakers.

Table 7.8: Speakers cited by ParliamentRAG and NotebookLM. α_{sys} and α_{base}: authority scores. Speakers selected by both are marked †. **Bold** $\Delta\alpha$ indicates ParliamentRAG advantage.

Group	**ParliamentRAG**	**NotebookLM**	$\boldsymbol{\alpha}_{\text{sys}}$	$\boldsymbol{\alpha}_{\text{base}}$	$\boldsymbol{\Delta\alpha}$
Gov.	G. Silli Undersec. Foreign Affairs	G. Meloni Prime Minister	–	–	–
FdI	G. Calovini III Comm.; NATO Parl. Ass.	S. Caiata III Comm.	**0.53**	0.50	**+0.03**
Lega	P. Formentini Vice-Pres., III Comm.; NATO	E. Zoffili Giunta delle Elezioni	0.52	0.52	±0.00
FI	A. Orsini III Comm.	A. Battilocchio XIV Comm. (EU)	**0.54**	0.51	**+0.03**
NM	I. Cavo	F.S. Romano	**0.55**	0.53	**+0.02**
PD	S. Graziano IV Comm. (Defence)	F. Porta Int'l Trade Comm.	**0.59**	0.46	**+0.14**
M5S	M. Pellegrini† IV Comm. (Defence)	M. Pellegrini†	0.64	0.64	±0.00
AVS	M. Grimaldi	L. Zanella Vice-Pres., XII Comm.	0.44	**0.62**	−0.18
Azione	E. Rosato† Comm. Republic Security	E. Rosato†	0.63	0.63	±0.00
IV	M. Del Barba†	M. Del Barba†	0.52	0.52	±0.00
Misto	B. Della Vedova†	B. Della Vedova†	0.61	0.61	±0.00

The systems agree on only 4 out of 11 speakers. Three cases illustrate how authority components interact.

Government: domain specialist vs. head of government. ParliamentRAG selects Giorgio Silli (Undersecretary for Foreign Affairs), whose institutional mandate directly covers the query topic. NotebookLM selects Prime Minister Meloni, more prominent but generalist. The authority model's role component assigns a high weight to a *sottosegretario* in a topically relevant ministry, while the committee component further boosts Silli via the III Commission embedding. This exemplifies the distinction between *prominence* and *query-specific authority*.

PD: defence committee vs. trade committee ($\Delta\alpha = +0.14$). The largest per-group gap. ParliamentRAG selects Stefano Graziano ($\alpha = 0.59$), IV Commission (Defence). NotebookLM selects Fabio Porta ($\alpha = 0.46$), International Trade Committee, with missing profession metadata and fewer defence-related speeches. The six-component design surfaces meaningful differences even between deputies of the same party.

AVS: a case favouring NotebookLM ($\Delta\alpha = -0.18$). NotebookLM selects Luana Zanella ($\alpha = 0.62$), Vice-President of XII Commission, with strong interventions and role components. ParliamentRAG selects Marco Grimaldi ($\alpha = 0.44$), whose speech content may be more topically relevant despite a lower authority score. This illustrates the design trade-off: ParliamentRAG prioritises topical relevance at retrieval and uses authority as a secondary signal.

Convergence. The four shared speakers (M5S, Azione, IV, Misto) are from smaller groups with narrow candidate pools, where semantic similarity and authority scoring converge. Three of these shared speakers have among the highest authority scores, confirming that both strategies identify the same voice when one speaker clearly dominates.

The full text of both responses is reproduced in Appendix H.

7.6 Synthesis per Research Question

All results reported below are based on evaluations by 6 domain experts (parliamentary journalists and parliamentary collaborators with 3–12 years of professional experience; Table 6.5). This is significant because the assessed dimensions (source authority, political balance, citation relevance) require familiarity with Italian parliamentary dynamics that general-purpose evaluators would lack.

RQ1 Multi-View Retrieval. The dual-channel retrieval architecture achieves $GC = 0.97$ (vs. 0.95) and $Compl = 0.99$ (vs. 1.00). Domain expert evaluators perceive ParliamentRAG as more balanced (Balance Perception: 4.66 vs. 4.48, $d = 0.28$) and rate its source coverage more favourably (4.64 vs. 4.39, $d = 0.35$). Balance Perception is non-negative on 14/15 topics.

RQ2 Speaker Authority. The central research question. The primary objective on this dimension is to assess whether the topic-dependent authority model—the core scientific contribution of this thesis—produces speaker selections that domain experts recognise as more authoritative than those of a system without dedicated authority modelling.

Mean Authority: 0.533 vs. 0.517 ($\Delta = +0.016$). The stronger evidence comes from blind expert evaluation: Source Authority (4.21 vs. 4.00, $d = 0.28$), Source Relevance (4.07 vs. 3.84, $d = 0.35$), and per-group authority votes favouring ParliamentRAG for 8/10 groups. These human-level results provide independent validation that the authority model produces meaningfully better speaker selections. Notably, the experts selected by the authority model are drawn from the retrieval pool in all evaluated queries, confirming that the broad retrieval strategy ($k = 200$) provides sufficient recall for post-retrieval authority scoring to operate effectively. As discussed in Section 7.2.3, this validation is particularly significant given the strength of the baseline: a commercially deployed, state-of-the-art RAG system operating under favourable retrieval conditions.

RQ3 Citation Integrity. $CF = 1.00$ across all queries, vs. 0.95 for NotebookLM. This is an architectural invariant: the offset-based extraction makes citation hallucination impossible by construction. NotebookLM's 5% error rate demonstrates that even a state-of-the-art commercial RAG system is not immune to citation misattribution.

RQ4 Balanced Synthesis. Near-perfect automated balance metrics. Favourable expert ratings on Balance Perception ($\Delta = +0.18$) and Source Coverage ($\Delta = +0.25$), with equivalent Balance Fairness ($\Delta = +0.01$). The trade-off is on Answer Quality ($\Delta = -0.25$) and Answer Clarity ($\Delta = -0.24$), reflecting the inherent tension between structural guarantees and generative fluency.

7.7 Limitations of the Experimental Evidence

Statistical power. With $N = 67$ paired observations and 9 tests under Holm–Bonferroni correction, the evaluation lacks power for small effects ($d < 0.3$). The absence of significance should not be interpreted as identical performance; effect sizes provide a more informative basis.

Low inter-rater agreement. Krippendorff's α values below zero on several dimensions indicate inconsistent scale interpretation. The protocol would benefit from detailed rubrics, anchor examples, and a calibration session. The analysis was conducted transparently despite unfavourable results, following the a priori plan.

Limited benchmark size. The 15 topics span 8 macro-areas but represent a small fraction of parliamentary debates. Results may not generalise to topics with different dynamics (e.g., highly technical regulatory matters or very low parliamentary activity).

Baseline non-reproducibility. NotebookLM is proprietary with opaque mechanisms that may change over time. Responses were pre-computed at a fixed point. To support independent evaluation, all experimental data (outputs from both systems and evaluator ratings) will be published alongside the thesis source code, enabling independent assessment of the outputs even though NotebookLM's generation cannot be replicated.

Authority is not used as a retrieval signal. The current architecture computes authority scores post-retrieval and does not use them to influence which chunks enter the evidence pool. The broad retrieval pool ($k = 200$ for the dense channel, augmented by graph-channel traversal) empirically contains the topically active speakers in all evaluated queries, making post-retrieval authority scoring effective in practice. Nevertheless, an alternative design could pre-compute and cache topic-cluster authority scores offline, enabling authority-weighted retrieval without per-query latency costs. This remains a direction for future work, though it would require careful design to avoid the feedback-loop risks discussed in Section 5.3.1.

Baseline advantage in source material. As detailed in Section 6.4 and Table 6.4, NotebookLM received manually curated source chunks (150 relevant + 150 distractors per topic), providing high-recall curated retrieval conditions. ParliamentRAG operates on the full knowledge graph (151K chunks) with automatic retrieval. The curated set ensures high recall (all perspectives are present) but not high precision (distractors reduce the signal-to-noise ratio). The term "near-ideal" used in the pre-print is therefore an overstatement; "high-recall curated retrieval" more accurately describes the setup. This asymmetry biases the comparison against ParliamentRAG: any advantage on retrieval-dependent dimensions (coverage, authority) is conservative.

Chapter 8

Conclusion

This chapter concludes the dissertation. Section 8.1 recapitulates the contributions. Section 8.2 reflects on the broader implications for democratic information access. Section 8.3 examines ethical considerations. Section 8.4 outlines directions for future research.

8.1 Summary of Contributions

This thesis set out to answer a motivating question: *when designing a RAG system for parliamentary debate, how can one ensure that the result is balanced, multi-perspective, and verifiable?* Four research questions were articulated in Chapter 1, and ParliamentRAG produced one contribution per question.

Topic-dependent authority scoring. The primary scientific contribution is the formalisation of a query-dependent authority model (Section 5.3):

$$\mathit{authority}(s, q) = \text{percentile}\left(\sum_{i=1}^{6} w_i \cdot c_i(s, q)\right)$$

Unlike static measures such as PageRank [12], the topic-dependent authority model produces a score that varies with the query topic, decays over time ($h = 548$ days; Equation 5.9), and is invalidated upon coalition crossings (Section 5.3.4). It integrates six components: profession ($w_{\text{prof}} = 0.15$), education ($w_{\text{edu}} = 0.10$), committee membership ($w_{\text{comm}} = 0.25$), legislative acts ($w_{\text{acts}} = 0.20$), previous speeches ($w_{\text{int}} = 0.25$), and institutional role ($w_{\text{role}} = 0.05$). The document-based design [7] computes authority only for speakers in the retrieved evidence set, ensuring efficiency while preserving the multi-view objective. The composite merging function (Equation 5.2) includes an authority term with weight 0.05; however, since the authority scorer requires the evidence pool as input, query-specific scores are not available at merge time, and all chunks receive a uniform constant value during merging. The authority weight is retained as a configuration

placeholder for a possible future feedback loop with pre-computed scores. In the current pipeline, authority scores are used operationally downstream: to select the top-authority speaker per group (expert computation) and, within the generation pipeline, to choose citing speakers and resolve semantic duplicates. Authority thus influences the final output as a soft signal without ever acting as a hard filter.

Dual-channel retrieval with diversity constraints. The retrieval module (Section 5.2) combines a dense vector similarity channel (Section 5.2.2) with a graph traversal channel (Section 5.2.3) that navigates from legislative acts to signatories to speeches. A five-component merging function (Equation 5.2) balances relevance, diversity, coverage, authority, and salience, with the coverage component (weight 0.20) ensuring all ten parliamentary groups are represented. While the individual techniques are established [1, 53], their coordination under parliamentary diversity constraints constitutes an engineering contribution.

Offset-based citation integrity. The citation system (Section 5.6) guarantees verbatim accuracy through deterministic extraction: $quote = speech.text[span_start : span_end]$, where character offsets are computed during preprocessing and stored in the knowledge graph (Section 4.3.2). The Citation Registry (Section 5.6.1), Coherence Validator (Section 5.6.3), and Integrator Guard (Section 5.5.4) together eliminate the citation hallucination problem [9, 63] by design.

Four-stage generation pipeline. The pipeline (Section 5.5) decomposes multi-view parliamentary topic summarisation into four stages: Claim Analyst (Section 5.5.2), Sectional Writer (Section 5.5.3), Narrative Integrator (Section 5.5.4), and Citation Surgeon (Section 5.5.5), with a citation-first discipline (Section 5.6.2) in which the language model writes around pre-selected evidence anchors. The separation enables per-group parallelism, intermediate integrity guards, and a fully deterministic final stage.

Together, these contributions provide a system that surfaces authoritative speakers in a topic-dependent manner (RQ2), retrieves evidence representative of the full political spectrum (RQ1), and generates multi-view summaries with verifiable citations (RQ3, RQ4).

8.2 Implications for Democratic Information Access

ParliamentRAG is situated at the intersection of NLP and democratic governance. Parliamentary proceedings are among the most important records of democratic deliberation,

yet they remain largely inaccessible due to their volume and specialised language.

From information access to informed citizenship. Traditional parliamentary portals offer keyword-based search and return documents, not synthesised positions. A citizen seeking to understand party positions must read multiple transcripts and reconstruct the political landscape manually. ParliamentRAG automates this process while preserving verifiability through deterministic citations. The shift from document retrieval to authority-aware, multi-view topic summarisation represents a qualitative change in parliamentary information access.

Structural balance as a design principle. Balance cannot be achieved through prompting alone. Language models reflect frequency disparities in corpora where some parties speak more. The architectural choices in this thesis (coverage component, mandatory per-group sections, post-retrieval authority scoring without filtering) enforce balance structurally. In particular, the post-retrieval placement of authority scoring is itself a balance mechanism: computing authority before retrieval and using it as a filter would risk systematically excluding smaller parliamentary groups whose deputies have lower aggregate visibility but may hold genuine domain expertise on specific topics (Section 5.3.1). By deferring authority to a ranking and selection role over a broad evidence pool ($k = 200$), the system preserves the multi-view objective while still surfacing the most authoritative voices within each group. This philosophy may be applicable to other domains requiring pluralistic representation: multi-stakeholder policy analysis, comparative journalism, or deliberative democracy platforms.

Authority as a double-edged instrument. The topic-dependent authority model amplifies speakers judged most relevant to a topic, which is desirable for the core use case. The experimental evaluation validates this design: in a blind A/B comparison against Google NotebookLM—a commercially deployed RAG system built on the Gemini model family and operating on manually curated high-recall source material (Section 6.4)—domain experts consistently rated ParliamentRAG's speaker selections as more authoritative (Source Authority: 4.21 vs. 4.00, $d = 0.28$) and more relevant (Source Relevance: 4.07 vs. 3.84, $d = 0.35$). That these advantages hold against a strong, well-resourced baseline reinforces the conclusion that dedicated authority modelling adds genuine value beyond general-purpose semantic retrieval. However, authority-based ranking risks reinforcing existing power structures [6]. The mitigations adopted (post-retrieval application, coverage weighting, low role weight $w_{\text{role}} = 0.05$) represent a first step toward responsible authority modelling. A more thorough fairness treatment remains an open challenge.

Verifiability as a precondition for trust. In political information, trust requires verifiability. The offset-based citation mechanism ensures every quote traces to a specific passage in an official transcript. This deterministic extraction provides a stronger guarantee than post-hoc verification, at the cost of limiting citations to verbatim passages.

Architectural complexity and user value. The experimental evaluation shows that ParliamentRAG achieves overall satisfaction parity with a strong commercial baseline (4.24 vs. 4.27) despite operating under more challenging conditions (automatic retrieval over 151K chunks vs. curated high-recall source material). This parity, however, masks a structured pattern of complementary strengths visible at the dimension level: ParliamentRAG is preferred on source quality and political balance, NotebookLM on prose fluency. The architectural complexity of ParliamentRAG—dual-channel retrieval, six-component authority model, four-stage generation pipeline—is not designed to maximise a single aggregate satisfaction score but to enforce properties (balanced representation, authority-aware speaker selection, citation integrity) that are functional requirements for professional users of parliamentary information. These properties cannot be achieved through prompting a general-purpose system, as demonstrated by NotebookLM's lower scores on Source Coverage (5.7:1 preference ratio), Source Authority (2.2:1), and its 5% citation error rate. The value of the proposed architecture is therefore best assessed not by aggregate preference but by the degree to which it delivers guarantees that its target users (journalists, parliamentary collaborators, policy analysts) recognise as superior. The evaluation data confirm that it does.

8.3 Ethical Considerations

Any system mediating access to political information carries ethical responsibility. This section examines potential risks and the design choices that mitigate them.

Can ParliamentRAG introduce bias? Yes, at multiple levels. The embedding model (`text-embedding-3-small`) may not represent all political traditions with equal fidelity. The authority model's heuristic weights could shift which speakers are surfaced. ParliamentRAG mitigates these risks structurally: mandatory per-group sections ensure no group is silently omitted, and the coverage component (weight 0.20) explicitly rewards underrepresented groups. These mechanisms do not eliminate bias in retrieved evidence but prevent it from causing systematic omission of any perspective.

Could ParliamentRAG be used to manipulate public opinion? The multi-view requirement is enforced structurally, not through prompting: the `require_all_parties` flag mandates sections for all groups, making selective omission architecturally difficult. The entire codebase is released as open-source software[1], with all configuration parameters documented (Appendix B). This transparency enables independent verification of the retrieval logic, authority computation, and generation prompts.

The risk of selective citation. Every citation is a verbatim extract from an official transcript, guaranteed by offset-based extraction. This eliminates hallucination but not *cherry-picking*: selecting only quotes supporting a particular narrative [106]. Three mechanisms reduce this risk: dual-channel retrieval broadens the evidence pool, diversity and coverage components penalise concentration, and per-group position briefs (Section 5.5.3) ensure writers receive representative evidence. These mechanisms reduce but do not eliminate the risk, as the retrieved pool is a sample (`top_k` = 200), not exhaustive.

LLM opacity. Stages 1–3 of the generation pipeline rely on `gpt-4o`, a proprietary model whose training data is not disclosed. The pipeline constrains what the LLM writes about (through claim decomposition and evidence anchoring) and verifies citation accuracy (through offset extraction), but cannot guarantee that framing and word choice are free from the political biases documented in the literature [5, 76]. ParliamentRAG ensures *structural* balance (all parties represented, all citations real) but cannot fully guarantee *tonal* balance as long as generation depends on an opaque model. Replacing `gpt-4o` with an open-weight model would reduce opacity at potential cost to generation quality.

Verifiability as an ethical safeguard. Despite these limitations, every factual claim is accompanied by a verbatim citation with speaker, date, and session identifier. The user is given the tools to check rather than asked to trust. In the context of democratic information access, the combination of structural multi-view enforcement, open-source transparency, and verbatim citation verifiability represents a principled approach to the ethical challenges of AI-mediated political information.

8.4 Future Work

The directions below are organised by increasing ambition.

[1] https://github.com/Emeierkeio/thesis-ParliamentRAG

8.4.1 Data-Driven Optimisation of Model Parameters

The topic-dependent authority model uses heuristically assigned weights (Table B.2), and the channel merger uses manually tuned coefficients (Equation 5.2). The A/B evaluation framework (Section 5.7) already collects human preference signals that could serve as supervision for a learning-to-rank approach [78] optimising component weights. Bayesian optimisation over the joint weight space is computationally feasible. This would transform the authority model from expert-designed to data-informed while preserving interpretable components.

8.4.2 Domain-Specific Embedding Models

ParliamentRAG uses `text-embedding-3-small` (Section 4.2.4) for all semantic operations. Parliamentary discourse exhibits specialised vocabulary (legislative terminology, procedural formulas, committee nomenclature) that may not be optimally captured by a general-purpose model [46]. Fine-tuning on the parliamentary corpus via contrastive learning, potentially augmented with the IPSA corpus [67] spanning 175 years, could improve retrieval recall on specialised queries.

8.4.3 Integration of Heterogeneous Sources

Parliamentary debate extends beyond the floor: deputies articulate positions through press agencies, social media, and interviews. Integrating these sources would require a *source reliability* dimension following Hwang et al. [79]:

$$score_{adjusted}(c) = authority(s, q) \cdot source_reliability(d_c)$$

Defining and calibrating reliability weights across source types constitutes a research challenge in its own right.

8.4.4 Temporal Extension: Beyond XIX Legislature

The knowledge graph schema already supports multiple legislatures through temporal properties (Section 4.4). Extending to earlier legislatures would enable longitudinal analysis of political positioning. The main challenges are conceptual: the temporal decay function ($h = 548$ days) is calibrated for intra-legislature dynamics and may require hierarchical parametrisation for cross-legislature spans. The coalition logic (Section 5.3.4) becomes more complex when deputies change party, coalition, and even chamber across legislatures.

8.4.5 Bicameral Extension: Integration of the Senate

Italy's bicameral system assigns equivalent legislative powers to both chambers. The current limitation to the Chamber of Deputies produces an incomplete picture. The Senate provides analogous open data, and the knowledge graph could be extended with a `chamber` property. The primary challenge is ensuring cross-chamber comparability of authority scores, avoiding systematic favouring of one assembly.

8.4.6 Cross-National Extension: European Parliamentary Families

The most ambitious direction is extending to a comparative, cross-national analysis. European parties are organised into transnational families (PES/S&D, EPP, ECR, Renew Europe). The ParlaMint corpus [64] provides uniformly annotated speeches from 29 European countries. The ideological compass module (Section 5.4) is language-agnostic when combined with multilingual embeddings: it operates on vectors, not raw text. The authority model's six components have structural equivalents in all European parliaments.

The enabled research question is compelling: *do parties within the same European family converge, or do national contexts prevail?* One could compare the Italian PD (S&D) with the Spanish PSOE (S&D) and the German SPD (S&D) on migration or fiscal policy, using the ideological compass to visualise convergence or divergence. Challenges include validating cross-lingual embedding comparability, adapting the authority model to institutional differences, and extending the knowledge graph with national and supranational layers.

ParliamentRAG demonstrates that treating *who speaks* as a first-class dimension of retrieval-augmented generation, alongside *what is said*, leads to measurable improvements in source authority, political balance, and citation integrity—dimensions that domain experts recognise as superior even when compared against a strong commercial baseline. The topic-dependent authority model, dual-channel retrieval, offset-based citation mechanism, and four-stage generation pipeline are elements of a coherent design in which authority awareness permeates the entire pipeline. The trade-off with generative fluency is real but addressable through future improvements to the generation stage, whereas the structural guarantees this architecture provides cannot be retrofitted onto systems that lack them. It is the hope of the author that this work contributes to making democratic deliberation more accessible to the citizens it is meant to serve.

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Appendix A

Knowledge Graph Schema

This appendix documents the complete Neo4j knowledge graph schema: all node types, properties, relationship types, and indexes. The graph contains 232,755 nodes and 488,487 relationships across 13 node labels and 15 relationship types, covering the XIX Legislature (13 October 2022 – 4 February 2026).

A.1 Node Types

Tables A.1–A.4 list all node types. Embedding properties (marked ^{J}) use `text-embedding-3-small` and are stored as JSON strings parsed at runtime. Only `Chunk.embedding` is stored as a native vector list, indexed for approximate nearest-neighbour search.

Table A.1: Node types: parliamentary actors.

Node Type	Count	Key Properties
`Deputy`	387	`id` (URI), `first_name`, `last_name`, `gender`, `education`, `profession`, `education_embedding`^{J}, `profession_embedding`^{J}, `photo` (URL), `deputy_card` (URL), `institutional_role`, `role_type`, `committee_role`, `term_of_office`, `term_of_office_start`
`GovernmentMember`	64	`id`, `first_name`, `last_name`, `gender`, `is_government` (true), `profession`, `education`, `photo`, `deputy_card`, `institutional_role`. Ministers, undersecretaries, and non-elected officials.[1]
`ParliamentaryGroup`	10	`name` (uppercase), `acronym`. The ten groups of the XIX Legislature (Table B.6).
`Committee`	80	`name`, `embedding`^{J} (name embedding). Permanent or special committees.

Table A.2: Node types: parliamentary proceedings.

Node Type	Count	Key Properties
`Session`	608	`id`, `date` (native Date), `number` (1–608), `legislature` (19), `chamber` (`"camera"`), `complete_date`, `year`, `month`, `day`
`Debate`	6,010	`id`, `title`, `order` (position within session)
`Phase`	6,515	`id`, `title`, `order` (position within debate)
`Speech`	40,416	`id`, `text` (raw transcript; used by Citation Surgeon), `preprocessed_text` (cleaned), `text_embedding`^{J}, `char_count`, `surname_name` (legacy), `merged_from_ids` (list)
`Chunk`	151,073	`id`, `text`, `index` (position within speech), `embedding` (List<Float>, native vector), `start_char_raw`, `end_char_raw` (offsets into `Speech.text`), `char_count`

Table A.3: Node types: legislative acts.

Node Type	Count	Key Properties
`ParliamentaryAct`	27,576	`uri`, `number`, `type` (interrogazione, mozione, odg, ...), `title`, `description`, `presentation_date`, `recipient` (ministry), `eurovoc` (EuroVoc descriptors), `title_embedding`^{J}, `description_embedding`^{J}, `eurovoc_embedding`^{J}

Table A.4: Node types: chat history and survey evaluation.

Node Type	Count	Key Properties
`ChatHistory`	16	`id`, `query`, `answer`, `preview`, `timestamp`, `citations` (JSON), `experts` (JSON), `commissioni` (JSON), `balance` (JSON), `compass` (JSON), `topic_stats` (JSON), `baseline_answer`, `ab_assignment` (JSON)
`SurveyEvaluation`	67	`chat_id`, `ratings` (JSON), `preference`, `timestamp`. A/B evaluation data.

A.2 Relationship Types

Table A.5 lists all relationship types with cardinalities. Relationships marked † carry temporal properties (`start_date`, `end_date`) essential for coalition-crossing detection (Section 5.3.4).

A.3 Indexes

Vector indexes. Two cosine-similarity vector indexes are defined (Table A.6). The chunk embedding index is queried by the dense retrieval channel via `db.index.vector.queryNodes`. At startup, a warmup query eliminates the cold-start latency (15–18 s). Both use `text-embedding-3-small` (1536 dimensions).

Table A.5: Relationship types in the knowledge graph with cardinalities.

Relationship	**Count**	**Source**	**Target**
Proceedings hierarchy			
`HAS_DEBATE`	6,010	Session	Debate
`HAS_PHASE`	6,515	Debate	Phase
`CONTAINS_SPEECH`	40,416	Phase	Speech
Speaker attribution			
`SPOKEN_BY`	40,372	Speech	Deputy / GovernmentMember
Text structure			
`HAS_CHUNK`	151,073	Speech	Chunk
`NEXT`	110,657	Chunk	Chunk (sequential chain)
Parliamentary membership			
`MEMBER_OF_GROUP`†	505	Deputy / GovernmentMember	ParliamentaryGroup
`MEMBER_OF_COMMITTEE`†	1,515	Deputy / GovernmentMember	Committee
Legislative engagement			
`PRIMARY_SIGNATORY`	27,373	Deputy	ParliamentaryAct
`CO_SIGNATORY`	103,959	Deputy	ParliamentaryAct
Institutional roles			
`IS_PRESIDENT`	23	Deputy	Committee
`IS_PRESIDENT`	10	Deputy	ParliamentaryGroup
`IS_VICE_PRESIDENT`	26	Deputy	Committee
`IS_SECRETARY`	26	Deputy	Committee
Government reference			
`GOVERNMENT_REFERENCE`	17	GovernmentMember	Committee

Table A.6: Vector indexes.

Index Name	**Node**	**Property**	**Dim.**
`chunk_embedding_index`	Chunk	`embedding`	1536
`act_description_embedding_index`	ParliamentaryAct	`description_embedding`	1536

Range indexes. Property-based range indexes support temporal and categorical filtering (Table A.7).

Table A.7: Range indexes for property-based filtering.

Node	**Indexed Properties**	**Index Names**
Session	id, date, number	session_id, session_date, session_number
Speech	id	speech_id
Chunk	id	chunk_id
Debate	id	debate_id
Phase	id	phase_id
Deputy	id	deputy_id
GovernmentMember	id	gov_id
Committee	name	committee_name
ParliamentaryGroup	name	group_name
ParliamentaryAct	presentation_date, type, uri, recipient	act_date, act_type, act_uri, act_recipient
ChatHistory	id	chat_history_id
SurveyEvaluation	chat_id	survey_chat_id

Appendix B

Configuration Parameters

Runtime behaviour is governed by `config/default.yaml`, modifiable at runtime via `PUT /api/config` without restart. Sensitive credentials are stored in environment variables and never exposed through the API.

B.1 Retrieval Parameters

Table B.1: Retrieval module parameters. Merger weights sum to 1.0 (Equation 5.2).

Parameter	Default	Description
`retrieval.dense.top_k`	200	Chunks returned by dense channel
`retrieval.dense.threshold`	0.30	Min. cosine sim. for inclusion
`retrieval.graph.lexical_min_match`	1	Min. keyword matches for acts
`retrieval.graph.semantic_threshold`	0.40	Min. cosine sim. for act matching
`retrieval.graph.chunk_sim_threshold`	0.30	Post-traversal chunk filter
`retrieval.graph.max_acts_per_query`	100	Max acts from graph channel
`retrieval.merger.relevance_w`	0.35	Relevance in composite score
`retrieval.merger.diversity_w`	0.15	Speaker diversity
`retrieval.merger.coverage_w`	0.20	Party coverage
`retrieval.merger.authority_w`	0.05	Authority score (placeholder)
`retrieval.merger.salience_w`	0.25	Chunk salience

B.2 Authority Scoring Parameters

Table B.2: Topic-dependent authority model parameters. Component weights sum to 1.0 (Table 5.2).

Parameter	Default	Description
`authority.weights.profession`	0.15	Profession (w_{prof})
`authority.weights.education`	0.10	Education (w_{edu})
`authority.weights.committee`	0.25	Committee (w_{comm})
`authority.weights.acts`	0.20	Legislative acts (w_{acts})
`authority.weights.interventions`	0.25	Interventions (w_{int})
`authority.weights.role`	0.05	Institutional role (w_{role})
`authority.decay.half_life`	548	Half-life h in days (Eq. 5.9)
`authority.relevance_threshold`	0.25	Min. cosine sim. for acts/interventions
`authority.normalization`	percentile	Normalisation method
`authority.max_component_contribution`	0.80	Cap on individual component scores

B.3 Compass Parameters

Table B.3: Ideological compass parameters (Section 5.4).

Parameter	Default	Description
`compass.pca.max_weight_per_frag.`	0.05	Max weight per fragment
`compass.pca.variance_threshold`	0.05	Min. total explained variance
`compass.pca.pc2_pc1_ratio`	0.40	Below this, only 1D is used
`compass.scr.min_threshold`	0.15	Min. SCR to retain fragment
`compass.kde.bandwidth_method`	scott	KDE bandwidth selection
`compass.kde.min_points`	3	Min. pts for KDE; else mean
`compass.tfidf_top_terms`	3	Top TF-IDF terms per pole
`compass.pole_purity_threshold`	0.30	Pole purity threshold
`compass.anchors.enabled`	false	Soft ideological anchoring
`compass.clip_sigma`	4.0	Z-score soft clipping
`compass.scatter_sample_size`	200	Max points in scatter plot
`compass.scatter_random_seed`	42	Fixed seed for reproducibility

B.4 Generation Parameters

Table B.4: Generation pipeline parameters (Section 5.5).

Parameter	Default	Description
generation.models.analyst	gpt-4o	LLM for Stage 1 (Analyst)
generation.models.writer	gpt-4o	LLM for Stage 2 (Sectional Writer)
generation.models.integrator	gpt-4o	LLM for Stage 3 (Integrator)
generation.temperature	0.3	Sampling temperature
generation.top_p	1.0	Nucleus sampling parameter
generation.max_tokens	4000	Max tokens per LLM call
generation.require_all_parties	true	Enforce all 10 group sections
generation.enable_synthesis	true	Convergence-divergence analysis
generation.position_brief.enabled	true	Enable position briefs
generation.position_brief.max_chunks	5	Top-N chunks per brief
generation.position_brief.chars_per_chunk	400	Max chars per chunk in brief
generation.position_brief.context_chars	500	Context chars per evidence item
citation.method	offset	Citation extraction method
citation.coherence_threshold	0.30	Min. sim. for coherence
citation.coherence_method	embedding	Validation method
citation.dedup_threshold	0.85	Sim. for citation dedup
citation.verify_on_insert	true	Re-extract and verify on insertion

B.5 Query Rewriting Parameters

Table B.5: Query rewriting parameters (Section 5.1).

Parameter	Default	Description
query_rewriting.enabled	true	Enable query expansion
query_rewriting.model	gpt-4o-mini	LLM model for rewriting
query_rewriting.max_query_words	5	Max words to trigger rewrite

B.6 Coalition Definitions

Table B.6: Coalition definitions (XIX Legislature). These drive the coalition-crossing invalidation logic (Section 5.3.4).

Coalition	**Parliamentary Groups**
Maggioranza	Fratelli d'Italia (FDI), Lega – Salvini Premier (LEGA), Forza Italia – Berlusconi Pres. – PPE (FI-PPE), Noi Moderati (NM-M-CP)
Opposizione	Partito Democratico – IDP (PD-IDP), Movimento 5 Stelle (M5S), Alleanza Verdi e Sinistra (AVS), Azione-PER-Renew Europe (AZ-PER-RE), Italia Viva – Il Centro – RE (IV-C-RE), Misto (MISTO)

Appendix C

Commissioni Topics Ontology

The committee relevance component of the authority scorer (Section 5.3.2) and the commission matching module (Section 5.2.3) rely on a curated ontology (`commissioni_topics.yaml`) mapping each permanent committee to domain-specific keywords. The ontology is used for:

1. **Authority scoring.** The `CommitteeComponent` embeds each committee's keyword list and computes cosine similarity with the query embedding.
2. **Retrieval filtering.** The `CommissionMatcher` maps queries to relevant committees, enabling the graph channel to privilege evidence from committee members.

The ontology covers the 14 permanent committees. Each entry maps 10–30 Italian-language keywords spanning the committee's jurisdiction. The full ontology is available in the source repository. Extending to other legislatures or the Senate would require updating this ontology.

Appendix D

API Specification

This appendix documents the REST API exposed by the FastAPI backend. All endpoints use JSON unless noted. The `/api/chat` endpoint uses SSE for streaming. The API is deployed at https://www.parliamentrag.it/.

D.1 Chat and Query

Table D.1: Chat and query endpoints.

Method	Path	Description
POST	/api/chat	Full multi-view pipeline with SSE streaming
POST	/api/query	Alternative endpoint (SSE or JSON)
GET	/api/chat/task/{id}	Task recovery for mobile clients

The SSE stream emits ten event types: `progress`, `commissioni`, `experts`, `citations`, `citation_details`, `balance`, `compass`, `chunk`, `baseline`, and `complete`.

D.2 Evidence

Table D.2: Evidence endpoints.

Method	Path	Description
GET	/api/evidence/{id}	Full evidence detail with quote extraction
GET	/api/evidence/{id}/verify	Deterministic citation integrity verification

D.3 Search and Graph Exploration

Table D.3: Search and graph exploration endpoints.

Method	Path	Description
GET	/api/search/results	Parliamentary search with filters
POST	/api/graph/query	Execute Cypher queries
GET	/api/graph/schema	Graph schema (labels, rel. types)
GET	/api/graph/stats	Node and relationship counts

D.4 Authority Ranking

Table D.4: Authority ranking endpoints.

Method	Path	Description
POST	/api/authority/ranking	Compute authority ranking for a topic

D.5 Configuration

Table D.5: Configuration endpoints.

Method	Path	Description
GET	/api/config	Current config (secrets excluded)
PUT	/api/config	Update config at runtime
GET	/api/config/parties	Parliamentary groups list
GET	/api/config/coalitions	Coalition definitions

D.6 History, Evaluation, and Baseline

Table D.6: History, evaluation, and baseline endpoints.

Method	Path	Description
GET	/api/history	Last 50 chat sessions
POST	/api/history	Save a chat session
GET	/api/history/{id}	Retrieve specific chat
DELETE	/api/history/{id}	Delete a chat session
POST	/api/history/{id}/baseline-experts	Extract and score NotebookLM speakers
POST	/api/surveys	Submit A/B evaluation
GET	/api/surveys/chats/pending	Chats without surveys
GET	/api/surveys/{chat_id}	Survey for specific chat
GET	/api/evaluation/dashboard	Aggregated eval. dashboard
GET	/api/evaluation/chat/{chat_id}	Metrics for specific chat

Appendix E

System Prompt Templates

This appendix documents the prompts used in the four-stage generation pipeline (Section 5.5). Full prompt texts are in the source repository[1]; descriptions below focus on structural elements essential for reproducibility.

E.1 Stage 1: Claim Analyst

Input: User query q.

System instruction: Decompose into atomic claims. For each, identify which parliamentary groups are expected to hold a salient position. Return structured JSON.

Output format:

```
{
  "claims": [
    {
      "claim": "text of atomic claim",
      "relevant_groups": ["FDI", "PD", "M5S", ...]
    }, ...
  ],
  "topic_summary": "brief description"
}
```

E.2 Stage 2: Sectional Writer

One prompt per parliamentary group (up to ten, executed in parallel).

[1] https://github.com/Emeierkeio/thesis-ParliamentRAG

Input: Atomic claims from Stage 1; retrieved evidence chunks ordered by authority score; Position Brief (top-5 chunks, 400 chars each, 500 chars context).

System instruction: Write a section on group G's position. Insert `[CIT:evidence_id]` placeholders after each supported claim. Do not generate quotation text; the Citation Surgeon inserts verbatim quotes. Cite the most authoritative speaker first.

Output format: Prose paragraph with `[CIT:id]` placeholders.

E.3 Stage 3: Narrative Integrator

Input: All party sections from Stage 2, concatenated with section delimiters.

System instruction: Merge into a single coherent narrative. Preserve *all* `[CIT:id]` placeholders exactly. Add transitional sentences. Maintain neutral tone. When synthesis is enabled, generate an "Analisi Trasversale" section.

Integrity guard: The Integrator Guard compares placeholder sets in input and output. Missing placeholders are re-inserted at section end, ensuring zero citation loss.

E.4 Stage 4: Citation Surgeon

Stage 4 is fully deterministic (no LLM).

Input: Integrated narrative with `[CIT:id]` placeholders; Citation Registry mapping each ID to source speech, offsets, and metadata.

Procedure: For each `[CIT:id]`: (1) look up evidence record; (2) extract verbatim quote via `speech.text[span_start:span_end]`; (3) adjust to sentence boundaries; (4) verify against stored `quote_text`; (5) format as `«quote» [Speaker, Party, Date, ID:id]`; (6) replace placeholder.

Coherence check: Pairwise embedding similarity between narrative and source sections; sections below $\theta_c = 0.30$ are flagged.

Appendix F

NotebookLM Baseline Prompt

This appendix reproduces the prompt used to generate NotebookLM baseline responses for the A/B evaluation (Section 6.4). The prompt imposes the same structural requirements as ParliamentRAG, ensuring that observed differences reflect architectural capabilities rather than output specification. The prompt was submitted identically for each topic (only the topic name varies). The original is in Italian.

```
Sei un editor parlamentare italiano esperto. Analizza i documenti
forniti e produci un rapporto strutturato sulle posizioni dei
partiti italiani sul tema "[TOPIC]"

STRUTTURA (rispetta questo ordine):

Introduzione
2-3 frasi con dati concreti: nomina il provvedimento in discussione,
cita il numero di interventi e deputati coinvolti, indica il periodo
temporale e le sedute specifiche quando disponibili. NON anticipare
le posizioni. NON usare grassetto per dati/statistiche.

Posizione del Governo (solo se presente)
Includi SOLO il Presidente del Consiglio (Meloni) o il ministro con
delega DIRETTAMENTE competente per il tema (es. Ministro della
Salute per sanità, della Difesa per questioni militari,
dell'Economia per bilancio). Ometti ministri non competenti. Se
nessun membro del governo è pertinente, ometti l'intera sezione.

Posizioni della Maggioranza
Fratelli d'Italia, Lega - Salvini Premier, Forza Italia, Noi Moderati

Posizioni dell'Opposizione
Partito Democratico, Movimento 5 Stelle, Alleanza Verdi e Sinistra,
```

```
Azione, Italia Viva, Misto

FORMATO:
NON usare header per i singoli partiti (no ###, no MAIUSCOLE)
Integra il partito nel testo: "Per [Partito], Cognome sostiene
che..."
Cognomi SEMPRE in grassetto
Un paragrafo per partito o ministro
Usa SEMPRE il nome completo del partito, mai abbreviazioni

CONTENUTO:
Posizioni CONCRETE: ogni frase deve indicare cosa il parlamentare
sostiene, propone o critica. Vietate frasi generiche come "ha
espresso la propria posizione" o "è intervenuto sul tema"
Citazioni dirette: Cognome (Partito, data) afferma che «testo
esatto dal documento»
Evita contenuto procedurale: no "annuncio il voto favorevole",
"ringrazio il Presidente", ecc.
Copri tutti e 10 i gruppi parlamentari. Se un partito non ha
interventi pertinenti scrivi: "Nei documenti disponibili non
emergono interventi specifici di [Partito] sul tema"
Bilanciamento: Maggioranza e Opposizione devono avere lunghezza
comparabile
Varia i verbi introduttivi: afferma / denuncia / propone / rileva /
contesta / auspica / ribadisce / evidenzia (zero ripetizioni)
Max 2-3 frasi per partito
```

The prompt encodes six requirements mirroring ParliamentRAG's structural constraints: (1) all ten groups covered with explicit no-evidence statements; (2) direct quotations with speaker attribution; (3) balanced length between coalition and opposition; (4) no procedural content; (5) concrete data in the introduction; (6) government position only when a competent minister intervened.

Appendix G

Evaluation Dataset Schema

The evaluation ground truth is stored in `backend/evaluation_set.json` as a JSON array. Each object contains:

`topic` (string): Topic name, used as query input.

`baseline_answer` (string): Pre-computed NotebookLM response.

`baseline_experts` (array): Pre-computed authority scores for NotebookLM-cited speakers. Each entry: `speaker_id`, `speaker_name`, `party`, `authority_score` (query-specific, computed with the same topic-dependent authority model as ParliamentRAG).

The `baseline_experts` scores are query-specific, ensuring methodological comparability with ParliamentRAG's authority computation.

Appendix H

Example: Full Responses

This appendix reproduces the complete responses generated by ParliamentRAG and NotebookLM for *Aiuti militari all'Ucraina* (Section 7.5.5). Both are presented verbatim in Italian. Verbatim citations are enclosed in guillemets (« »).

H.1 ParliamentRAG Response

Introduzione

Le comunicazioni del Ministro della Difesa riguardanti la proroga dell'autorizzazione alla cessione di mezzi, materiali ed equipaggiamenti militari all'Ucraina sono state al centro del dibattito parlamentare, con 73 interventi analizzati e 44 deputati coinvolti. Il periodo di discussione si estende dal 29 novembre 2022 al 15 gennaio 2026, coprendo sedute come la N. 15, 16, 24, 41, 42 e altre 21. Questo dibattito riflette l'importanza strategica e politica del sostegno militare all'Ucraina nel contesto dell'aggressione russa.

Posizione del Governo

Il Governo italiano ha espresso un chiaro sostegno all'Ucraina, condannando fermamente l'aggressione russa. Giorgio Silli afferma che «il punto di partenza chiaro e inequivocabile è la nostra ferma condanna dell'aggressione russa all'Ucraina, che consideriamo una gravissima violazione della Carta delle Nazioni Unite e dell'ordine internazionale. Per le stesse ragioni, sosteniamo convintamente l'Ucraina, la sua popolazione, la sua integrità territoriale, la sua sovranità, la sua indipendenza e la sua libertà». Il Governo si impegna a fornire supporto in vari ambiti, inclusi quelli politico, umanitario e finanziario, per garantire la stabilità e la sicurezza nella regione.

Posizioni della Maggioranza

Per Fratelli d'Italia, il sostegno militare all'Ucraina è essenziale per contrastare la strategia russa di sopraffazione. Giangiacomo Calovini sottolinea che «la strategia russa è chiara: è infliggere sofferenze morali e materiali insopportabili al popolo

ucraino per poi piegarne la resistenza; e noi, davanti a questa strategia, non possiamo permettere che questa sopraffazione abbia successo». Il partito sostiene la necessità di estendere il contributo militare fino al 31 dicembre 2025.

La Lega – Salvini Premier si mostra favorevole al sostegno militare all'Ucraina, evidenziando l'importanza di distinguere tra aggressore e aggredito. Paolo Formentini osserva che «proprio per questo motivo, nella consapevolezza che in territorio ucraino si fronteggiano un aggressore potente, la Russia, ed un aggredito, che ha bisogno di essere assistito, l'Ucraina, riteniamo corretta la scelta del Governo di confermare la politica di sostegno attuata a partire dallo scorso 28 febbraio». Il gruppo sottolinea l'importanza di mantenere un equilibrio tra gli aiuti all'Ucraina e le esigenze delle Forze armate italiane.

Per Forza Italia – Berlusconi Presidente – PPE, il sostegno all'Ucraina è parte di una strategia più ampia per garantire la sicurezza in Europa. Andrea Giorgio Felice Maria Orsini dichiara che «quindi, non truppe, ovviamente, in Ucraina – altra cosa sono gli aiuti militari già decisi –, ma personale civile di magistratura per collaborare a costruire l'Ucraina di domani, un'Ucraina libera e sicura, in un'unione di popoli liberi». Il partito ritiene che il sostegno militare sia essenziale per la difesa dell'Ucraina.

Noi Moderati (Noi con l'Italia, Coraggio Italia, UDC e Italia al Centro) – MAIE – Centro Popolare si concentra sull'importanza di sostenere l'Ucraina in un momento di crisi internazionale. Ilaria Cavo puntualizza che «sì, Ministro, stiamo aiutando il popolo ucraino a difendersi e a sopravvivere». Il gruppo riconosce la complessità della situazione e la necessità di un sostegno continuo e articolato.

Posizioni dell'Opposizione

Il Partito Democratico – Italia Democratica e Progressista sottolinea l'importanza di sostenere l'Ucraina con forniture militari per garantire un equilibrio sul campo. Stefano Graziano rileva che «e gli equilibri sul campo, oggi, se non sosteniamo l'Ucraina, se non sosteniamo il popolo ucraino, ovviamente non è che cambieranno, saranno unilaterali, cioè saranno tutti a favore di una parte». Il partito insiste sulla necessità di un'azione diplomatica più forte.

Il Movimento 5 Stelle esprime preoccupazione per il crescente coinvolgimento dell'Italia nel conflitto ucraino. Marco Pellegrini denuncia che «tale circostanza delinea, in tutta evidenza, un coinvolgimento sempre più grande del nostro Paese nel conflitto in corso, che potrebbe accrescere i rischi di una ulteriore e gravissima escalation militare, la quale ostacolerebbe, di fatto, ogni tentativo diplomatico». Il gruppo critica la strategia attuale, che privilegia l'invio di armamenti.

Alleanza Verdi e Sinistra esprime preoccupazione per la mancanza di trasparenza riguardo agli aiuti militari. Marco Grimaldi contesta che «presidente, colleghe e

colleghi, stiamo per votare il decimo pacchetto di aiuti militari all'Ucraina. Anche questa volta, a differenza di ciò che avviene in altri Paesi europei – Presidente, mi faccia dire –, l'elenco del materiale da inviare è secretato». Il partito evidenzia la necessità di maggiore trasparenza.

Azione – Popolari Europeisti Riformatori – Renew Europe sottolinea l'importanza di continuare a sostenere l'Ucraina con aiuti militari. Ettore Rosato assicura che «noi non possiamo permettercelo, come non possono permetterselo gli ucraini. Quindi il sostegno a quel Paese va continuato fino in fondo». Il partito si impegna a sostenere le missioni nel quadro delle alleanze internazionali.

Italia Viva – Il Centro – Renew Europe sottolinea l'importanza di sostenere l'Ucraina attraverso aiuti su diversi fronti. Mauro Del Barba conferma che «fondamentale è che, accanto a questa immediata condanna, siano seguiti gli aiuti su più fronti, compreso quello delle armi». Il partito evidenzia l'impegno nel supportare l'Ucraina con azioni concrete.

Il gruppo Misto si concentra sull'importanza di continuare a sostenere l'Ucraina per creare le condizioni per un negoziato di pace. Benedetto Della Vedova caldeggia che «la risposta che noi dobbiamo dare è, invece, quella di un proseguimento del sostegno, finché si possa arrivare, attraverso il rafforzamento della difesa ucraina, a creare le condizioni per un negoziato vero di pace». Il gruppo ritiene fondamentale continuare l'assistenza militare per evitare la resa del paese.

H.2 NotebookLM Response

Introduzione

Il dibattito parlamentare in esame verte sulle misure urgenti per la proroga dell'autorizzazione alla cessione di mezzi, materiali ed equipaggiamenti militari in favore delle autorità governative dell'Ucraina, affrontate nei disegni di legge S. 389, S. 974 e S. 1335. Le discussioni si estendono dal novembre 2022 al gennaio 2026, comprendendo gli interventi di oltre trenta deputati in svariate assemblee, tra cui le sedute 15, 240 e 594. I documenti delineano il quadro normativo ed esecutivo sull'impegno dell'Italia rispetto al conflitto in corso senza fornire valutazioni o anticipare gli schieramenti.

Posizione del Governo

Per il Governo, la Presidente del Consiglio Meloni (Fratelli d'Italia, 17/12/2025) evidenzia che «in gioco non ci sono solo la dignità, la libertà e l'indipendenza dell'Ucraina, ma anche la sicurezza dell'Europa nel senso più ampio del termine». La Premier garantisce inoltre l'impegno dell'esecutivo a mantenere un fronte compatto

a livello europeo per ostacolare attivamente le risorse finanziarie che alimentano la macchina bellica della Federazione Russa.

Posizioni della Maggioranza

Per Fratelli d'Italia, Caiata (Fratelli d'Italia, 30/11/2022) rileva che «il nostro impegno, oltre a essere un sostegno militare, deve innanzitutto essere un sostegno umanitario». Il deputato ritiene fondamentale limitare gli effetti della crisi per la popolazione e rinforzare i legami interni all'Alleanza Atlantica per giungere alla fine delle ostilità.

Per la Lega – Salvini Premier, Zoffili (Lega – Salvini Premier, 15/01/2026) dichiara che «la Lega, e lo ha detto anche il Ministro Crosetto, non accetterà mai l'invio di militari italiani in Ucraina». Il gruppo difende la fornitura di soli sistemi difensivi orientati a proteggere la popolazione civile e le infrastrutture sensibili dai raid missilistici.

Per Forza Italia, Battilocchio (Forza Italia, 23/01/2023) ribadisce che «è il momento di non cedere e di non esitare, confermando, sia l'aiuto umanitario, sia quello militare». Il parlamentare insiste sulla necessità di supportare costantemente Kiev per impedire una resa, in attesa che si concretizzi un valido e giusto tavolo negoziale.

Per Noi Moderati, Romano (Noi Moderati, 15/01/2026) nota che «non esiste un negoziato serio se una delle parti è costretta a trattare sotto i bombardamenti». La forza politica giudica l'assistenza armata come presupposto fondamentale e ineludibile per poter avviare la diplomazia in condizioni di reale equilibrio.

Posizioni dell'Opposizione

Per il Partito Democratico, Porta (Partito Democratico, 07/02/2024) ritiene che «nessuno di noi crede che l'Ucraina non debba esercitare il suo diritto a difendersi e che la comunità internazionale, Europa in primis, non possa aiutare con tutti i mezzi possibili l'eroica resistenza del popolo ucraino». La fazione chiarisce che gli sforzi tattici sul campo devono progredire di pari passo con un'attiva e autorevole iniziativa di pace.

Per il Movimento 5 Stelle, Pellegrini (Movimento 5 Stelle, 22/01/2025) denuncia che «se la fornitura di armi non è accompagnata da una penetrante, convincente, direi quasi ossessiva attività diplomatica e negoziale, l'unico risultato di inviare le armi sempre più potenti è di alimentare la guerra, non certo di circoscriverla». Il partito esorta il Parlamento a bloccare ulteriori pacchetti bellici per tentare di superare l'attuale stallo internazionale.

Per Alleanza Verdi e Sinistra, Zanella (Alleanza Verdi e Sinistra, 23/01/2023) contesta che «l'azione di questo Governo, sulla scia di quello precedente, si sta limitando solo alle forniture militari». Il gruppo condanna la carenza di sforzi

tangibili per ottenere il cessate il fuoco e critica fermamente i profitti miliardari accumulati dalle industrie delle armi.

Per Azione, Rosato (Azione, 28/02/2024) afferma che «purtroppo la pace, la libertà e la democrazia si difendono proprio così, anche con le armi». Il raggruppamento sollecita le istituzioni nazionali ed europee ad accelerare la produzione di artiglieria per bilanciare l'economia di guerra russa e proteggere l'intero continente.

Per Italia Viva, Del Barba (Italia Viva, 15/10/2024) propone che «accanto a questa immediata condanna, siano seguiti gli aiuti su più fronti, compreso quello delle armi». Il deputato insiste affinché l'Unione Europea non arretri e invoca l'utilizzo diretto delle ingenti risorse russe congelate a garanzia dei fondi per la difesa ucraina.

Per il Misto, Della Vedova (Misto, 08/02/2024) precisa che «è ancora più importante garantire agli ucraini la possibilità di difendersi dall'aggressione russa, perché Putin scommette esattamente sulla nostra fatica». L'esponente invita a impiegare legalmente gli immensi asset finanziari sottratti alla Federazione Russa in Occidente come contromisura per finanziare l'assistenza.

H.3 Key Differences

1. **Speaker selection.** Seven of eleven speakers differ. ParliamentRAG selects domain specialists (Silli, Calovini, Formentini, Orsini, Graziano); NotebookLM selects higher-profile or less specialised deputies (Meloni, Caiata, Zoffili, Battilocchio, Porta). See Section 7.5.5 for authority score analysis.

2. **Narrative structure.** ParliamentRAG provides contextual framing before each citation; NotebookLM follows a more formulaic pattern (speaker, date, quote, brief comment).

3. **Citation style.** ParliamentRAG's citations tend to be longer and more policy-substantive, reflecting authority-weighted evidence selection that favours substantive policy argumentation.

4. **Metadata.** ParliamentRAG reports precise quantitative metadata (73 interventions, 44 deputies, specific sessions) from the full knowledge graph. NotebookLM reports similar but less precise data, consistent with its reliance on a curated document subset.